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ZOO502 - Animal Physiology and Behavior

Topic-133 Viscosity of Blood

Viscosity of Blood

- Plasma has a viscosity of about 1.8 relative to water.
- Viscosity increases with the addition of blood cells.
- Relative viscosity of mammalian and bird blood at 37°C is between 3 and 4. Thus, blood is three or four times more viscous than water.

Effects of Blood Viscosity on Flow

- According to Poiseuille's law, the flow of blood is inversely related to its viscosity.
- Due to high viscosity, larger pressure gradients are required to maintain flow of blood through vascular system.

Blood Viscosity and Flow in Arterioles

Blood flowing through small vessels (arterioles) behaves as if its relative viscosity is much reduced.

In vessels less than 0.3 mm in diameter, the apparent relative viscosity of blood approaches the viscosity of plasma i.e. 1.8 only.

It happens due to plasma skimming i.e.

Plasma Skimming

RBCs accumulate in the center of blood vessel during flow and the walls remain relatively free of cells. This causes difference in viscosity between the center and the walls and results in increase in velocity of blood flow.

Fahraeus-Lindqvist Effect

This phenomenon is called the Fahraeus-Lindqvist effect.

- Phenomenon of plasma skimming and resultant increased blood flow velocity is called Fahraeus-Lindqvist effect

It reduces the energy required to drive blood through microcirculation.

Blood Viscosity and Flow in Capillaries

In very small vessels (diameters 5 to 7 µm) Fahraeus-Lindqvist effect is inverted i.e. apparent viscosity of blood increases. In these vessels, red blood cells move in a queue, pressing against the walls. So blood flow is slowest in capillaries.

Topic-134 Compliance in Circulatory System

Compliance

- Blood vessels contain elastic fibers that enable them to distend. As the pressure in a vessel increases, its walls are stretched and the volume of the vessel increases.
- The ratio of change in volume to change in pressure is termed as compliance of the system.
- The compliance of a system is related to its size and elasticity of its walls.
- The greater the initial volume and elasticity of the walls, the greater is the compliance of a system.

Compliance in Venous System

- Venous system is very compliant because a small change in pressure produces large change in volume. Similarly large changes in volume have little effect on venous pressure.

Effect of Compliance in Venous System

- Due to greater compliance, venous system acts as a volume reservoir without causing a significant change in blood pressure.

Compliance in Arterial System

- Arterial system is less compliant than the venous system. It acts as a pressure reservoir in order to maintain capillary blood flow.
- Some portions of arterial system are more elastic e.g. near the heart. They help to reduce the oscillations in pressure generated by contractions of the heart and to maintain flow in distal arteries during diastole.

Topic-135 Peripheral Circulation

Vascular Pathways

Cardiovascular system includes two pathways of blood circulation:

- Pulmonary circuit that circulates blood through lungs
- Systemic or peripheral circuit

Peripheral Circulation

The pathway followed by blood pumped from the left ventricle of the mammalian heart till its return to the right atrium is known as peripheral or systemic circulation.

Pathway of Peripheral Circulation

- The oxygenated blood is pumped by the left ventricle into the aorta which is the largest artery in the peripheral circulation.
- Branches from aorta form the arterial system and supply blood to the organs and major body regions.
- Arteries branch to form capillary beds in the tissues where oxygen is exchanged for carbon dioxide.
- The deoxygenated blood returns through the venous system to the superior and inferior vena cavae which open into the right atrium.

Blood Vessels

Three types of blood vessels carry blood during circulation:

- Arteries which carry blood away from the heart to organs and tissues.
- Veins which return blood to the heart.
- Capillaries which have thin walls formed of only a single layer of endothelial cells and permit exchange of materials between blood and tissues.

Topic-136 Blood Pressure

Contraction of the ventricle of heart generates blood pressure. This pressure causes the blood to flow in the arteries away from the heart from higher pressure to lower pressure areas.

Changes in Blood Pressure during Circulation

The pressure is highest in the aorta and decreases during circulation through tiny arterioles and capillaries which offer substantial resistance to flow due to narrow diameters. Blood pressure is almost nonexistent in the venous system.

Changes in Blood Pressure With Cardiac Cycle

Arterial blood pressure is highest during ventricular systole. The pressure at this time is called systolic pressure.

This pressure causes the arteries to stretch.

During diastole, when the ventricles are relaxed, pressure drops. However, a lower pressure (diastolic pressure) is maintained in the arteries due to contraction of elastic walls of arteries.

Regulation of Blood Pressure

Blood pressure fluctuates with vasoconstriction and vasodilation due to contraction or relaxation of smooth muscles in arteriole walls.

Vasoconstriction narrows the arterioles and increases the blood pressure while vasodilation causes an increase in diameter of arterioles that causes blood pressure to fall.

Contraction of smooth muscles of arteriole walls is regulated through chemical, nervous and hormonal controls. These controls are also affected by physical or emotional states e.g. exercise and stress

Chemical Control of Blood Pressure

- Hormones that affect body state during stress also act to cause vasodilation or vasoconstriction and, thereby, regulate blood pressure. These hormones include cortisol, epinephrine and norepinephrine.
- The renin-angiotensin-aldosterone system of the kidneys that regulates blood volume also regulates blood pressure.
- Nitric oxide (NO) serves as a major inducer of vasodilation in the cardiovascular system.
- A peptide, endothelin is a potent inducer of vasoconstriction.

Topic-137 Effect of Gravity on Pressure and Flow

Effect of Gravity on Arterial blood Pressure

Gravity affects the column of blood in larger arteries. So pressure in limbs and head region varies in different positions of the body.

In lying down position, the heart is at the same level as the limbs and head with respect to gravity. So, blood pressure in the arteries of head and limbs is equal.

In a standing position, the relationship between the head, heart, and limbs changes with respect to gravity.

In this position heart is below the head and about a meter above the lower limbs. So, the arterial pressure in lower limbs increases while in the head region, it decreases.

Effect of Gravity on Venous Flow

Because the vascular system is elastic, an increase in absolute pressure expands blood vessels, particularly the veins which are more compliant.

Thus, pooling of blood tends to occur in veins, in different regions of the body.

Pooling of Blood

Pooling of blood causes difficulty in maintaining capillary flow in those regions that may have severe physiological consequences e.g. change in kidney filtration rates.

Such problems are particularly acute in species with long necks and limbs e.g. giraffe, camels and dinosaurs.

Regulation of Blood Pressure in Distant Body Regions

The pressure generated by heart is regulated with position of the neck.

When the neck is raised, heart pumps blood at much higher pressure through aorta (195-300 mm Hg) to maintain arterial pressure in the brain at around 98 mm Hg.

When the neck is lowered to the ground, aortic pressure is reduced to maintain a relatively constant blood flow to the brain.

This wide variation in aortic pressure as the giraffe moves its head position can lead to extensive pooling of blood (head raised) or decreased flow (head lowered) in arterioles of regions between head and chest.

In these regions, pooling is prevented by vasoconstriction of peripheral vessels when the head is raised. When the head is lowered, extensive vasodilation of arterioles leading to capillary beds maintains flow despite the lower aortic pressure.

Arterial pressures in the legs of giraffe become higher than aortic pressures due to gravity and cause pooling.

To deal with pooling of blood, the giraffe has large quantities of connective tissue surrounding the limb vessels.

Topic-138 Counter Current Exchangers

Counter Current Exchangers

- Circulatory system controls heat flow between the interior and exterior of the body through special arrangements of blood vessels.
- In these special arrays of blood vessels, arterial and venous blood flows in opposite direction in close proximity.
- Such systems are known as counter current exchangers.

Significance of Exchangers

Counter current exchange systems are adaptive features that play a significant role in thermoregulation.

These help to exchange heat between the bloods flowing in arteries and veins and regulate the rate of heat loss via the limbs.

Additionally, they serve to exchange gases and ions between arterial and venous systems.

Principle of Counter Current Exchangers

They work by:

- Regulating the extent of blood flow near the body surface
- Trapping heat within the body core.

Presence of Exchangers

Countercurrent exchangers are formed in specific parts of an animal due to an arrangement of small arteries and veins in which blood flows in opposite directions.

They are common in the limbs of birds and mammals. In fishes they are extensively distributed in body.

In humans, this type of countercurrent exchanger is found only in the kidney.

Rete Mirabile

A countercurrent network arrangement of arterial and venous capillaries is referred to as a rete mirabile.

Before entering a tissue, an artery divides into a large number of small capillaries that parallel a series of venous capillaries leaving the tissue.

The arterial capillaries are surrounded by venous capillaries.

Rete Mirabile—Functions

This arrangement forms an extensive exchange surface between inflowing and outflowing blood.

Examples

Tuna (fish) have a large number of rete mirabile, which are used to regulate the temperature of the brain, muscles, and eyes.

The rete mirabile leading to the swimbladder of some fish (e.g. eels) function as a carbon dioxide countercurrent exchanger.

Topic-139 Capillaries and Microcirculation

Capillaries

- Capillaries are microscopic vessels about 1 mm long and 3 - 10 µm in diameter.
- They have very thin, porous walls which are completely devoid of connective tissue and smooth muscle.
- They consist of a single layer of endothelial cells surrounded by a basement membrane of collagen and mucopolysaccharides.

Capillary Beds

- In the tissues, small terminal arteries subdivide to form arterioles, which in turn subdivide to form metarterioles which subsequently form capillaries.
- Networks of these capillaries are called capillary beds or microcirculatory beds.
- The microcirculatory beds infiltrate each tissue such that any tissue cell is not more than three or four cells away from a capillary.

Material Transfer across Capillary Walls

- The endothelium composing the capillary wall is highly permeable.
- It allows substances to move with relative ease in and out of capillaries.
- Due to this, exchange of chemicals, nutrients, waste products and gases between the blood and tissue cells occurs across the walls of capillaries by diffusion.

Permeability of Capillaries

The capillaries in various tissues differ considerably in permeability.

These permeability differences are associated with the structure of endothelium.

Types of Capillaries

Based on their wall structure, capillaries are classified into three types

- Continuous capillaries
- Fenestrated capillaries
- Sinusoidal capillaries

Continuous capillaries: They are least permeable and have a continuous basement membrane. Their endothelium is about 0.2-0.4 µm thick. They are located in muscle, nervous tissue, lungs, connective tissue and exocrine glands.

Fenestrated capillaries: They are perforated by pores in some regions and exhibit intermediate permeability. They are found in the renal glomerulus, intestines and endocrine glands.

Sinusoidal capillaries: They are the most permeable as they have large paracellular gaps that extend through the basement membrane. They are present in the liver, bone marrow, spleen, lymph nodes and adrenal cortex.

Topic-140 Capillary Pressure and Flow

Precapillary Sphincter

The arterioles have smooth muscle that becomes discontinuous in the metarterioles and ends in a smooth muscle ring, the precapillary sphincter.

Control of Circulation in Capillary Beds

The innervated smooth muscle of the arterioles and the precapillary sphincter control blood distribution to each capillary bed and thus alter the blood distribution within the tissue.

Capillaries Open to Flow

Different tissues have varying numbers of capillaries open to flow. Most issues have about 30%-50% capillaries that are open at a time.

In many tissues, most of the capillaries tend to be open (e.g., in the brain) while in many other tissues e.g., in the skin, they remain closed for considerable periods.

Pressure and Flow in Capillary Beds

All capillaries in a network are only a short distance from an arteriole.

So, the pressure and flow are fairly uniform throughout the capillary bed.

Transmural pressures in capillaries are about 10 mm Hg.

Higher pressures inside a capillary result in the filtration of fluid from blood into the interstitial space.

In most tissues, there is only a small net movement of fluid across capillary walls and tissue volume remains constant

Rise in Capillary Pressure

A rise in capillary pressure, owing to a rise in either arterial or venous pressure, will result in increased loss of fluid from the blood, thus causing tissue edema.

In general, arterial pressure remains fairly constant to prevent large oscillations in tissue volume.

High Capillary Pressure in Kidneys

In the kidney, capillary pressure is high. It is called filtration pressure that helps formation of an ultrafiltrate in the kidney tubule that eventually forms urine. The kidney is encapsulated to prevent swelling of the tissue in the face of ultrafiltration

Topic-141 Regulatory Controls of Circulation

Requirement of Circulatory Regulation

Regulation of circulation is necessary to:

- Assure adequate supply of blood to the vital organs: brain and heart
- Supply adequate blood to other organs of the body according to their activity level
- Control capillary pressure to maintain tissue volume and maintain composition of interstitial fluid within reasonable ranges

Regulatory Mechanisms

The regulatory mechanisms focus on controlling cardiac output and blood pressure.

They include:

- Autoregulation
- Neural Mechanisms
- Endocrine Mechanisms

Autoregulation

- Autoregulation causes immediate, localized homeostatic adjustments.
- Chemical factors in interstitial fluids change the pattern of blood flow locally at the tissue level within capillary beds.
- These factors include local vasodilators and vasoconstrictors that act on precapillary sphincters causing them to open and close.

Local vasodilators include:

  - Decreased tissue oxygen levels or increased CO_2 levels.
  - Lactic acid or other acids generated by tissue cells.
  - Nitric oxide (NO) released from endothelial cells.
  - Rising concentrations of potassium ions or hydrogen ions in the interstitial fluid.
  - Elevated local temperature.

Local vasoconstrictors include:

  - Prostaglandins and thromboxanes released by activated platelets and white blood cells
  - Endothelins released by damaged endothelial cells.
- If autoregulation fails to normalize conditions at the tissue level, neural mechanisms and endocrine factors are activated.

Neural and Endocrine Mechanisms

- If autoregulation fails to normalize conditions at tissue level, these mechanisms are activated

Neural Mechanisms

Neural mechanisms respond to changes in arterial pressure or blood gas levels sensed at specific sites.

When those changes occur, the cardiovascular centers in the medulla oblongata and pons adjust cardiac output and peripheral resistance to maintain blood pressure and ensure adequate blood flow.

Endocrine Mechanisms

The endocrine system releases hormones that enhance short-term adjustments and that direct long-term changes in cardiovascular performance.

Topic-142 Central Control of Cardiovascular System

Cardiovascular Center

The sensory inputs of cardiovascular system are integrated in the cardiovascular center of the brain.

This center is located in the medulla oblongata and pons.

It induces reflex effects to maintain an adequate arterial pressure.

Sensory Receptors of Cardiovascular System

The body employs a variety of sensory receptors for monitoring the status of cardiovascular system.

(1) Baroreceptors: monitor blood pressure at various sites in the cardiovascular system
(2) Chemoreceptors: monitor CO_2, O_2, and pH of the blood.
(3) Cardiac mechanoreceptors
(4) Thermoreceptors

Sensory input from these receptors is transmitted to the medullary cardiovascular center.

The medullary cardiovascular center also receives inputs from:

- Medullary respiratory center
- Hypothalamus
- Amygdala nucleus
- Cortex

Role of Cardiovascular Center

- The output from the medullary cardiovascular center is fed to sympathetic and parasympathetic autonomic motor neurons.
- These neurons innervate the heart and smooth muscles of arterioles and veins. They also link to the other areas of brain e.g. medullary respiratory center.

Sympathetic Control

Stimulation of sympathetic nerves increases the rate and force of contraction of the heart and causes vasoconstriction.

It results in a marked increase in arterial blood pressure and cardiac output.

Parasympathetic Control

Stimulation of parasympathetic nerves causes reverse effects.

It results in a drop in arterial blood pressure and cardiac output.

Topic-143 Arterial Baroreceptors

Baroreceptors

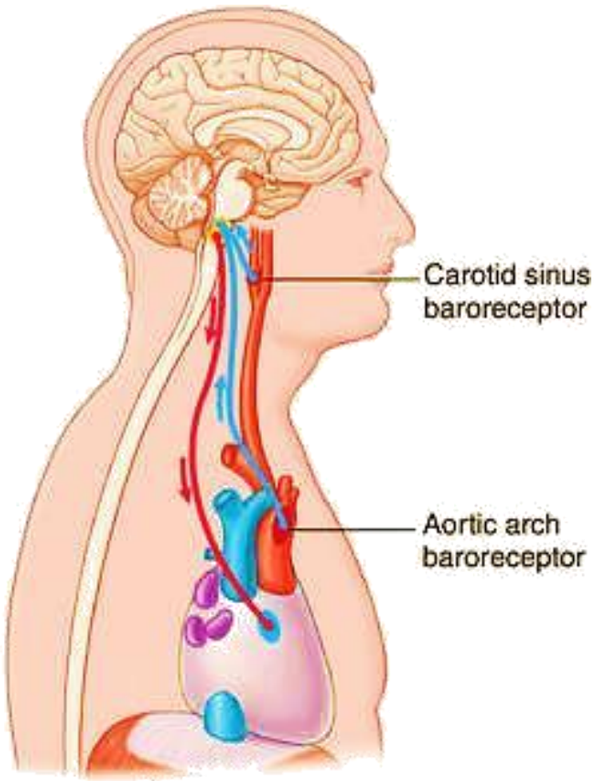
The baroreceptor system is a simple and rapidly acting control mechanism that contributes to the regulation of arterial blood pressure.

Baroreceptors in Vertebrates

Baroreceptors are widely distributed in the arterial system of vertebrates.

- The central cardiovascular system of amphibia, reptiles and mammals has unmyelinated baroreceptors.
- These unmyelinated baroreceptors only respond to pressures above normal, initiating reflexes that reduce arterial blood pressure and thus protect the animal from damaging increases in blood pressure.
- Mammals also have myelinated baroreceptors which respond to blood pressures below normal, thus protecting the animal from prolonged periods of reduced blood pressure.
- The myelinated baroreceptors are present in the mammalian carotid sinus, aortic arch, subclavian artery, pulmonary arteries and the wall of right atrium.

Myelinated Baroreceptors

- Respond to B.P. below normal
- Protect from prolonged periods of reduced B.P.
- Aortic baroreceptors monitor B.P in ascending aorta
- Respond by triggering aortic reflex—adjusts B.P. in systemic circuit
- Aortic baroreceptors monitor blood pressure within the ascending aorta. Any changes trigger the aortic reflex, which adjusts blood pressure in the systemic circuit.
- Carotid sinus baroreceptors are extremely sensitive and trigger reflexes that maintain adequate blood flow to the brain.
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Working Mechanism of Baroreceptors

Baroreceptors are stimulated by stretch of the arterial wall.

- Stimulated by stretch of arterial wall due to rise in arterial pressure
- Send nerve impulses to medulla of the brain
- Medulla responds to bring blood pressure to normal levels by:
- Decreasing cardiac output
- Widespread peripheral vasodilation

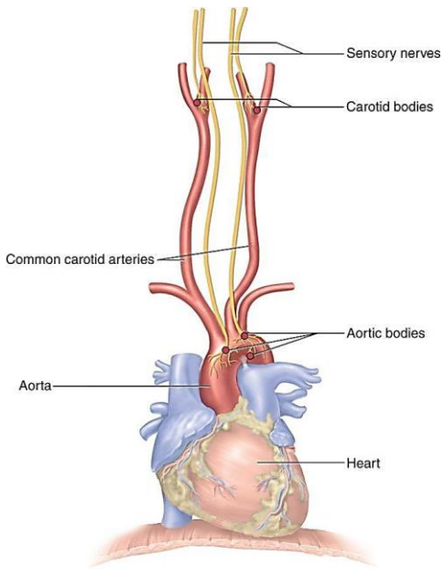
When the arterial pressure rises too high, the baroreceptors send nerve impulses to the medulla of the brain which produces two major effects that lead to a decline in blood pressure to normal levels:

(1) A decrease in cardiac output, due to parasympathetic stimulation and the inhibition of sympathetic activity. This happens due to the release of acetylcholine at the sinoatrial node.
(2) Widespread peripheral vasodilation, due to the inhibition of excitatory neurons in the cardiovascular center.

Topic-144 Arterial Chemoreceptors

Arterial Chemoreceptors

Arterial chemoreceptors are located in the carotid and aortic bodies.



They are primarily important in regulating ventilation, but they are important in the function of cardiovascular system which requires close relationship with the respiratory system.

Chemoreceptors in the cardiovascular system monitor chemical characteristics of the blood to help regulate function of both cardiovascular and respiratory systems

These chemoreceptors respond to an increase in CO_2 or to decrease in O_2 and pH of the blood.

Arterial Chemoreceptors: Effects

Chemoreceptor stimulation has a direct effect on heart rate and breathing rate.

They enhance both heart rate and breathing rate if levels of carbon dioxide are high. This helps to eliminate CO_2 from the blood.

They slow down the heart rate, if the animal is not breathing (e.g. during dive).

They also cause peripheral vasoconstriction while the cardiac output is reduced.

It increases arterial blood pressure.

It also ensures maintenance of blood pressure in arteries leading to the brain, ensuring blood flow to brain.

Topic-145 Cardiac Receptors

Cardiac Receptors

- The information about the state of the heart is collected by mechanoreceptive and chemoreceptive afferent nerve endings which innervate the atria and ventricles of heart.
- They transmit signals via the spinal cord to the medullary cardiovascular center and other regions of the brain.
- In addition, stimulation of some cardiac receptors causes hormone release either directly from the atria or from other endocrine tissues within the body.
- These hormones regulate heart rate and blood pressure.

Atrial Receptors

The atrial walls contain many mechanoreceptive afferent fibers. These include:

- Myelinated A-type afferent fibers
- Myelinated B-type afferent fibers
- Unmyelinated C-type afferent fibers
- Stretch-sensitive secretory cells

Myelinated A-type Afferent Fibers

Myelinated A-type afferent fibers respond to changes in heart rate

Myelinated B-type Afferent Fibers

- Myelinated B-type afferent fibers respond to increases in the rate of filling and volume of the atria.

- The central cardiovascular centers, in return, generate two major effects:
  (i) increased heart rate
  (ii) increase in urine excretion (diuresis) mediated by a decrease in antidiuretic hormone (ADH) levels in the blood.

Unmyelinated C-type Afferent Fibers

- Unmyelinated C-type afferent fibers innervate the junction of the veins and atria.
- Their stimulation affects both heart rate and blood pressure.
- If heart rate is low, distension of this region results in an increase in heart rate.
- If heart rate is high, stimulation results in a fall in heart rate and blood pressure.

Stretch-sensitive Secretory Cells

The stretch-sensitive secretory cells produce atrial natriuretic peptide (ANP).

This hormone causes an increase in urine production and sodium excretion, thereby effectively reducing blood volume and therefore blood pressure.

Mode of Action of ANP

- ANP inhibits release of renin by the kidney and production of aldosterone by the adrenal cortex.
- It thus diminishes the renin-angiotensin-aldosterone system, which stimulates sodium resorption and an increase in blood volume
- ANP also inhibits release of ADH and acts directly on the kidney to increase water and sodium excretion.

Ventricular Receptors

- The endings of both myelinated and unmyelinated sensory afferent fibers are also embedded in the ventricle.
- They are stimulated by interruption of coronary blood flow.
- At low stimulation levels, these fibers cause increased sympathetic outflow and decreased vagal outflow to the heart, raising cardiac contractility as well as blood pressure.
- At higher stimulation levels, these fibers are necessary for the perception of pain in the heart.

Topic-146 Nervous Control of Microcirculation

Role of Nervous System in Maintaining Blood Pressure

Nervous control serves to maintain arterial pressure by:

- Adjusting resistance to blood flow in the peripheral circulation.
- controlling the number of capillaries open at any moment

The nervous control of capillary flow ensures that if arterial pressure falls, blood flow to the gut, liver, and muscles is reduced to maintain flow to the brain and heart.

Nervous Innervations of Arterioles

- Most arterioles are innervated by sympathetic nerves, which release norepinephrine at their endings.
- Some arterioles are innervated by parasympathetic nerves, which release acetylcholine at their endings.

Sympathetic Nerve Stimulation

- Stimulation of sympathetic nerves causes release of norepinephrine.
- Binding of the norepinephrine to α-adrenoreceptors in the smooth muscle of arterioles causes vasoconstriction and therefore a decrease in diameter of the arterioles.

- This decrease in diameter causes an increase in resistance to flow, thus reducing blood flow through that capillary bed.

Microcirculation of Heart

- The action of norepinephrine is extensively modulated by neuropeptide Y in the microcirculation of heart.
- The atrial and ventricular myocardium and the coronary arteries are surrounded by nerve fibers that contain neuropeptide Y.
- Neuropeptide Y decreases coronary blood flow and the contraction of cardiac muscle.

Parasympathetic Stimulation

Arterioles of the brain and lungs are innervated by parasympathetic nerves.

These nerves contain cholinergic fibers, which release acetylcholine from their nerve endings.

In mammals, parasympathetic nerve stimulation causes vasodilation in arterioles.

Topic-147 Local Control of Microcirculation

Many compounds influence capillary blood flow locally within a tissue.

These can be grouped into three categories:

- Endothelium-produced compounds
- Inflammatory mediators
- Metabolites and other secretions

Endothelium-produced Compounds

The vascular endothelium releases nitric oxide, endothelin, and prostacyclin that affect vascular smooth muscle and, therefore, capillary blood flow.

- Nitric oxide is produced and released continuously by the vascular endothelium. It causes relaxation of vascular smooth muscle and acts as vasodilator.
- Endothelins are proteins that cause vasoconstriction.
- Prostacyclin causes vasodilation and acts as an anticoagulant.

Inflammatory Mediators

- Thromboxane A, formed in the plasma by platelets causes vasoconstriction.
- Histamine released from connective tissue and white blood cells in injured tissues in mammals causes a marked vasodilation of vessels
- Plasma kinins (e.g. bradykinin) are another group of potent vasodilators in damaged tissues.

Metabolites and other Secretions

- Norepinephrine released from sympathetic nerves acts as vasoconstrictor.
- Angiotensin II formed primarily in the lungs and circulates in the blood acts as vasoconstrictor.
- Serotonin acts as a vasoconstrictor or vasodilator, depending on the vascular bed and concentration. It is found in high concentration in the gut and blood platelets.

Respiratory Physiology

Topic-148 Respiration: General Considerations

Gas Exchange

Animals take in oxygen from the environment and give off CO_2. This exchange of oxygen and carbon dioxide with the environment is called gas exchange or external respiration or breathing.

Oxygen is used for cellular respiration. Carbon dioxide is generated as waste during this process. Removal of CO_2 is important because its accumulation in the body causes the pH to fall that may be lethal.

Gas Exchange Mechanism

Oxygen and carbon dioxide are transferred passively by diffusion from the environment across the respiratory surface which may be skin or special respiratory epithelium.

Features of Respiratory Surfaces

A respiratory surface has specialized features that facilitate gas exchange.

1. Large Surface Area and Low Diffusion Distance
2. Thin epithelium
3. High vascularization

Surface Area of Respiratory Epithelium

An animal's O_2 requirement and CO_2 production increase as a function of mass.

However, the rate of gas transfer across the body surface is related primarily to surface area and diffusion distance.

To facilitate the rate of gas transfer, surface area of respiratory epithelium should be large and diffusion distances as small as possible.

In very small animals the distances for diffusion are small, and the ratio of surface area to volume is large. For this reason, diffusion across the body surface is sufficient in small animals, such as rotifers and protozoa, which are less than 0.5 mm in diameter.

Need for Specialized Respiratory Organs

Increase in size results in increase in diffusion distances and reductions in the ratio of surface area to volume.

Surface Area of Respiratory Organs

Large surface-area-to volume ratios are maintained in larger animals with the evolution of specialized respiratory organs with extensive respiratory epithelium for the exchange of gases.

This surface comprises the major portion of the total body surface. In humans, for instance, the respiratory surface area of the lung is between 50 and 100 m^2.

Thin Epithelium

The surface specialized for gas exchange is made up of a thin layer of cells, the respiratory epithelium, which is 0.5 to 15 µm thick.

Highly Vascular Epithelium

Vascularization

A respiratory surface should have an extensive capillary network just beneath it. Blood flowing through the capillaries reduces diffusion distances required for the diffusion of gases.

Gas Transfer System

A gas-transfer system in animals takes in oxygen from the environment takes it to the issues through blood and brings CO_2 from tissues, transporting it to the gas exchange surface and removes it by diffusion.

Such a system involves four basic steps:

1. Breathing movements, which assure a continual supply of air or water to the respiratory surface (e.g., lungs or gills)
2. Diffusion of O_2 and CO_2 across the respiratory epithelium
3. Bulk transport of gases by the blood
4. Diffusion of O_2 and CO_2 across capillary walls between blood and mitochondria in tissue cells

Topic-149 Respiratory Pigments

Respiratory Pigments

- Respiratory pigments are the substances that transport respiratory gases, oxygen and CO_2 in the blood.
- They are complexes of proteins and metallic ions.
- Each respiratory pigment has a characteristic color that changes with its O_2 content.
- The color of the pigment is reflected in the color of blood.

Types of Respiratory Pigments

Various respiratory pigments found in animal kingdom include:

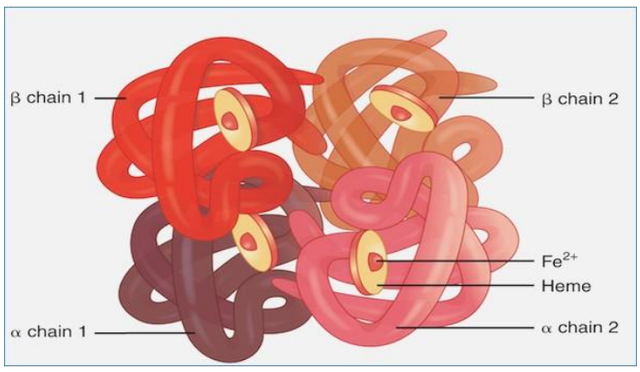
hemoglobin, myoglobin, hemocyanin, hemerythrin and chlorocruorin.

Hemoglobin

- Hemoglobin is the most common respiratory pigment found in all vertebrate and many invertebrate groups.
- Its color is bright red when loaded with O_2 and dark maroon when deoxygenated.
- If O_2 is bound, the molecule is referred to as oxyhemoglobin; if O_2 is absent, it is called deoxyhemoglobin.

Structure of Hemoglobin

- A Hemoglobin molecule contains four iron containing heme groups, associated with a tetrameric globin protein.
- Iron, in hemoglobin, is in the ferrous state (Fe^{2+}).



Oxygen Carrying Capacity of Hemoglobin

- Hemoglobin (and other respiratory pigments) increases the O_2 binding capacity of the blood.

- The concentration of oxygen in blood with hemoglobin is 20 ml per 100 ml blood at 37°C at atmospheric pressure.
- It is a 70 times increase in capacity as compared to a blood without hemoglobin i.e. 0.3 ml O_2 per 100 ml blood.

Formation of Methemoglobin

- Binding of O_2 to hemoglobin to form oxyhemoglobin does not oxidize ferrous iron (Fe^{2+}) to ferric iron (Fe^{3+}).
- Certain substances oxidize the ferrous iron in hemoglobin to the ferric state. This oxidation produces methemoglobin.
- Methemoglobin is nonfunctional and does not bind O_2. So, higher concentration of methemoglobin in blood impairs oxygen transport.
- Red blood cells contain an enzyme methemoglobin reductase, which reduces methemoglobin to the functional ferrous form, hemoglobin.
- Certain compounds (e.g., nitrites and chlorates) act either to oxidize hemoglobin or to inactivate methemoglobin reductase, thereby increasing the level of methemoglobin.

Hemoglobin Inhibition by Carbon Monoxide

Carboxyhemoglobin

- The affinity of hemoglobin for carbon monoxide is about 200 times greater than its affinity for oxygen.
- Carbon monoxide displaces oxygen and saturates hemoglobin, even at very low partial pressures of carbon monoxide. This causes a marked reduction in oxygen transport to the tissues.
- Hemoglobin saturated with carbon monoxide is called carboxyhemoglobin.
- Saturation of hemoglobin with CO has effects similar to that of oxygen deprivation on oxidative metabolism. This condition is called anoxia which may impair brain functions and prove lethal.

Myoglobin

- Myoglobin is a respiratory pigment that stores O_2 in vertebrate muscles.
- It has a much higher oxygen affinity than hemoglobin.

Hemocyanin

- Hemocyanin is a large, copper-containing respiratory pigment found in many molluscs and Arthropods.
- Unlike hemoglobin, hemocyanin is not packaged in cells.
- In its oxygenated form, it is light blue and colorless in its unoxygenated form.

Hemerythrin

- Hemerythrin is used for oxygen transport by members of two phyla of marine invertebrates (Priapulida and Brachiopoda) and only one member of Annelida.
- It is colorless when deoxygenated, but turns violet-pink in the oxygenated state

Chlorocruorin

- It is found free floating in the blood plasma of many annelids, particularly marine polychaetes.
- It is a dichromatic compound that appears green in dilute solutions and light-red in concentrated solutions.

Topic-150 Transport of Oxygen in Blood

Transport of Oxygen

Oxygen is absorbed from the respiratory epithelium by the blood. In the blood most of oxygen binds with hemoglobin (97%) which carries and transports oxygen as oxyhemoglobin.

A small proportion (3%) of O_2 also gets dissolved in the plasma.

Oxygen Capacity of Hemoglobin

The ability of hemoglobin to bind with oxygen is called oxygen carrying capacity of the blood.

This capacity is directly proportional to the partial pressure of oxygen, P_{O2}.

Each hemoglobin molecule can combine with four oxygen molecules, each heme combining with one molecule of oxygen.

If all sites on the hemoglobin molecule are occupied by O_2, the blood is 100% saturated and the oxygen content of the blood is equal to its oxygen capacity.

Hemoglobin can absorb maximum O_2 at the sea level. The maximum amount of O_2 which normal human blood absorbs and carries at the sea level is about 20 ml/100 ml of blood, which is the oxygen capacity.

Liberation of Oxygen in Tissues

Oxyhemoglobin is unstable and dissociates rapidly into hemoglobin and oxygen at low oxygen concentration and less pressure.

This forms the basis of liberation of oxygen in the tissues.

When oxygen pressure falls below 60 mm Hg, as in many cells and tissues, the splitting of oxygen from hemoglobin occurs very sharply.

Carbonic anhydrase enzyme present in RBC's facilitates this activity.

Factors Affecting Oxygen Carrying Capacity

The hemoglobin-oxygen affinity is labile and depends on three important factors:

1. **Effect of Temperature on O_2 Carrying Capacity**
   Temperature: Rise in temperature causes a decrease in oxygen carrying capacity of blood.
2. **Effect of CO_2 on O_2 Carrying Capacity**
   Carbon dioxide Concentration: O_2 tension decreases as CO_2 pressure increases and the capacity of hemoglobin to hold oxygen becomes less. In this way increased CO_2 tension favors the greater liberation of O_2 from the blood in the tissues.
3. **Effect of pH on O_2 Carrying Capacity**
   pH: As the pH of blood declines, the affinity of oxygen to bind hemoglobin also declines. This phenomenon is termed as Bohr effect, or Bohr shift.
   This occurs because decreased pH results from an increase in hydrogen ions and the hydrogen ions combine with the protein parts of the hemoglobin molecules, causing a decrease in the ability of hemoglobin to bind oxygen.
   Conversely, an increase in blood pH results in an increased ability of hemoglobin to bind oxygen.

Bohr effect

- Low pH means increased hydrogen ion conc.
- H^+ combine with protein parts of hemoglobin molecules
- This decreases ability of hemoglobin to bind O_2

- Conversely: Increase in blood pH increases hemoglobin ability to bind oxygen—causing failure in liberating O_2

Topic-151 Transport of Carbon Dioxide in Blood

Carbon Dioxide Release

Cells release CO_2 generated by cellular metabolism into intracellular spaces

Carbon dioxide dissolves freely in the tissue fluid surrounding the cells. From the tissue fluid, dissolved Carbon dioxide passes to the plasma within the blood capillaries.

Carbon Dioxide Transport in Blood

Carbon dioxide is transported in the blood, to the capillaries of the lungs, in several different states.

- CO_2 is transported in blood, from tissues to the capillaries of lungs
- Transport occurs in three forms:
- As carboxy-hemoglobin
- By binding to plasma proteins
- As bicarbonate ions

1. **Transport of CO_2 as Carboxyhemoglobin**
   About 23% of Carbon dioxide is carried as carboxyhemoglobin. Carboxyhemoglobin is formed when carbon dioxide combines with amino group of globin part of hemoglobin.
2. **Transport by Binding to Plasma Proteins**
   About 7% Carbon dioxide is carried by binding to plasma proteins.
3. **Transport of CO_2 as Bicarbonate Ions**
   About 70% Carbon dioxide is carried as bicarbonate ion in the plasma.
   As carbon dioxide from tissue fluid enters the capillaries, it combines with water to form carbonic acid.

$$CO_2 + H_2O \rightleftharpoons H_2CO_3$$

The carbonic acid is an unstable compound and splits quickly and ionizes to produce hydrogen ions and bicarbonate ions.

$$H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

HCO3- ions travel through circulation to the lungs
In the lungs, all these reactions occur in reverse order

Bicarbonate ions combine with hydrogen ions to form carbonic acid which splits into water and carbon dioxide.

$$HCO_3^- + H^+ \longrightarrow H_2CO_3 \longrightarrow H_2O + CO_2$$

Carbon dioxide diffuses out from the capillaries of the lungs into the space of alveolar sac.

Topic-152 Vertebrate Lungs

Lungs in Vertebrates

Terrestrial vertebrates (amphibians, reptiles, birds, and mammals) use a pair of lungs to exchange oxygen and carbon dioxide between their tissues and the air.

Basic Plan of Vertebrate Lung

Considerable structural variations occur among the four groups.

The generalized vertebrate lung consists of a complex network of tubes and sacs or air spaces.

The sizes of terminal air spaces become progressively smaller in the order: amphibians, reptiles, and mammals. However the total number of air spaces per unit volume becomes greater.

Alveoli make the respiratory epithelium of lungs. Respiratory epithelium is thin and well vascularized.

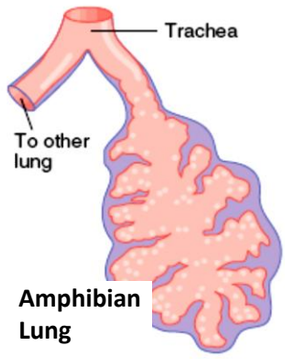
Amphibian Lungs

The structure of amphibian lungs vary in sub-groups.

In some urodeles, it is just like a smooth-walled pouch.

Gas exchange is limited to the outer surface of the lung.

Lungs in frogs and toads are balloon-like which are subdivided by septa and folds into numerous interconnected air sacs.

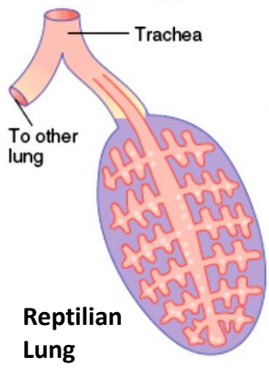


Amphibian Lung

Reptilian Lungs

The lungs of most reptiles have a single bronchus running down the centre, from which numerous branches reach out to individual pockets throughout the lungs.

These pockets are similar to alveoli in mammals, but much larger and fewer in number. These give the lung a sponge-like texture.



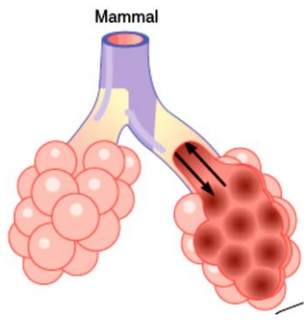
Reptilian Lung

Mammalian Lungs

Subdivision of lungs in mammals is much more than any other group.

The trachea subdivides to form bronchi, which branch repeatedly, leading eventually to terminal bronchioles and finally respiratory bronchioles, each of which is connected to a terminal spray of alveolar ducts and sacs.

Subdivision of lungs, forming smaller sacs, results in an increase in respiratory surface area per unit volume of lung.

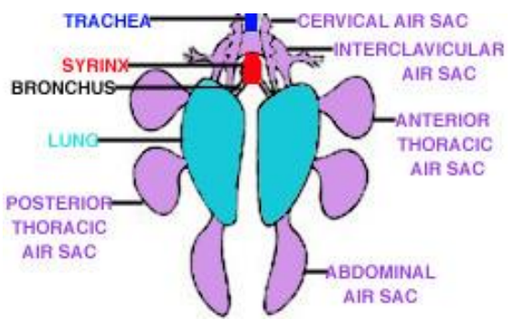


Birds Lungs

Lungs of birds are relatively small, but are connected to 8 or 9 air sacs that extend through much of the body. These air sacs are connected to air spaces within the bones.

The lungs contain millions of tiny parallel passages called parabronchi. Like the alveoli in other lungs, they are the site of gas exchange.

The air sacs, which hold air, do not contribute much to gas exchange, as they are poorly vascularised.



Parabronchi allow a unidirectional flow of air. This enables the bird's lungs to remove 90% of oxygen from air with each breath, whereas a mammal can remove only 25%.

Topic-153 Functional Anatomy of Mammalian Lungs

Air Pathway to Lungs

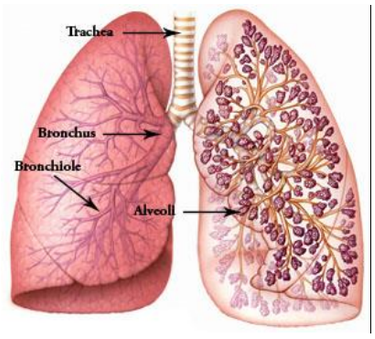
Lungs are located in the thoracic cavity.

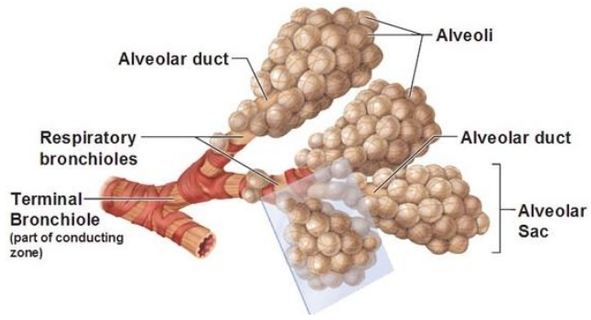
Air enters through the nostrils and enters trachea through the pharynx.

A system of branching ducts conveys air to the lungs.

Functional Anatomy of Mammalian Lungs

Trachea subdivides to form bronchi, which branch repeatedly, leading eventually to terminal bronchioles and finally respiratory bronchioles. Each respiratory bronchiole is connected to a terminal cluster of alveolar ducts and sacs (alveoli). The terminal bronchioles, the respiratory bronchioles, the alveolar ducts, and the alveolar sacs constitute the respiratory portion of the lung.





Cross Sectional Area of Airways

Although the diameter of individual air ducts decreases from the trachea to the terminal bronchioles, total cross-sectional area of the airways increases rapidly as a result of extensive branching.

Gas Exchange Surface

Gases are transferred across the thin-walled alveoli. The airways leading to the terminal bronchioles constitute the non-respiratory portion of the lung.

Prevention from Collapse

Air ducts leading to the respiratory portion of the lung contain cartilage and a little smooth muscle. This prevents air ways from collapse.

Topic-154 Lung Volumes and Capacities

Lung Volumes and Capacities

Lung volumes and lung capacities refer to the volume of air associated with different phases of the respiratory cycle.

Lung volumes are directly measured; Lung capacities are inferred from lung volumes.

Lung Volumes

Lung volumes are of five types:

1. Resting Tidal volume
2. Inspiratory reserve volume

3. Expiratory reserve volume
4. Residual volume
5. Minimal volume

Resting Tidal Volume

The amount of air that moves into or out of lungs during a single respiratory cycle under quiet, resting conditions. The resting tidal volume averages about 500 mL in human male and female both.

Inspiratory Reserve Volume

The amount of air that can be inhaled by maximum inspiratory effort. On average, the lungs of males are larger than those of females. For this reason, the inspiratory reserve volume of males averages 3300 mL, while in females, it is 1900 mL.

Expiratory Reserve Volume

The amount of air that can be voluntarily expelled after tidal volume by maximum expiratory effort. A male human can expel an additional 1000 mL of air with maximum use of the accessory muscles. Female expiratory reserve volume averages 700 mL.

Residual Volume

The amount of air that remains in lungs even after a maximal exhalation.

It is typically about 1200 mL in males and 1100 mL in females.

Minimal Volume

The minimal volume is a component of residual volume. It is the amount of air that would remain in lungs if they are allowed to fully collapse. The minimal volume ranges from 30 to 120 mL.

Unlike other volumes, it cannot be measured in a healthy person.

Lung Capacities

Respiratory capacities can be calculated by adding the values of various volumes. These include:

Inspiratory Capacity

The sum of tidal volume and inspiratory reserve volume. It is the amount of air that can be drawn into lungs after a complete respiratory cycle.

Functional Residual Capacity

FRC is the sum of the expiratory reserve volume and the residual volume.

It is the amount of air remaining in lungs after a complete respiratory cycle.

Vital Capacity

It is the maximum amount of air that can be moved into or out of lungs in a single respiratory cycle.

It is the sum of the expiratory reserve volume, the tidal volume, and the inspiratory reserve volume. It averages around 4800 mL in males and 3400 mL in females.

Total Lung Capacity

It represents the total volume of lungs.

It is calculated by adding the vital capacity and the residual volume.

The total lung capacity averages around 6000 mL in males and 4200 mL in females.

Topic-155 Mechanism of Lung Ventilation in Mammals

Lung Ventilation Breathing

Lung ventilation or breathing is a mechanical process that consists of two phases: Inspiration and expiration.

During inspiration, fresh air moves in and in expiration air with low oxygen and high CO2 content is moved out of the lungs.

Anatomical Features for Breathing in Mammals

Certain anatomical features help the lungs of mammals to carry out ventilation.

1. Lungs are spongy in nature. They neither pull air in nor push it out. During inspiration, passive expansion of elastic lungs occurs and expiration is due to a passive contraction of lungs.
2. At the floor of thoracic cavity, in which lungs are situated, is a diaphragm. Diaphragm is a muscular sheet. The shape of diaphragm is domelike when its muscles are relaxed. When the muscles of diaphragm contract, its shape becomes less domelike.
3. Walls of the thoracic cavity are composed of ribs. Between the ribs, external and internal intercostal muscles are present which act as antagonistic pair. Contractions or relaxations of these muscles cause the rib cage to elevate or drop and increasing or decreasing pressure on the lungs.

Mechanism of Inspiration

During inspiration, the space inside the thoracic cavity is increased in two ways:

Firstly, the external intercostal muscles contract and cause the ribs to move forward and upward, elevating the rib cage.

Secondly, the muscles of diaphragm also contract, making the diaphragm less domelike.

The combined effect of these two actions is the expansion of thoracic cavity, resulting in decrease of air pressure inside the lungs which, in turn, expand.

With the expansion of lungs, vacuum is created inside the lungs and air rushes passively into them from outside due to higher atmospheric pressure i.e. inspiration happens.

Mechanism of Expiration

During expiration, the external intercostal muscles relax and the ribs move downward and inward.

At the same time, muscles of the diaphragm also relax making it more domelike.

The combined effect of both these actions is the reduction in the volume of thoracic cavity.

This exerts pressure on the lungs, causing higher pressure in them resulting in moving out of the air passively i.e. expiration.

Topic-156 Mechanism of Ventilation in Birds

Birds Respiratory System

Birds' respiratory system consists of lungs and large air sac (numbering 7 or 9).

Air Sacs of Birds

- Two posterior (abdominal)
- Two anterior thoracic (cranial)
- Two posterior thoracic (caudal)
- Two cervical (these are not present in some species)
- One interclavicular

The air sacs have branches which also extend into the bones.

Respiratory cycle of a bird

Respiration in birds requires two ventilation cycles to move the air through the entire respiratory system. A breath of inhaled air remains in the respiratory system for two complete inhalation and exhalation cycles before it is fully used and exhaled out the body.

First Inspiration

During the first inspiration, the air travels through the nostrils, and passing through the larynx enters the trachea.

From the trachea air does not go directly to the lung, but instead travels to the posterior abdominal air sacs.

Simultaneously, a small amount of air passes through the posterior abdominal air sacs to the lung through the ventrobronchi and dorsobronchi.

In the lungs, these bronchi continue to divide into smaller diameter parabronchi and air capillaries.

Blood capillaries flow through the air capillaries and this is where the oxygen and carbon dioxide are exchanged.

First Expiration

During the first expiration, the spent air in the lungs flows out the body through the trachea. This air is replaced by the fresh air from the posterior air sacs.

Second Inspiration

When the bird inspires the second time fresh air again enters both the posterior sacs and the lungs. Spent air in the lungs is again displaced by incoming air, but it cannot exit through the trachea because fresh air is flowing inward. Instead, the spent air from the lungs enters anterior thoracic (cranial) air sacs.

Second Expiration

On the second expiration, the air moves out of the cranial air sacs and the lungs, and flows out through the trachea. Fresh air in the posterior sacs again enters the lungs for gas exchange.

Maintenance of Air Flow: Ventilation

Birds do not have a diaphragm.

Air is moved in and out of the respiratory system through pressure changes in the air sacs.

Muscles in the chest cause the sternum to be pushed outward. This creates a negative pressure in the air sacs, causing air to enter the respiratory system.

Expiration is not passive, but requires certain muscles to contract to increase the pressure on the air sacs and push the air out.

Efficiency of Birds Lung Ventilation System

The respiratory system of birds is more efficient than that of mammals, transferring more oxygen with each breath.

This pattern of airflow through the respiratory system creates unidirectional (one-way) flow of fresh air over the gas exchange surfaces in the lungs. Furthermore, fresh air passes over the gas exchange surfaces during both inhalation and exhalation, resulting in a constant supply of fresh air enabling the bird to experience a near-continuous state of gas exchange within the lungs.

Topic-157 Mechanism of Lung Ventilation in Reptiles

Thoracic Cavity in Reptiles

The ribs of reptiles like snakes, lizard and crocodilians, form a thoracic cage around the lungs like those of mammals. In tortoises and turtles, the ribs are fused to a rigid shell.

Reptiles do not possess a diaphragm. However, pressure differences between the thoracic and abdominal cavities occur, indicating a functional separation of these cavities.

Breathing in Snakes and Crocodilians

Inhalation

During inhalation, posterior movement of the ribs causes expansion of thoracic cavity.

Expansion of thoracic cavity reduces pressure in the lungs below atmospheric pressure. Air is drawn passively into the lungs by passing through open nares and glottis.

Exhalation

Elastic recoil of the lungs and the movement of ribs and body wall, compress the lungs and air is expelled.

Breathing in Tortoises and Turtles

The ribs of turtles are a part of the shell; thus, movements of the body wall to which they attach are impossible.

Turtles exhale by contracting muscles that force the viscera upward, compressing the lungs.

They inhale by contracting muscles that increase the volume of the visceral cavity, creating negative pressure to draw air into the lungs.

Topic-158 Mechanism of Ventilation of Lungs in Frogs

Respiratory Apparatus of Frog

In frogs, the nose opens into a buccal cavity. Buccal cavity is connected to paired lungs via the glottis.

The lungs are balloon shaped. Each lung consisting of small thin walled chambers called alveoli. Alveoli increase the surface area of the lungs.

Inspiration

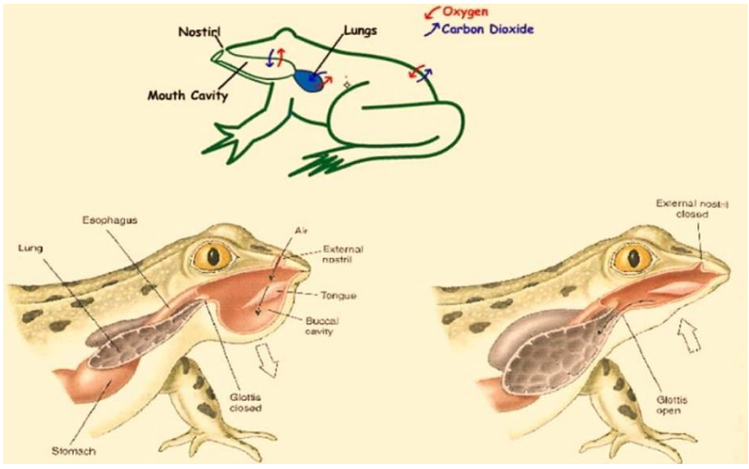
Frogs use positive pressure pumping mechanism in which air is pushed into the lungs by the pumping pressure of buccal cavity.

The frog can open and close its nares and glottis independently.

Steps of Inspiration

- Air is drawn into the buccal cavity with the nares open and glottis closed. Floor of the mouth is lowered.
- Then the nares are closed, glottis is opened, and the buccal floor is raised.
- This forces the air from the buccal cavity into the lungs.

This lung-filling process may be repeated several times in sequence, before ventilation cycle begins.



Expiration

Expiration is complex and occurs in steps. The lungs release air in portions to the buccal cavity.

Expiration may not be complete, so that some of the air from the lung is mixed with ambient air in the buccal cavity and then pumped back into the lungs.

During repeating sequence of inspirations, when fresh air is drawn into the buccal cavity, opening of glottis occurs. This opening of glottis causes back flow of air from lungs into the buccal cavity. As the floor of buccal cavity is raised and fresh air is pushed into the lungs, the exhaled air also pumped back into the lungs.

That is, a mixture of pulmonary air, presumably low in O_2 and high in CO_2 is mixed with fresh air in the buccal cavity and returned to the lungs.

The reason for this complex method of lung ventilation may be directed toward reducing oscillations in CO_2 levels in the lungs in order to stabilize and regulate blood P_{CO2} and control blood pH.

After few cycles, a full ventilation cycle (exhalation) happens that repeats the inhalation steps in reverse order.

Topic-159 Pulmonary Surfactants and Alveolar Collapse

Tension in Lung's Wall

Lungs have the ability to stretch and expand due to tension in the lung wall.

The tension in lung wall depends on:

- Properties of the alveolar wall that allow stretching
- Surface tension at the liquid-air interface that causes resistance to stretch

Pulmonary Surfactants

The alveoli of a healthy lung have relatively low surface tension due to the presence of surfactants.

Surfactants are lipoprotein complexes that bestow a very low surface tension on the liquid-air interface of alveoli.

Lung surfactants not only reduce the effort associated with breathing but also help prevent collapse of alveoli.

Surfactants are produced by type II cells within the alveolar lining and have a half-life of about 12 hours in mammals.

Surfactants are found in the lungs of all terrestrial vertebrates i.e. amphibians, reptiles, birds, and mammals.

Alveolar Collapse

Alveoli are very small bubble-like structures that are inflated and deflated alternately.

The continuous inflation and deflation creates mechanical problems that might cause them to collapse.

The air pressure inside the alveoli with thin walls may cause them to burst or join with other larger alveoli with lower pressure, forming larger sacs. This is known as alveolar collapse.

It results in reduced surface area available for gas exchange.

Prevention of Alveolar Collapse

Alveolar collapse normally does not occur in the lung for two reasons:

- surrounding tissue helps prevent overexpansion of alveoli
- presence of pulmonary surfactants

Role of Pulmonary Surfactants in Preventing Collapse

Due to presence of surfactant on the surface of alveoli, the wall tension increases when the surface film is expanded and decreases when it is compressed.

When their volume decreases (deflation), alveoli fold and the layer of surfactant between the folds becomes thick.

Thick layer of surfactant reduces the surface tension. This permits easy inflation of folded alveoli.

This effect of surfactants also minimizes pressure differences between large and small alveoli, thus reducing the chance of collapse.

Topic-160 Neural Control of Breathing

Central Processor of Breathing

The integration of breathing movements in all air-breathing vertebrates results from the central processing of many sensory inputs.

The central processor consists of:

- a pattern generator that determines the depth and amplitude of each breath
- a rhythm generator that controls breathing frequency

Medullary Respiratory Centers

The central rhythm generator that maintains breathing rhythm in human is located in the ventral medulla.

Rhythmic activity is enhanced by neurons in the pons and medulla. Some neurons just anterior to the medulla cause prolonged inspiration in the absence of rhythmic drive from the pons.

Thus the medulla contains a central rhythm generator that drives the pattern generator within the medullary respiratory center to cause breathing movements.

Neural Control of Mammalian Lung

Mammalian lung is ventilated by the action of diaphragm and muscles between the ribs.

These muscles are activated by spinal motor neurons and the phrenic nerve, which receive inputs from medullary respiratory centers.

The control of respiratory muscles can be very precise, allowing extremely fine control of air flow, as is required for such complex actions in humans as singing, whistling, and talking, as well as simply breathing.

Inspiratory and Expiratory Neurons

The medullary respiratory center contains inspiratory neurons, whose activity coincides with inspiration, and expiratory neurons, whose activity coincides with expiration.

Respiratory rhythm depends primarily on the activity of inspiratory neurons.

Exhalation is a passive process, which does not depend on activity in expiratory neurons.

Expiratory neurons become active only when inspiratory neurons are inactive, and then they show a burst pattern somewhat similar to that of inspiratory neurons.

Roles of Central Processor

- Receives pH input and adjusts ventilation to maintain blood pH.
- Integrate breathing movements with feeding, talking and singing, or other body movements.
- Generate coughing or sneezing reflexes to protect respiratory epithelium from environmental hazards.
- Optimize breathing patterns to minimize energy expenditure.

Topic-161 Factors Affecting Breathing Rate and Depth

Factors Affecting Breathing Rate and Depth

Chemical stimuli affecting ventilation are changes in pH and concentration of O_2,and CO_2.

Rate and depth of ventilation is also changed with emotions, sleep, lung inflation and deflation, lung irritation, variations in light and temperature and requirements for speech.

Breathing rate and depth can also be controlled consciously.

Receptors of Respiratory Stimuli

Several types of receptors respond to these stimuli and send information to the medullary respiratory center for integration. Medullary respiratory center causes reflex changes in the rate and/or depth of breathing.

Monitoring O_2, CO_2 and pH Levels

Three types of chemoreceptors monitor changes in O_2 and CO_2 and pH levels in arterial blood:

- Peripheral Chemoreceptors
- Central Chemoreceptors
- Type-J Chemoreceptors

Peripheral Chemoreceptors

They are found in the carotid bodies and aortic bodies of mammals, in the carotid body of birds, and in the carotid labyrinth of amphibians.

In teleost fishes, chemoreceptors are located in the gills.

In all cases, the chemoreceptors are innervated by branches of ninth (glossopharyngeal) or tenth (vagus) cranial nerve.

Peripheral chemoreceptors respond to falling O_2 levels in water and blood and are important in increasing ventilation during periods of hypoxia.

Central Chemoreceptors

They are found in mammals and many other air breathing vertebrates. They are located in the medulla and respond to decrease in the pH of the cerebrospinal fluid (CSF) due to high CO_2 levels. Stimulation of these H^+-sensitive receptors causes reflex increase in breathing.

Type-J Chemoreceptors

A third group of receptors are present in the lung. They are positioned close to the pulmonary capillaries in interstitial spaces. These are called juxtapulmonary capillary receptors, or type J receptors.

Stimulation of type J receptors elicits a sensation of breathlessness. Violent exercise results in a rise in pulmonary capillary pressure and an increase in interstitial volume, which could cause stimulation of type J receptors and therefore breathlessness.

Topic-162 Respiratory Responses to Hypoxia

Hypoxia is a condition in which the body or a region of the body is deprived of adequate oxygen supply.

Respiratory Responses of Aquatic Animals to Hypoxia

Oxygen level in aquatic environment varies frequently. Aquatic animals are subjected to more frequent and rapid changes in oxygen levels.

Many aquatic animals can withstand very long periods of hypoxia. Many of these animals utilize a variety of anaerobic metabolic pathways to survive the period of reduced oxygen availability.

Many animals adjust the respiratory and cardiovascular systems to maintain oxygen delivery during reduced oxygen availability. For example, many fishes increase gill ventilation rate in hypoxic conditions. Ram ventilators (e.g. tuna) swim forward with their mouth open, increase the size of gap during hypoxia to increase water flow over the gills.

Responses to Hypoxia by Animals at High Altitudes

There is a gradual reduction in oxygen concentration with altitude.

Animals vary in their capacity to climb to high altitudes and withstand the reduced ambient oxygen levels.

A reduction in environmental oxygen results in a decrease in blood P_{O2}.

This stimulates the carotid and aortic body chemoreceptors, causing an increase in lung ventilation in mammals.

Rise in lung ventilation leads to an increase in CO_2 elimination and a decrease in blood P_{CO2} that leads to an increase in pH of the CSF.

Increase in CSF pH tends to reduce ventilation, thereby attenuating the hypoxia-induced increase in lung ventilation.

If hypoxic conditions are maintained, both blood and CSF pH are returned to normal levels by the excretion of bicarbonate.

This acclimatization takes about one week in humans.

Topic-163 Respiratory Responses to Hypercapnia

Hypercapnia

Hypercapnia is a condition of abnormally elevated carbon dioxide levels in the blood.

Animal Responses to Hypercapnia

In many animals, hypercapnia results in an increase in ventilation, in proportion to the rise in CO_2. A marked increase in ventilation occurs almost immediately in response to a rise in CO_2 and is maintained for long periods till compensatory mechanisms take control.

Receptors of Hypercapnia

Increase in ventilation rate is mediated by the activity of several receptors that send messages to the medullary respiratory center. These receptors include the chemoreceptors of the aortic and carotid bodies and the mechanoreceptors in the lungs.

Hypercapnia and pH Fall

An increase in carbon dioxide levels in the blood also cause the pH to fall. This change in pH is responded by the central H^+ receptors which also cause increase in ventilation rate.

Normalization: Role of Bicarbonates

Ventilation rate eventually returns to a level slightly above the level that prevailed before hypercapnia.

This return to a value only slightly greater than the initial ventilation level is related to increases in levels of plasma bicarbonate and CSF bicarbonate.

Increase in bicarbonate level brings the pH to normal even though the raised CO_2 levels are maintained. Correction of CSF pH is very important in the return of ventilation to normal.

Topic-164 Respiratory Responses to Diving

Diving

Many air-breathing vertebrates (e.g. dolphins, whales and water birds) that live in water may go under water (submerge or dive) for varying periods of time.

They have to rise to the surface to breathe. The time between breaths varies with the diver, but is mostly around 10-20 minutes for many diving vertebrates.

Dealing with Hypoxia and Anoxia during Dive

Diving mammals and birds are subjected to periods of hypoxia during submergence. The mammalian central nervous system (CNS) cannot withstand anoxia and must be supplied with oxygen throughout the dive.

Diving animals solve the problem by utilizing oxygen stores in the lungs, blood, and tissues.

Many diving animals have high hemoglobin and myoglobin levels, and their total oxygen stores generally are larger than those in nondiving animals.

To minimize depletion of available stores, oxygen is preferentially delivered to the brain and the heart during a dive. Blood flow to other organs may be reduced, and these tissues may adopt anaerobic metabolic pathways.

During prolonged dives, metabolic rates and thus oxygen needs often are reduced in such animals (e.g., elephant seals).

Control of Inhalation and Exhalation during Dive

Receptors that detect the presence of water and that inhibit inspiration during a dive are situated near the glottis and near the mouth and nose (depending on the species).

The decrease in blood O_2 levels and increase in CO_2 levels that occur during a dive do not stimulate ventilation because inputs from the chemoreceptors of the carotid and aortic bodies are ignored by the respiratory neurons while the animal is submerged.

Topic-165 Gas Exchange across Gills of Fishes

Gills as Respiratory Organs of Fishes

Fishes live in an environment that contains less than 2.5% of the oxygen present in air.

To maintain adequate levels of oxygen in their bloodstream, fishes must pass large quantities of water across gill surfaces and extract the small amount of oxygen present in the water.

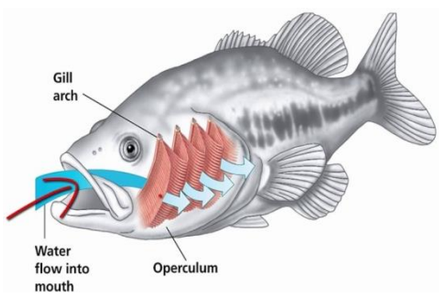
Fishes have gills which are efficient respiratory organs for gas exchange in water.

Flow of Water over Gills in Bony Fishes

Most of bony fishes (teleosts) have a muscular pumping mechanism to move water over the gills.

This branchial pump is powered by the skeletal muscles surrounding the pharynx and opercular (branchial) cavity.

Water is drawn into the mouth, passes over the gills, and exits through the gill clefts.



Flow of Water over Gills in Cartilaginous Fishes

Cartilaginous fishes don't have an operculum.

Many of them have gill bars with external flaps that can close and form a cavity functionally similar to the opercular cavity of bony fishes.

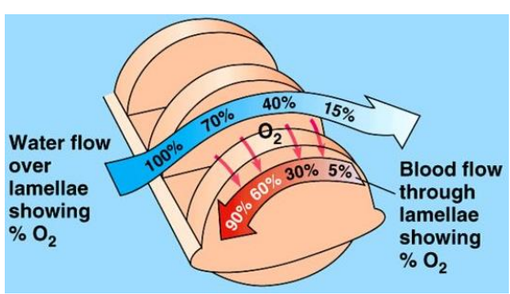
Some elasmobranchs and open-ocean bony fishes (e.g. tuna), maintain water flow over gills by ram ventilation. They keep their mouths open while swimming so as to ventilate the gills by the forward motion of the body. These fishes must keep moving to survive.

Gas Exchange across Gills

Gas exchange across gill surfaces is very efficient.

Gas exchange occurs as blood and water move in opposite directions on either side of the lamellar epithelium.

This countercurrent exchange mechanism provides very efficient gas exchange.



It maintains oxygen concentration gradient between the blood and water over the entire length of the capillary bed.

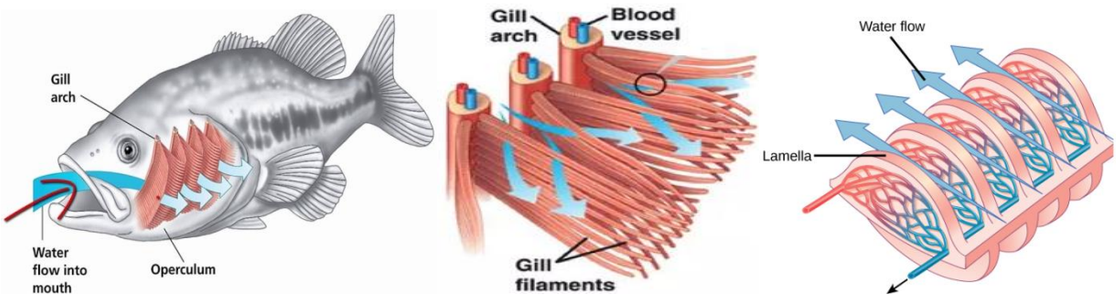
This arrangement provides the greatest possible extraction of oxygen from water.

Topic-166 Functional Anatomy of Gills of Fishes

Gills

Gills are effective respiratory organs for gas exchange in water.

They have a thin, moist and highly vascularized layer of epidermis that permits gas exchange through water medium.



Anatomy of Fish Gills

Each gill consists of two rows of thin, dorsoventrally flattened filaments supported by a gill arch.

Each filament has an upper and a lower row of vascular folds of epithelium, called pharyngeal lamellae.

Pharyngeal Lamellae

Pharyngeal lamellae represent the respiratory portion of the gill.

Each gill lamella is covered by a thin sheet of epithelial cells.

The inner lamellar wall is formed by pillar cells. Pillar cells are associated with an extensive collagen network, which gives them support.

Branchial arteries that carry blood to the gills break into capillary beds in pharyngeal lamellae.

The lamellae of successive filaments in a row are in close contact so that the whole gill forms a sievelike structure.

Lamellar Channels

Successive lamellae form slitlike channels between them through which water flows.

These channels are about 0.02-0.05 mm wide and about 0.2-1.6 mm long.

Due to the small size of channels, water flows in thin sheets between the lamellae. As a result diffusion distance between water and blood flowing in capillaries is reduced to a maximum of 0.01 -0.025 mm.

Topic-167 Osmoregulation and Excretion

Osmoregulation and Excretion

Water is the solvent in the cells that makes solutions with solutes i.e. salts, electrolytes and ions. Each cell is adapted to a defined quantity of water in relation to salts in it to perform its functions. Changes in the concentration of either water or solutes in it disturb the homeostatic balances. Osmoregulation and excretion are the homeostatic mechanisms that maintain water and solute balance.

The mechanism of regulation of solutes and concentration of water in the cells and body fluids of animal is called osmoregulation.

The mechanism which eliminates nitrogenous wastes is referred as excretion.

Correlation of Osmoregulation and Excretion

Metabolic nitrogenous wastes e.g. ammonia and urea disturb the osmotic balance of the cells at toxic levels. Their removal is called excretion and is necessary to maintain osmotic balances. So, osmoregulation and excretion are correlated processes.

Osmolarity

Cells consistently encounter changing extracellular environment.

An external medium that is dilute as compared to the cell concentration is designated as hypotonic environment.

The more concentrated external environment is termed as hypertonic environment.

The environment whose solute concentration resembles to that of cell's internal solution is termed as isotonic.

Effects of Osmolarity

Hypotonic environments osmotically cause entry of water into the cell and dilute the cell solution. The cell may become turgid or swells and even burst.

Hypertonic environment causes the cells to lose water and renders cell solutions concentrated.

Importance of Osmoregulation

Animal cells require critical balance of water and solutes in the body as they cannot survive a net water gain or loss. Water continuously enters and leaves the cells, however, the quantity of water and solutes is kept in balance through osmoregulation.

Topic-168 Osmoregulators and Osmoconformers

Osmoregulators

Animals whose body fluid concentrations differ noticeably from the outside environment are called osomoregulators.

They actively regulate water and salt to maintain an internal osmolarity different from the medium in which they are living.

They discharge excess water in hypotonic environment and excrete salts in hypertonic conditions.

Their internal tissues are sensitive to minor changes in extracellular osmolarity. So they need to actively regulate water and electrolytes to maintain cell volume.

Animals living in different environments e.g. marine, freshwater and terrestrial environments have distinct adaptations to regulate osmotic balance.

Most vertebrates are osmoregulators. Only exceptions are elasmobranchs and hagfishes.

Marine vertebrates keep their blood solute concentration hyposmotic to seawater while fresh water fishes maintain hyperosmotic composition as compared to freshwater.

Among invertebrates, some fresh water and many terrestrial invertebrates are osmoregulators.

Osmoconformers

Animals whose body fluids are kept isotonic to the external environment are called osmoconformers.

They do not actively control the osmotic condition of their body fluids and conforms to the osmolarity of the ambient medium.

All marine invertebrates are osmoconformers i.e. in osmotic balance with seawater. Ionic concentrations in their body fluids parallel the seawater in which they live.

Among vertebrates elasmobranchs and hagfishes are osmoconformers.

Osmoconformers display a high degree of cellular osmotic tolerance.

Their cells can cope with high plasma osmolarities and maintain cell volume.

They do so by increasing their intracellular osmolarities with intracellular organic osmolytes. Osmolytes e.g. urea and trimethylamine oxide, utilized by marine elasmobranchs, are substances which reduce the need to maintain osmotic pressure with inorganic ions.

Topic-169 Osmoregulation in Aquatic Environments

Environmental Osmolarities and Osmotic Problems

Animals face distinct osmotic problems related to water and salt exchange in aqueous environments.

The osmolarities of aqueous environments range from a few milliosmoles per liter in freshwater lakes to about 1000 mosm/L in ordinary seawater, or even more in landlocked salt seas.

Intermediate environments, such as brackish bogs, marshes, and estuaries, have salinities ranging between these extremes.

As a rule, animal body fluids (i.e. interstitial fluids and blood) tend away from the environmental osmotic extremes.

The nature of osmotic problems faced by freshwater and marine animals and their mechanisms for dealing with them are different.

Stenohaline Animals

Most oceanic invertebrates can tolerate only a narrow osmotic range and have very limited abilities to withstand osmotic change.

They cannot tolerate substantial changes in external osmolarity and are said to be stenohaline.

If they are exposed to dilute seawater, they die quickly because their body’s cells cannot tolerate dilution and are helpless to prevent it. So, these animals are restricted to living in a narrow salinity range.

An example is the marine spider crab.

Euryhaline Animals

Euryhaline animals can survive large fluctuations in external osmolarity. They can tolerate a wide range of salinities and demonstrate varying powers of osmotic regulation with changing salt concentration in water.

Euryhaline animals are found along coasts and in estuaries and river mouths where conditions are much less constant than those of the open ocean. Here animals must be able to withstand large and often abrupt changes in salinity as the tides ebb and flow and mix with fresh water draining from rivers.

They include certain osmoconformers and osmoregulators. Many barnacles and mussels are euryhaline osmoconformers while striped bass and the various species of salmon are euryhaline osmoregulators.

Topic-170 Osmoregulation in Fresh Water Animals

Osmolarity of Freshwater Animals

The body fluids of freshwater animals, including invertebrates, fishes, amphibians, reptiles, and mammals, are generally hyperosmotic to their aqueous surroundings.

Freshwater vertebrates have blood osmolarities in the range of 200 to 300 mosm/L, while the osmolarity of freshwater is generally less than 50 mosm/L.

Osmotic Problems of Freshwater Animals

They face two kinds of osmoregulatory problems in hyposmotic aqueous surroundings:

- They are subject to osmotic flooding of body fluids which results in swelling.
- They are subject to continuous loss of body salts to the surrounding medium.

Osmoregulatory Needs of Freshwater Animals

Freshwater animals need to:

1. Prevent net gain of water and get rid of excess water
2. Prevent loss of salts and gain salts to replace for net loss

Osmoregulatory Strategies of Freshwater Animals

Almost all fresh water animals are osmoregulators.

- Freshwater protozoa (Amoeba, paramecium) have contractile vacuoles for pumping out excess water.
- As a general rule, vertebrates living in freshwater refrain from drinking water, to reduce excess water entry into the body.
- Most freshwater vertebrates (fishes and amphibians) produce a copious amount of dilute urine to avoid a net gain of water. The nephrons of freshwater fishes possess large glomeruli for more filtration and relatively short tubule systems that allow little water reabsorption.
- The useful salts are largely retained by reabsorption into the blood from the ultrafiltrate in the tubules of the kidney.
- Yet, some salts pass out in the urine that causes a potential problem of loss of biologically important salts such as KCl, NaCl, and $CaCl_2$.
- Lost salts are replaced, in part, from ingested food.
- Salts are also actively transported across the epithelium of skin (in amphibians) or gills (in fishes) from the external dilute medium.
- In some freshwater animals, especially reptiles, birds, and mammals, water uptake and salt loss are minimized by an integument having low permeabilities to both salts and water.

Topic-171 Osmoregulation in Marine Animals

Osmoregulation in Marine Invertebrates

The body fluids of marine invertebrates and ascidians (primitive chordates) are maintained at similar levels (isosmotic) to seawater both in osmolarity and in the plasma concentrations of individual major inorganic salts.

Such animals need not expend much energy in regulating the osmolarity of their body fluids.

Osmoregulation in Marine Vertebrates: Hagfishes

Plasma of hagfishes is isosmotic with marine water.

However, the concentration of individual ions is regulated. They maintain their blood Ca^{2+}, Mg^{2+} and SO_4^- ions at significantly lower concentrations than seawater. They also maintain Na^+ and Cl^- at higher levels than sea water.

Osmoregulation in Marine Vertebrates: Cartilaginous Fishes

Cartilaginous fishes such as sharks, rays, and skates have plasma that is approximately isosmotic to seawater.

However they maintain far lower concentrations of electrolytes (i.e., inorganic ions).

They excrete excess inorganic electrolytes such as NaCl via the kidneys and also by means of a special excretory organ, the rectal gland, located at the end of the alimentary canal.

They maintain an isotonic osmolarity with sea water due to organic osmolytes such as urea and trimethylamine oxide (TMAO).

High urea concentration tends to cause the breakup of proteins, whereas trimethylamine oxide has the opposite effect. So, it stabilizes protein structure in the face of high urea levels.

Osmoregulation in Marine Vertebrates: Teleost Fishes

The body fluids of marine teleosts (modern bony fishes) are hypotonic to seawater.

These fishes have a tendency to lose water, especially across the gill epithelium.

To replace the lost volume of water, they drink saltwater that also results in excess salt uptake.

Most of the excess salts are subsequently eliminated from the blood. Na^+, Cl^- and some K^+ are eliminated by active transport across the gill epithelium. Ca^{2+}, Mg^{2+}, and SO_4^- are excreted in urine through the kidneys.

The urine produced is isotonic to the blood, but rich in salts.

The net result of the combined osmotic work of gills and kidneys in the marine teleosts is a net retention of water.

Topic-172 Osmoregulation in Terrestrial Animals

Osmoregulatory Problems in Terrestrial Environment

The terrestrial animals often face the problem of both water and salt losses.

Water loss occurs due to:

- Water permeable epidermis
- Respiratory epithelia
- Sweat and urine excretion

Salt loss is caused due to:

- Sweat
- Urine excretion

Maintaining Water and Salt

- Terrestrial animals maintain a balance between water and salt loss and gain.
- These animals feed on food materials which are moist and rich in salts to compensate the deficiency of water and salts.
- They either drink large amount of water to replenish the lost water or conserve water by physiological and behavioral means.

Physiological Adaptations

Physiological adaptations to meet the osmotic problems in terrestrial animals (e.g. reptiles, birds, mammals, some crustaceans and annelids) are:

- The body is usually covered by a water proof material which prevents the evaporation of water from the body. This covering may be of horny scales (as in certain reptiles), feathers (as in birds) and hair or fur (as in mammals).
- The loss of water through urine and feces is checked by various mechanisms. Water is absorbed from feces by rectum and cloaca and water from urine is absorbed by loop of Henle in the kidney.
- Many terrestrial animals, which do not drink water directly, such as birds, snakes and lizards excrete a semisolid urine containing uric acid crystals, thus minimizing water loss.
- The animals of extreme terrestrial habitat, the dry hot desert, have developed special means of water conservation.

Topic-173 Osmoregulation in Desert Living Animals

Challenges of Desert Environment

The animals living in desert are faced with excessive heat, evaporative water loss and near absence of free freshwater.

Such animals have special physiological adaptations for survival in hot and dry environment.

Kangaroo Rat—Model of Desert Adaptations

Kangaroo rats live in some of the hottest and driest desert environments. They eat only dry seeds and never drink water. Their survival strategies include a variety of osmoregulatory physiological adaptations characteristic of many small desert mammals.

- They have adapted a nocturnal lifestyle i.e. remaining in burrows during daylight and coming out only at night. This reduces evaporative water loss. The cool burrows reduce the animal's temperature load and also reduce respiratory water loss.
- Their nasal epithelium has cooling properties and has a countercurrent mechanism for absorbing respiratory moisture.
- They have adapted to utilize metabolic water in place of drinking water.
- They excretes a highly concentrated urine and also absorb water from the feces in the rectum to produce essentially dry fecal pellets.

By using all these adaptations for desert survival, the kangaroo rat greatly reduces its potential water loss, required for survival.

Camels—Large Desert Mammals

- Large mammals like camels can go for long periods without water in hot and dry deserts.
- Their primary adaptation is their ability to withstand extreme body dehydration. A camel can lose as much as 40 per cent of the water in its body fluids and still survive.
- When deprived of drinking water, camels allow their body temperature to rise during hot day time rather than losing water by evaporative cooling or sweating.

- During the cooler night, their body temperature drops.
- The body temperature of a dehydrated camel may vary from 35°C at night to 41°C during the day.
- This strategy of heating during the day and cooling at night is impossible in small rodents, whose body temperatures oscillate much more rapidly than in the larger animals.
- The camel also reduces heating by orienting to give minimal surface exposure to direct sunlight.
- The camel, like other desert animals, produces dry feces and concentrated urine.
- When water is not available the camel does not produce urine but stores urea in the tissues.
- They can tolerate not only dehydration but also high urea levels in the body.
- When water becomes available, they rehydrate by drinking larger amounts of water.

Topic-174 Osmoregulation in Marine Mammals

Osmoregulatory Challenges of Marine Mammals

Marine mammals, such as whales, porpoises, sea lions and seals face osmoregulatory problems similar to those of desert animals because they live in an environment without available drinking water.

They avoid drinking seawater as they lack extrarenal salt-secreting glands. Their physiological responses, although different in detail, are generally similar to those of desert mammals.

Osmoregulatory Strategies of Marine Mammals

- They lose water through excretion and through moisture in the exhaled air. So, their basic osmoregulatory emphasis is on water conservation.
- They get their water entirely from their food intake and its subsequent metabolism.
- They depend primarily on their kidneys for maintaining osmotic balance. They have highly efficient kidneys capable of producing very hypertonic urine.
- To reduce water loss via ventilation, seals have a characteristic labyrinth-like proliferation of epithelial surfaces in the nasal passages. Whales and dolphins have a blow hole which cools the air passing through it. Many have an ability to suspend breathing for quite longer time (up to 40 minutes) with and alternating deep breathing period of about 5 minutes. This strategy reduces respiratory water loss.
- The females have an additional water loss in their milk when they are nursing young, but this loss is counterbalanced by a concentration of milk. The milk of these mammals is about ten times as concentrated in fat as is cow’s milk.

Topic-175 Osmoregulation in Terrestrial Arthropods

Osmoregulatory Challenges

Terrestrial environments are highly desiccating. The major challenges for insects and other arthropods are:

- Acquisition of water
- Conservation of water

Acquisition of Water

- Insects do not take water directly through drinking.
- Most insects obtain water from their food.
- Some insects and arthropods have the ability to extract water vapor directly from the air.

Extraction of Water Vapors from Air

- The terrestrial arthropods having the ability to extract water vapors from the air include certain arachnids (ticks, mites) and a number of wingless forms of insects, primarily larvae.
- In these arthropods, the site of entry of water vapors is the rectum and mouth.
- The principle of absorption of water in vapor form is that highly concentrated salt solutions absorb water from air.
- Insects create very concentrated salt solutions in their rectum and mouth that can absorb water from air.
- In the rectum this role is played by NaCl while in the mouth, salivary glands secrete a highly concentrated solution of KCl to absorb water from the air.

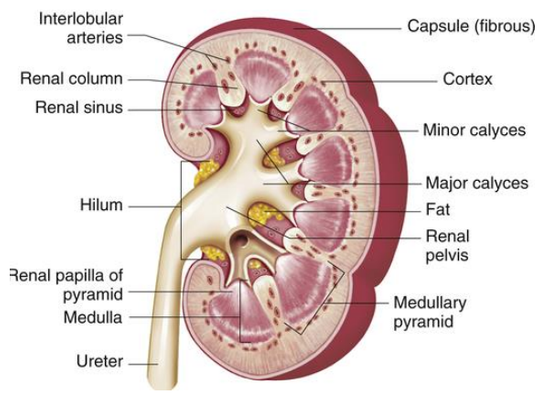
Water Conserving Strategies

Arthropods have developed better water conservation strategies in terrestrial habitats. These include:

- Presence of exoskeleton of chitin that prevents evaporative water loss.
- Production of highly concentrated or nearly dry digestive excreta in the rectum.
- Use of uric acid as metabolic excretory product. The association of excretory malpighian tubules with the gut prevents water loss involved in nitrogenous waste excretion.
- Prevention of respiratory water loss through evolution of tracheal system for respiration.

Topic-176 Mammalian Kidney: Anatomy

Kidney is the osmoregulatory organ of mammals.



Anatomy of the Mammalian Kidney

A kidney is a reddish brown bean shaped organ.

A fibrous connective tissue layer called the fibrous capsule encloses each kidney.

Hilus (Hilum)

The lateral surface of each kidney is convex but its medial side is deeply concave. This forms a medial depression known as the hilus.

Through the hilus, the renal artery and nerves enter and the renal vein and the ureter exit.

Cortex and medulla

The kidney is divided into two functional layers that contain the functional urine forming units—nephrons. The outer layer is called renal cortex while the inner layer is renal medulla.

Renal Pyramids

The renal medulla consists of a number of cone shaped renal pyramids. The base of each pyramid is located at the boundary between renal cortex and the renal medulla.

The tips of the conical pyramids are called renal papillae. They are pointed towards the center of the kidney, projecting into the pelvis.

The parts of cortex which are projected between the renal pyramids are called renal columns.

The spaces between any two renal columns are called minor calyces.

Each renal papilla is pierced by tiny opening that leads into the minor calyx.

The minor calyces from several pyramids join together to form large funnel shaped spaces called major calyces.

The major calyces converge to form an enlarged channel called renal pelvis.

Urine formed within the kidneys passes from renal papillae into the minor calyces, then into the major calyces.

From the major calyces, urine is collected in the renal pelvis and exits the kidney through ureter.

Topic-177 Mammalian Kidney: Anatomy of Nephron

Nephron

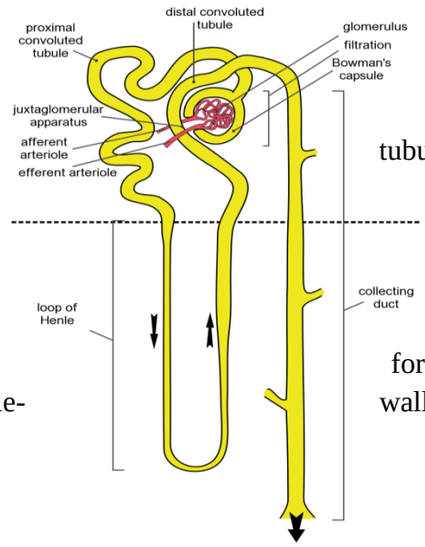
The nephron is the functional unit of mammalian kidney. The number of nephrons per kidney varies from several hundred in lower vertebrates to many thousands in small mammals, and a million or more in humans and other large species.

Structure of Nephron

The nephron consists of a renal corpuscle and a renal tubule.

Renal Corpuscle

The renal corpuscle is an expanded part at the closed end of renal tubule. It is composed of a network of capillaries called glomerulus which is responsible for the first step in urine formation i.e. ultrafiltration. Glomerulus is surrounded by a thin, double-walled, cup-shaped structure called Bowman's capsule.



https://www.kuensting.org/school/bb/systems/excretion/1000px-Nephron.svg.png

Renal Tubule

It leads away from the Bowman's capsule.
The wall of the renal tubule is only one cell thick.

It has three portions:

- Proximal convoluted tubule
- Loop of Henle
- Distal convoluted tubule

The first portion of the tubule called proximal convoluted tubule is highly coiled.

The proximal convoluted tubule dips towards the renal pelvis into the medulla forming a sharp loop called loop of Henle.

It consists of a descending limb and an ascending limb.

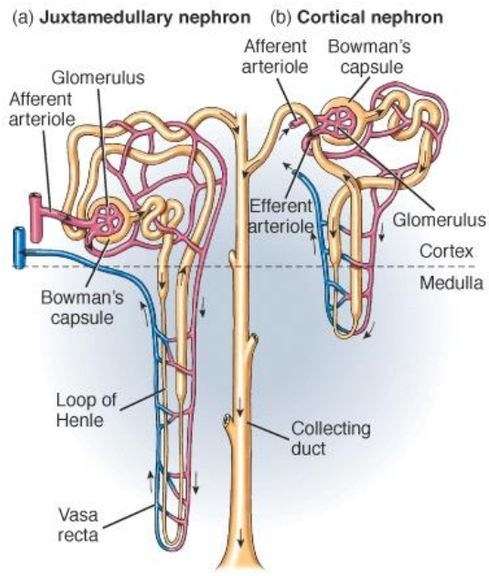
The ascending limb returns to the region of the renal corpuscle, where it becomes highly coiled again, and is called the distal convoluted tubule which is connected to the collecting duct.

Arrangement of Nephrons in Kidney

The glomeruli are found in the renal cortex, and the loops of Henle reach down into the papillae of the medulla; thus the nephrons are arranged in a radiating fashion within the kidney.

Based on the location of glomeruli and loop of Henle, nephrons are divided into two groups:

- Juxtamedullary nephrons, which have their glomeruli in the inner part of the cortex and long loops of Henle that plunge deeply into the medulla. These nephrons are specifically instrumental in the production of concentrated urine.
- Cortical nephrons, which have their glomeruli in the outer cortex and relatively short loops of Henle that extend only a short distance into the medulla



Topic-178 Blood Supply to the Nephron

Blood Supply to Kidneys

Kidneys are about 1 % of total body weight in humans. In comparison to their small size, the kidneys receive a remarkably large blood flow, i.e. 20%-25% of the total cardiac output.

Due to high blood supply, kidneys filter the equivalent of the blood volume every 4-5 minutes.

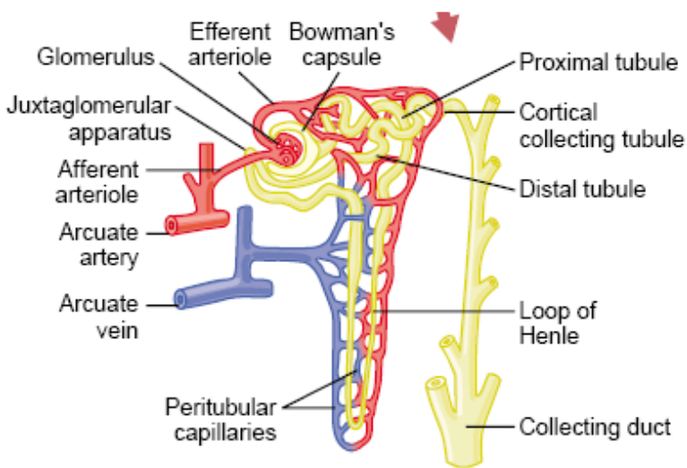
Blood Supply to the Nephron

Blood is supplied to the kidneys through renal artery.

Renal artery gives rise to branches called interlobular arteries which project into the cortex.

Interlobular arteries subdivide to form a series of short afferent arterioles.

Afferent arterioles supply blood to the renal corpuscle of each nephron and form glomerular capillaries.



Text book of medical physiology, Guyton. 11th ed. 2005

The capillaries of the glomerulus recombine to form an efferent arteriole; unlike most other vessels, which join to form veins.

Flow in the efferent arteriole is less than that in the afferent arteriole. So the glomerular capillaries have higher pressures than other capillaries because of the low-resistance input pathway and high resistance output pathway.

The efferent arteriole then subdivides once again to form another plexus of capillaries called peritubular capillaries that surround proximal and distal convoluted tubules and the loop of Henle.

The hairpin-loop like specialized part of peritubular capillaries that surround the loop of Henle is called vasa recta. It starts in the cortex region and then enters medulla along the descending loop of Henle and then re-enters cortex with ascending loop.

The peritubular capillaries drain into interlobular veins, which drain into renal vein which leaves the kidney.

Topic-179 Urine Production: Overview

Role of Kidneys

Kidneys function as excretory and osmoregulatory organs.

Their excretory functions include the filtration of nitrogenous wastes from blood and their removal in the form of urine.

Being osmoregulatory organs, kidneys control the amount of water excreted in the urine.

They produce dilute urine when body is flooded with water and form concentrated urine when body is dehydrated.

Urine Formation by Kidneys

The formation of urine involves three main processes:

- Glomerular filtration
- Tubular reabsorption
- Tubular secretion

Glomerular Filtration (Ultrafiltration)

- Glomerular filtration of plasma to form an ultrafiltrate is the first step in urine formation.
- It takes place in the renal corpuscle under pressure.
- The pressure comes from glomerular capillaries which have exceptionally high blood pressure than any other capillary bed in the body.
- Due to high filtration pressure, water and small solute molecules are filtered out of the glomerular capillaries.
- This filtrate is collected in the Bowman's capsule.
- Large molecules like proteins as well as blood cells and platelets are left behind in the blood.
- The filtered fluid in the Bowman's capsule is called glomerular filtrate.
- It has a chemical composition similar to that of blood plasma.

Tubular Reabsorption (Selective Reabsorption)

Approximately 99% of the water and most of the useful substances are selectively reabsorbed from the ultrafiltrate in the tubular part of the nephron into the blood in surrounding peritubular capillaries.

The major reabsorbed molecules include glucose, amino acids, vitamins and inorganic salts.

Due to reabsorption, the filtrate is converted into urine that has only left over nitrogenous and other waste substances when it leaves the tubule.

Tubular Secretion

- The epithelial cells of convoluted tubule secrete certain substances by active transport into the ultrafiltrate passing through the nephron's tubular part.
- e.g. ammonium and hydrogen ions
- Main purpose of this secretion is to maintain the pH of the urine.

Topic-180 Glomerular Filtration

Urine formation begins when a large amount of fluid that is virtually free of protein is filtered from the glomerular capillaries into Bowman's capsule.

Most substances in the plasma, except for proteins, are freely filtered, so that their concentration in the glomerular filtrate in Bowman's capsule is almost the same as in the plasma.

Filtration takes place in the renal corpuscle as fluids move across the three layered wall of the glomerulus and into the capsular space.

The process of ultrafiltration in the glomerulus is entirely passive and depends on three factors:

(1) Hydrostatic pressure difference between the lumen of the capillary and the lumen of Bowman's capsule, which favors filtration. The reason for hydrostatic pressure is the small diameter of efferent arteriole which offers considerable resistance. Relatively higher pressures are needed to force blood into it. As a result, glomerular pressures are high i.e. about 50 mm Hg, instead of 35 mm Hg typical of peripheral capillaries.

(2) The colloidal osmotic pressure that arises because of the separation of proteins during the filtration process. This pressure opposes filtration.

(3) The permeability of the three-layered glomerular filtration membrane separating these two compartments. This membrane acts as a molecular sieve, excluding almost all proteins from the ultrafiltrate based mainly on molecular size, but also on shape and charge. There is a bulk flow of water through the sieve carrying with it ions, glucose, urea, and many other small molecules.

Topic-181 Tubular Reabsorption

Tubular Reabsorption

As the glomerular filtrate enters the renal tubules, it flows sequentially through the successive parts of the tubule.

Along this course, some useful substances are selectively reabsorbed from the tubules back into the blood.

Reabsorption takes place in every segment of the renal tubule, however, the substances absorbed and their relative concentration changes from segment to segment.

Reabsorption at the PCT

The cells of the proximal convoluted tubule normally reabsorb 60–70 percent of the volume of the ultrafiltrate.

The PCT reabsorbs more than 99 percent of the glucose, amino acids, and other organic nutrients in the fluid. This reabsorption involves a combination of facilitated transport and cotransport.

The PCT actively transports several ions, including sodium, potassium, magnesium, phosphate, bicarbonate and sulfate ions. It also absorbs some ions passively.

Much of water is also reabsorbed from PCT.

The reabsorbed materials enter the peritubular fluid, diffuse into peritubular capillaries, and are quickly returned to the circulation.

Reabsorption in Loop of Henle

Reabsorption of water continues in the descending limb of the loop of Henle that is freely permeable to water due to numerous water channels formed by aquaporin proteins.

In contrast, there is a near absence of channels for salt and other small solutes, resulting in a very low permeability for these substances.

Unlike the descending limb, the ascending limb contains ion channels, but not water channels.

Overall, the loop of Henle reabsorbs about half of the water and two-thirds of the sodium and chloride ions from the filtrate entering it.

This reabsorption takes place efficiently by countercurrent exchange.

Reabsorption at the DCT

Only 15–20 percent of the initial filtrate volume reaches the distal convoluted tubule.

Throughout most of the DCT, the tubular cells actively transport Na^+ and Cl^- out of the tubular fluid.

Tubular cells along the distal portions of the DCT also contain ion pumps that reabsorb tubular Na^+ in exchange for K^+.

DCT also reabsorbs bicarbonate ions to regulate the pH of urine.

Topic-182 Tubular Secretion

The blood entering the peritubular capillaries still contains a number of potentially undesirable substances that did not cross the filtration membrane at the glomerulus.

Such substances are secreted directly through the tubular cells into the tubules into the out going urine.

Secretion takes place in every segment of the renal tubule. However, most substances are secreted in the PCT and DCT.

Secretion at the PCT

The proximal tubule is an important site for secretion of organic acids and bases, bile salts, oxalate, urate, and catecholamines.

Many end products of metabolism, harmful drugs and toxins that must be rapidly removed from the body through urine are also secreted through the tubular cells into the tubules.

Secretion at the DCT

The distal convoluted tubule is involved in the secretion of potassium and hydrogen ions.

Ammonium ions are also pumped into the tubular fluid by sodium-linked countertransport.

Secretion of Potassium Ions

Tubular cells secrete excess potassium ions in body fluids by exchanging them with sodium ions which are reabsorbed. These potassium ions diffuse into the lumen of the DCT through potassium leak channels at the apical surfaces of the tubular cells.

Secretion of Hydrogen Ions

Hydrogen ion secretion is also associated with the reabsorption of sodium. H^+ secretion involves generation of carbonic acid by the enzyme carbonic anhydrase. Carbonic acid dissociates to generate hydrogen ions which are secreted in exchange for Na^+ in the tubular fluid.

Hydrogen ion secretion acidifies the tubular fluid while elevating the pH of the blood. Hydrogen ion secretion speeds up when the pH of the blood falls.

Topic-183 Glomerular Filtration Rate

The Glomerular Filtration Rate (GFR)

The glomerular filtration rate (GFR) is the amount of filtrate the kidneys produce each minute.

Each kidney contains about 6 m^2 of filtration surface, and the GFR averages an astounding 125 mL per minute. This means that roughly 10% of the fluid delivered to the kidneys by renal arteries leaves the bloodstream and enters the capsular spaces.

In the course of a single day, the glomeruli generate about 180 liters (48 gal) of filtrate, roughly 70 times the total plasma volume. But as filtrate passes through the renal tubules, about 99 percent of it is reabsorbed.

Factors Affecting GFR

1. **Net filtration pressure**
   The glomerular filtration rate depends on the net filtration pressure across glomerular capillaries.
   Any factor that alters the net filtration pressure also alters the GFR and affects kidney function.
   One of the most significant factors is a drop in renal blood pressure.
   If blood pressure at the glomeruli drops by 20 percent (from 50 mm Hg to 40 mm Hg), kidney filtration ceases, because the net filtration pressure is 0 mm Hg.
   For this reason, the kidneys are sensitive to changes in blood pressure that have little or no effect on other organs.
2. Hemorrhaging, shock, and dehydration are relatively common clinical conditions that can cause a dangerous decline in the GFR and lead to acute renal failure.

Importance of GFR

- Glomerular filtration is the vital first step for all other kidney functions.
- If filtration does not take place, waste products are not excreted and pH control is lost.

Control of the GFR

A variety of regulatory mechanisms ensure that GFR remains within normal limits.

Three interacting levels of control stabilize GFR:

(1) Autoregulation, at the local level. It involves changes in the diameters of afferent arterioles, efferent arterioles, and glomerular capillaries to maintain GFR.
(2) Hormonal regulation, initiated by the kidneys through renin–angiotensin system and the natriuretic peptides.
(3) Autonomic regulation, primarily by the sympathetic division of the autonomic nervous system.

Topic-184 Regulation of pH by Kidneys

Importance of pH Regulation

Within the human body, blood must be maintained within the narrow alkaline pH range of 7.35 to 7.45. Outside that range, pH becomes incompatible with life as proteins are denatured and enzymes lose their ability to function.

Role of Kidneys in pH Regulation

Kidneys play important role in maintaining pH by two ways:

1 Bicarbonate buffering system
2 Removal of fixed acids and H^+

1 Bicarbonate Buffering System

- Buffer systems minimize changes in pH.
- Most important buffer system is the CO_2/HCO_3^- buffer.
- By adjusting the level of HCO_3^- the kidney is able to adjust pH, without elimination of excess hydrogen
- Under normal circumstances all filtered HCO_3^- is reabsorbed from the filtrate in kidney.

Bicarbonate Reabsorption

- Kidneys reabsorb 4,000 to 5,000 mmol of the filtered bicarbonate ions daily which is equivalent to the removal of the same amount of H^+.
- About 85 to 90% of the filtered bicarbonate is reabsorbed in the proximal tubule while the rest is reabsorbed by the thick ascending limb, distal tubule and collecting ducts.

Bicarbonate Reabsorption Mechanism

- Absorption of bicarbonate is facilitated by the secretion of H^+ in the PCT lumen by the A-type cells.
- H^+ ions are secreted by three mechanisms:
  - Via a Na^+-H^+ antiporter in the PCT and ascending limb
  - Via H^+-ATPase (proton pump) in the PCT and DCT
  - H^+-K^+ ATPase in the collecting duct
- Secreted H^+ ions combine with filtered HCO_3^- to form CO_2 and H_2O. The CO_2 is lipid soluble and diffuses into the tubular cell. In the cell, it combines with OH^- to produce bicarbonate.
- The HCO_3^- crosses the basolateral membrane via a Na^+-HCO_3^- symporter that transfers three HCO_3^- into the blood for every one Na^+.

This mechanism does not lead to the net excretion of any H^+ from the body as the secreted H^+ is consumed in the reaction with the filtered bicarbonate in the tubular lumen.

Removal of Fixed Acids and H^+

- Kidneys are also responsible for the excretion of fixed acids i.e. acidic anions (citrate, acetate, gluconate, lactate) and associated H^+.
- Kidneys excrete about 70-100 mmols of fixed acids per day. This helps to maintain the plasma $[H^+]$ of only 40 nanomoles/litre.

Mechanisms for Removal of H^+

H^+ ions (acids) are removed as:

1. Ammonium ions
2. Bound to filtered buffers

Excretion of Ammonium Ions

- Ammonium ions (NH_4^+) are produced from glutamine.
- Filtered glutamine is absorbed into proximal tubular cells and metabolized to NH_4^+ (ammonium) and HCO_3^-.
- HCO_3^- diffuses into blood, and the NH_4^+ is secreted into the tubule via the NH_4^+-Na^+ antiporter and eliminated in urine.
- As NH_4^+ is formed of NH_3+H^+, so causes removal of acidic H^+ ions.

Removal of H^+ Bound to Filtered Buffers

- Secreted H^+ may also combine with a filtered buffer (e.g. PO_4^{3-}).
- These H^+ ions are not reabsorbed.
- About 36 mmol of H^+ is eliminated with filtered PO_4^{3-} each day, with each PO_4^{3-} binding two H^+ ions.

Topic-185 Urine Concentrating Mechanisms

Production of Hypertonic Urine

- Mammalian kidneys can produce hypertonic urine as compared to blood plasma.
- The loop of Henle is considered to be of central importance in concentrating the urine.
- Vertebrates that lack the loop of Henle are incapable of producing hyperosmotic urine.
- The animals which produce very concentrated urine (such as desert animals) have very long loops of Henle capable of creating a very large osmotic gradient.

Countercurrent Multiplication in Loop of Henle

- Efficiency of reabsorption in loop of Henle is due to a countercurrent exchange system between its descending and ascending limbs.
- Both limbs lie very close together and are separated by a peritubular fluid.
- Filtrate moves in opposite directions in both limbs.
- During the movement, exchange of water and salts occurs between the filtrate in two limbs and peritubular fluid.
- This exchange is referred to as countercurrent multiplication.

Countercurrent Multiplication: Mechanism

- The two parallel limbs of the loop of Henle have very different permeability characteristics.
- The descending limb is permeable to water but impermeable to solutes. Thus water moves through the tubular wall into the peritubular space. This makes the filtrate hypertonic. This filtrate enters the ascending limb.
- The ascending limb is impermeable to water but permeable to solutes.
- From the filtrate in ascending limb, Na^+, Cl^-, and K^+ are actively transported out into the peritubular space.
- This makes fluid in peritubular space hypertonic and creates an osmotic pressure.
- Osmotic pressure causes more water to be drawn out of the descending limb into the peritubular space.
- This makes filtrate in the descending limb more hypertonic.
- Arrival of a highly concentrated solution in the thick ascending limb speeds up transport of sodium and chloride ions out into the peritubular fluid.
- Thus countercurrent flow within the descending and ascending limb increases, or multiplies the osmotic gradient between tubular fluid and interstitial space, causing osmotic removal of more and more water from the filtrate.

Role of Vasa Recta

Absorbed water enters into the circulatory system from the interstitium via the vasa recta, which surround the loop of Henle.

Role of ADH

The amount of water reabsorbed and concentration of urine is controlled by antidiuretic hormone, which affects the water permeability of renal tubules.

Topic-186 Renal Regulatory Mechanisms

The osmoregulatory function of mammalian kidney is regulated by a combination of nervous and hormonal controls.

Nervous Controls

The kidney is innervated with efferent sympathetic nerve fibers reaching the renal vasculature, the tubules, the juxtaglomerular granular cells, and the renal pelvic wall.

Stimulation of efferent renal sympathetic nerve reduces renal blood flow, urinary sodium excretion and increase renin secretion.

Kidneys are also supplied with renal sensory nerves which mainly innervate the renal pelvic wall.

Activation of the sensory nerves elicits an inhibitory reflex that causes a decrease in efferent renal sympathetic nerve activity leading to natriuresis (excretion of sodium in the urine).

Hormonal Controls

The Hormones affecting kidney function include:

- Antidiuretic hormone (ADH)
- Renin-Angiotensin-Aldosterone System
- Atrial natriuretic peptide (ANP)

Antidiuretic Hormone (ADH)

Osmoreceptor cells in the hypothalamus monitor the osmolarity of blood and regulate release of ADH from the posterior pituitary.

ADH is released in response to an increase in osmolarity above the set point of 300 mosm/L.

When ADH reaches the kidney, its main targets are the distal tubules and collecting ducts. ADH makes the epithelium more permeable to water, resulting in increased water reabsorption and concentration of urine, reduced urine volume, and lowering blood osmolarity back toward the set point.

A reduction in blood osmolarity below the set point has the opposite set of effects.

Renin-Angiotensin-Aldosterone System (RAAS)

The RAAS is present in a specialized tissue called the juxtaglomerular apparatus (JGA), located near the afferent arteriole that supplies blood to the glomerulus.

When blood pressure or blood volume in the afferent arteriole drops JGA releases the enzyme renin. Renin initiates chemical reactions that yield a peptide angiotensin II.

Angiotensin II raises blood pressure by constricting arterioles, decreasing blood flow to the nephrons. It also stimulates the adrenal glands to release aldosterone. Aldosterone acts on distal tubules, making them reabsorb more sodium (Na^+) and water, thus increasing blood volume and pressure.

Atrial Natriuretic Peptide (ANP)

The walls of the atria of the heart release ANP in response to an increase in blood volume and pressure.

Atrial natriuretic peptide (ANP), opposes the RAAS.

It inhibits the release of renin from the JGA, inhibits NaCl reabsorption by the collecting ducts, and reduces aldosterone release from the adrenal glands.

These actions lower blood volume and pressure.

Topic-187 Nitrogenous Wastes in Animals

Nitrogenous Wastes

Catabolism of nitrogen containing macromolecules results in formation waste and toxic byproducts.

The major nitrogenous waste is the amino group ($–NH_2$) which is produced by the catabolism of amino acids during deamination. It is highly toxic and is removed by the animals in three forms: Ammonia, urea and Uric acid.

Many other nitrogenous waste metabolic byproducts are produced in lower quantities. These include: creatinine, creatine and trimethylamine oxide.

Metabolism of purine and pyrimidine bases produces significant amount of nitrogenous wastes: hypoxanthine, xanthine and allantoin in addition to uric acid, urea and ammonia.

Major Nitrogenous Wastes

The byproducts of amino acid metabolism i.e. ammonia, urea and uric acid are considered as the major nitrogenous wastes which are excreted by animals in large quantities.

Ammonia

Ammonia is highly toxic and needs to be immediately excreted out of the body. It requires large amount of water for its excretion. Excretion of one gram of ammonia requires about 0.5 liters of water. So, only aquatic animals use it as major excretory product.

Urea

Terrestrial animals convert ammonia into urea or uric acid which can be excreted with less water loss.

Urea is less toxic than ammonia. It requires only 0.05 liters of water to excrete 1 g of urea, i.e. only 10% as compared to ammonia. Urea synthesis, however, consumes ATP that is energetically costly.

Uric Acid

Excretion of uric acid requires even less water i.e. only 0.001 liter for 1 g uric acid which is only 1% as compared to ammonia.

Uric acid is only slightly soluble in water and is excreted as a white pasty precipitate.

Evolutionary Trends in Excretion of Nitrogenous Wastes

In the course of evolution, different animal groups have adapted to produce mainly one of these forms for excretory purposes during whole or part of their life cycles.

In general, water availability and habitat determines the nature and pattern of nitrogenous waste excretion.

Topic-188 Excretion of Ammonia: Ammonotelic Animals

Ammonotelic Animals

- Animals which excrete their nitrogenous wastes principally as ammonia are called ammonotelic animals.
- Most teleost fishes and aquatic invertebrates are ammonotelic. They produce little or no urea, as they have plenty of water for removal of ammonia.

Formation of Ammonia

- The deamination of amino acids in cells removes amino group (NH_4^+).
- The amino groups are transferred to glutamate with the aid of a transaminase enzyme.
- Glutamate is deaminated in the liver to form ammonium ions and α-ketoglutarate.
- In mammals and urea excreting animals, most of these ammonium ions are converted to urea by urea cycle in liver. However small amounts of ammonia are left which are excreted in their urine.
- In fishes and other ammonotelic animals, ammonium ions dissolve in plasma water to form ammonia which is excreted either through gills or kidneys.
- Liver also converts glutamate to glutamine, which is much less toxic than ammonia. Glutamine is released from the liver into the blood.

Excretion of Ammonia through Kidneys

- Glutamine is taken up by the kidneys.
- Cells of the kidney tubules deaminate glutamine, liberating ammonia into the tubular fluid.
- The excreted ammonia in tubular fluid takes up a proton to form the NH_4^+, which cannot diffuse back into the tubular cells and thus leaves the body via the urine.
- In this way, glutamine, which is nontoxic, acts as the amino-group carrier through blood and tissues until its deamination in the kidney.

Excretion of Ammonia through Gills

- Most of ammonia excretion in fishes occurs by passive diffusion through gill surfaces. Gill membranes permit diffusion as they are highly permeable to ammonia.
- As H^+ and carbon dioxide are also excreted through gill surface, they acidify the water next to gill surface. Acidified water traps NH_3, so ammonia excretion is enhanced.

Toxicity of Ammonia

- A blood concentration of only 0.05 mmol/L ammonia is toxic to most animals, causing convulsions, coma and death.
- The toxicity of NH_3 is due in part to the elevation of pH it produces, which causes changes in the tertiary structure of proteins.
- Ammonia also interferes with some ion-transport mechanisms, because NH_4^+ substitutes for K^+ in some cases.
- Ammonia can also affect brain blood flow and some aspects of synaptic transmission, particularly glutamate metabolism.

Topic-189 Excretion of Urea: Ureotelic Animals

Ureotelic Animals

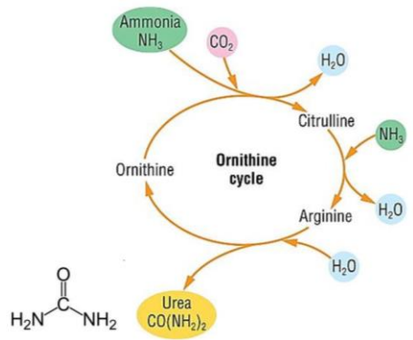
Animals which excrete nitrogenous wastes as urea are known as ureotelic animals.

Urea is far less toxic than ammonia, and requires much less water for excretion.

A urea molecule contains two nitrogen atoms, so excreting urea has double efficiency in removal of nitrogenous waste as compared to ammonia.

Formation of Urea

Urea is formed from ammonia. Ureotelic animals utilize one of two pathways for urea formation:

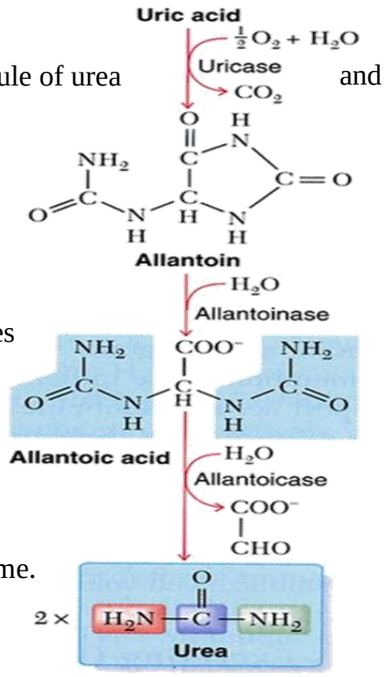


- Ornithine-urea cycle
- Uricolytic pathway

Ornithine-Urea Cycle

- All vertebrates except most teleost fishes, synthesize urea primarily in the liver via the ornithine-urea cycle.
- In this cycle, two ammonia and one CO_2 molecule are added to ornithine to form arginine.
- Arginine is cleaved by the enzyme arginase, to remove a molecule of urea and regenerate ornithine.

Uricolytic Pathway



- Most teleosts and many invertebrates utilize the uricolytic pathway.
- In this pathway, urea is produced from uric acid that either arises from a transamination via aspartate or is produced during nucleic acid metabolism.
- Uric acid is first converted to allantoin with the help of enzyme uricase. Allantoin is converted to allantoic acid with the help of enzyme allantoinase.
- Allantoic acid is then converted to urea using allantoicase enzyme.

Most mammals don't have uricase, allantoicase and allantoinase enzymes and thus excrete uric acid as an end product of nucleic acid metabolism.

Topic-190 Excretion of Uric Acid: Uricotelic Animals

Uricotelic Animals

Animals which excrete nitrogen chiefly in the form of uric acid or the closely related compound guanine are known as uricotelic animals.

In general, uricotelic animals are adapted to conditions of limited availability of water.

They include: birds, reptiles, and most terrestrial arthropods.

Uric acid and guanine have the advantage of carrying away four nitrogen atoms per molecule.

Formation of Uric Acid

Uric acid is formed by the conversion of ammonia and urea in uricotelic animals.

It is also formed during the metabolic breakdown of the amino acids glycine, aspartate, and glutamine.

It is also formed by the breakdown of purine and pyrimidine bases of nucleic acids.

Transport and Excretion of Uric Acid

Uric acid is transported from the blood into the cells of the renal tubule via a urate-anion exchanger or via a urate uniporter.

It then moves from the cells into the lumen of the tubule down an electrochemical gradient and is excreted in the urine.

Uric acid, because of its low solubility, precipitates and is excreted with very little urinary water.

Physiology of the Digestive Tract

TOPIC-191 Overview of the Alimentary Systems

Roles of Alimentary Systems

Alimentary systems procure food for the animal and convert it to usable form by the body of an animal through the process collectively called as nutrition.

The major steps of animal nutrition include:

- Digestion of food
- Absorption of digested food
- Egestion (Removal of indigestible wastes)

Types of Alimentary Systems

There are two basic plans of alimentary systems observed in animals:

1. **Sac-like Gastrovascular Cavity**
2. **Tubular Alimentary Canal or Gut**

1. Sac-like Gastrovascular Cavity

The sac-like gastrovascular cavity is characteristic feature of Cnidarians.

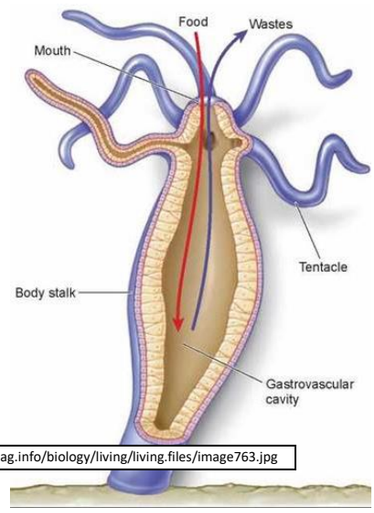
This cavity has only one opening, the mouth that serves for both ingestion as well as egestion.

2. Tubular Alimentary Canal or Gut

The complex bilateral animals have a hollow, tubular cavity which is open at both ends with two openings: mouth for ingestion and anus for egestion.

Such a tube is called a complete digestive tract or alimentary canal.

Alimentary Canal allows unidirectional flow of food materials, increasing the efficiency of digestive and absorptive processes.



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The food passes through the canal through various regions that have specializations for different types of digestive actions involving:

- mechanical, chemical and bacterial treatment of the food for digestion
- absorption of products of digestion
- storage of undigested waste materials

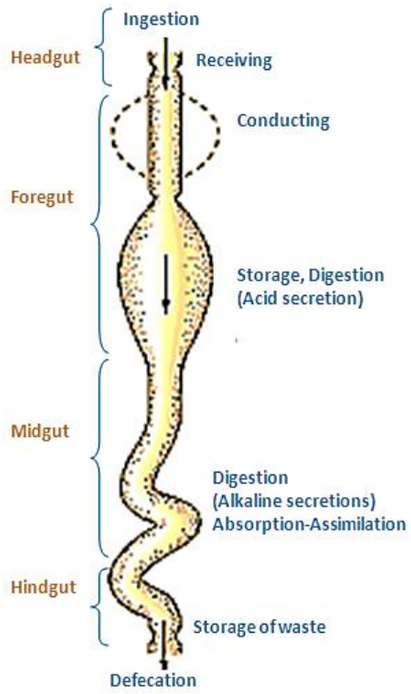
TOPIC-192 Generalized Alimentary Canal

The alimentary canal of all animals has a basic tubular organization that extends between two openings, a mouth and an anus.

Mammalian Digestive Tract

The mammalian digestive tract is a representative example of an alimentary canal.

Regions of the Alimentary Canal



In general, alimentary canals can be divided on a structural and functional basis into four major divisions:

(1) Headgut: specialized for receiving ingested material. Includes Buccal cavity and pharynx

(2) Foregut: specialized for conducting, storing, and digesting ingested material. Includes esophagus and stomach. Also includes crop in some animals.

(3) Midgut: specialized for digesting and absorbing nutrients. Includes small intestine: duodenum, jejunum and ileum.

(4) Hindgut: specialized for absorbing water and defecating. Includes large intestine: colon, cecum, rectum and anus.

Regional Specialization

Alimentary canal's specialized regions allow particular digestive tasks to be carried out in a stepwise fashion.

Ingested material is subjected to various mechanical, chemical, and bacterial treatments as it passes through this canal, and digestive juices, primarily enzymes and acids, are mixed with the ingested material at appropriate regions in the alimentary canal.

It also allows both acidic and alkaline secretions to occur at the same time in different parts.

Accessory Glands

In mammals, the alimentary canal is associated with various accessory glands that secrete digestive juices through ducts into the gut.

The accessory glands of the mammalian digestive system are:

- Three pairs of salivary glands
- Gastric glands
- Pancreas
- Liver
- Gallbladder

TOPIC: 193 Headgut

Headgut is the anterior region of the alimentary canal that receives food through ingestion.

It consists of organs and structures for feeding and swallowing.

The major organs of headgut include:

- Buccal cavity with associated structures mouthparts, teeth, tongue, and salivary glands
- Pharynx

Buccal Cavity

Buccal cavity is the space behind mouth. It has teeth, tongue, and salivary glands.

Functions of Buccal Cavity

Buccal cavity performs many important functions:

1. Food selection

Buccal cavity selects food through taste and feeling.

The food entering the buccal cavity is selected and swallowed if its taste and feeling is good. Otherwise it is rejected.

2. Food handling and Mastication

Teeth and tongue play role in food handling and mastication.

Teeth are present in all vertebrates. They are used for holding or tearing the prey. In most mammals, they also grind the food by the process known as mastication.

Tongue is also present in vertebrates and assists in grasping, handling and rolling the food during mastication.

Tongue is also important in the swallowing process.

Tongue is also used in chemoreception as it bears taste buds which are gustatory receptors.

3. Lubrication and chemical digestion:

Salivary glands produce saliva that lubricates the food material. Lubrication assists swallowing.

Saliva also contains digestive enzyme, salivary amylase, which helps in semi-digestion of starch content in food.

Swallowing

Swallowing is the process that allows the bolus of food to pass from the buccal cavity through the pharynx into the esophagus.

It involves integrated movements of muscles in the tongue and pharynx, as well as peristaltic movements of the esophagus.

Process of Swallowing

Swallowing occurs in following steps:

(a) Tongue pushes the bolus to the back of mouth.
(b) Soft palate moves upwards and to the rear. This action closes the nasal cavity.
(c) Larynx moves upward and forces the cartilaginous epiglottis into horizontal position. Thus glottis, the opening of trachea is closed. This step is very crucial to prevent the entry of bolus into the trachea----which may be life-threatening.

The beginning of swallowing action is voluntary, but once food reaches the back of buccal cavity, swallowing becomes involuntary, under direct neural control of the medulla oblongata of the brain.

TOPIC-194 Foregut: Esophagus and Crop

The foregut consists of the food conducting region i.e. esophagus and storage region i.e. stomach. In some animals it also includes a crop.

Esophagus

After being swallowed, food enters the esophagus which is a tube that leads from the pharynx to the stomach which is a storage as well as digestive region.

Esophagus does not produce any digestive secretion or enzymes and only the previous digestive actions of saliva continue during this conduction.

Peristalsis

The esophagus conducts the bolus by peristaltic movement.

Peristaltic movements are the waves of contraction and relaxation in the smooth muscles of alimentary canal walls.

These movements start from oral cavity and end at the distal end of rectum.

Crop

In some animals, esophagus contains a sac-like expanded section, the crop, which is used to store food before digestion.

Crop is generally found in those animals that feed infrequently. It allows quantities of food to be stored for digestion at a later time.

Crop—Examples:

Many birds e.g. chicken.

Leeches, which feed very infrequently, with weeks or months between feeding periods.

In some animals crops are also used to ferment or digest foods for purposes other than normal digestive process. For example some parent birds prepare food in this way to be regurgitated for their nestlings e.g. pigeon's milk.

Topic-195 Foregut: Stomach

Functions of Stomach

1. Stomach is the part of foregut that serves as a food storage organ.
2. In many vertebrates it also carries out initial stages of protein digestion by secreting the enzyme pepsinogen (pepsin) in a highly acidic environment (due to hydrochloric acid) required for pepsin activation.
3. It is also involved in mechanical grinding and mixing of food contents by contraction of its muscular walls.

Types of Stomach in Vertebrates

Based on the number of chambers, stomachs are classified as:

- Monogastric stomachs
- Digastric stomachs

Monogastric Stomachs

A monogastric stomach consists of a single strong muscular tube or sac.

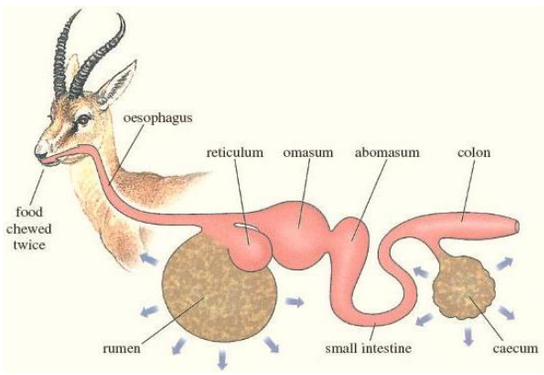
Carnivorous and omnivorous vertebrates have a monogastric stomach.

Digastric Stomachs

Digastric stomachs are multichambered and are found in ruminant mammals of order Artiodactyla e.g. deer, giraffe, sheep, cattle, camel, llama etc.

These stomachs have four chambers which are separated into two divisions.

- The first division consists of fermentation chambers called rumen and reticulum.
- The second division comprises the omasum and abomasum (true stomach).



Fermentation Chambers

The rumen and reticulum receive grazed vegetation and act as fermentation chambers.

These chambers have bacteria and protozoans that cause extensive digestive breakdown by fermentation of carbohydrates.

Omasum

Symbiotic microorganisms grown in the rumen, along with undigested particles, are passed into the omasum and then into the abomasum.

Abomasum

Abomasum is homologous to the monogastric stomach of non-ruminants that secretes digestive enzymes.

Stomach of Birds

The stomach of birds has two chambers:

- A highly muscular grinding chamber, the gizzard (also known as the muscular stomach or ventriculus). It also contains swallowed sand, pebbles, or stones which aid in the grinding of seeds and grains.
- The secretory true stomach called the proventriculus

Some fishes, insects and decapod crustaceans also have variation of this pattern.

Topic-196 Midgut: Small Intestine

In vertebrates, midgut comprises of the small intestine which is the major site for chemical digestion and absorption of food.

While digestion occurs in acidic medium in stomach, in small intestine, it occurs in an alkaline environment.

Divisions of Midgut

The vertebrate midgut or small intestine is typically divided into three distinct portions:

- **Duodenum.** The first, shorter section (25 centimeters in human). It receives food from stomach through pyloric sphincter. It secretes mucus and enzymes and receives bile salts from gall bladder to assist in emulsification of fats and enzyme-containing pancreatic juice from pancreas which aid in digestion.
- **Jejunum.** A longer part (about 2.5 meters long in human) Also secretes enzymes and is involved in digestion and absorption.
- **Ileum.** Acts primarily to absorb nutrients digested in the duodenum and jejunum. It also secretes some enzymes which complete the digestion of food.

Length of the Small Intestine

The length of human small intestine is about 6 meters.

Carnivorous vertebrates have shorter and simpler intestines than herbivores.

It reflects the shorter time required to digest meat than vegetation.

For example, a tadpole, which is herbivore, has a longer intestine than the adult frog, which is carnivorous.

Intestinal Symbionts

Intestine of most animals contains large numbers of bacteria, protozoans, and fungi.

These microorganisms contribute enzymatically to digestion.

Another important function of some intestinal symbionts is the synthesis of essential vitamins.

Topic-197 Absorptive Epithelium of Intestine

Absorption

Digested nutrients in the lumen of alimentary canal need to be absorbed to reach body tissues.

Most of the absorption occurs by crossing the epithelial lining of ileum of small intestine.

The epithelial lining of ileum is specialized for absorptive functions. These specializations include an increase in the surface area for absorption and supply of transport system to carry the absorbed nutrients.

Adaptations to Increase Surface Area

Intestinal folds

The luminal wall of small intestine has numerous circular folds.

These folds increase the surface area of the intestine.

Villi

Each circular fold has thousands of finger-like projections called villi.

Each villus is about 1 mm long.

Due to villi, internal surface of ileum exhibits velvety appearance.

Epithelial Cells of Villi

Each villus has a covering of epithelial cells which make the actual absorptive surface of the small intestine.

Microvilli

Each epithelial cell has several thousand, closely packed cylindrical processes on its apical surface called microvilli, exposed to the intestinal lumen. Each microvillus is 0.5 to 1.5 µm long and about 0.1 µm wide.

Large number of microvilli, laying side-by-side, gives the intestinal epithelium a brush-like appearance that is generally referred to as brush border.

The microvilli greatly increase the total capacity for nutrient absorption.

Supply of Transport System

Internally, it has a network of blood vessels-arterioles, capillaries and venules-and a network of lymph vessels, the largest of which is the central lacteal.

Nutrients taken up from the intestine are transferred into these blood and lymph vessels for transport to other tissues; the central lacteal can, in addition, take up larger particles.

Surface Area for Absorption

The intestinal folds, villi and microvilli collectively increase the surface area available for absorption more than 500 times. The total area for absorption in small intestine of human is 200-300 m^2 as compared to the small intestine's gross cylindrical area of 0.4 m^2.

Topic-198 Hindgut: Large Intestine

Hindgut includes the large intestine which is comprises of colon, cecum, and rectum.

Junction of Small and Large intestine

The small intestine connects to the large intestine through the ileocolic sphincter that controls the movement of material entering the large intestine. It opens and closes time to time to allow small amount of residue from ileum to enter the large intestine.

The junction of small and large intestine forms a T, having a sac-like cecum on one side and the longer colon on the other side.

Cecum

The cecum is a blind sac that is large and functional in non-ruminant herbivore mammals. It harbors bacteria and microorganisms for fermenting plant material.

Humans have a relatively small cecum with a finger-like extension, the appendix which is about 9 cm long.

Appendix has no role in digestion or absorption but has an undefined role in immunity as it has many lymph nodes.

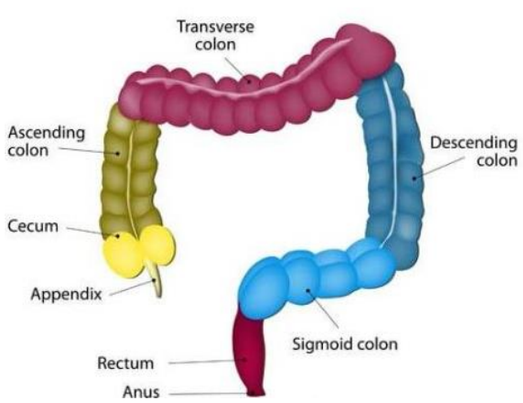
Colon

The colon is about 1.5 m long and consists of four parts: ascending colon, transverse colon, descending colon and sigmoid colon.

Colon receives the undigested part of the food along with inorganic ions and excess water.

Ions and water are absorbed by the colon.

As water is absorbed, indigestible wastes become increasingly solid and are converted into feces as they move along the colon by peristalsis.



It takes approximately 12-24 hours for material to travel the length of the colon and reach the rectum for elimination through the anus.

Rectum

The terminal portion of the large intestine is the rectum which is a straight muscular tube that begins at the termination of sigmoid colon and ends at the anal canal.

In the rectum, feces are stored until they can be eliminated.

Anal Canal

The last 2-3 cm of the digestive tract forms the anal canal.

It begins at the inferior end of the rectum and ends at the anus.

Anal Sphincters

There are two sphincters between the rectum and the anus. The inner sphincter is formed from smooth muscle and is involuntary and the outer one is formed from skeletal muscle and is voluntary.

Defecation Reflex

Periodic strong contractions of the colon cause defecation reflex, opening of the anal sphincters that creates an urge to defecate.

Topic-199 Motility of the Alimentary Canal

The ability of the alimentary tract to contract and propel ingested material along its length is called motility.

The motility depends on the activity of smooth muscle walls and/or ciliary lining of the digestive tract.

Motility of alimentary canal is important for many digestive functions to happen.

It is significant for:

1. Conduction/translocation of food
2. Mechanical treatment of food
3. Prevent Scumming

Conduction of Food

Correctly timed translocation of food along the entire length of the alimentary canal is necessary.

Due to this translocation, food reaches the various parts that have specialized surfaces for secretion of digestive enzymes, absorption of the products of digestion and absorption of water and ions.

It is also important for the expulsion of fecal material.

Mechanical treatment of food

Ingested food materials include larger components. It is necessary to breaking them into smaller soluble forms on which enzymes and bile salts can act.

This involves grinding and kneading actions due to the muscular activity of the alimentary canal.

Muscular activity is also necessary to mix the digestive juices with the food contents.

Mixing of the luminal contents is achieved primarily by a process called segmentation.

Segmentation consists of asynchronous contractions of the circular muscle layer that occur along the wall of gut at various points without participation of the longitudinal muscle.

Prevent Scumming

Motility ensures mixing of the contents so that there is continuous renewal of material in contact with the absorbing and secreting surfaces of the epithelial lining of alimentary canal.

This is important to prevent any layer formation and clogging of the surfaces that may reduce the absorptive or secretory area.

Topic-200 Muscular and Ciliary Motility

Motility of alimentary canal is achieved by two different mechanisms:

- Muscular motility
- Ciliary motility

Muscular Motility

Muscular motility is involved in the transport of food content by muscle contraction of the walls of the alimentary canal.

Muscular mechanisms permit handling of harder and larger pieces of food.

It is the only mechanism found in arthropods and chordates.

In chordates, motility is achieved strictly by smooth muscle fibers. The rhythmic contraction and relaxation of smooth muscles of the alimentary canal is called peristalsis that pushes the food throughout the digestive tract.

In many arthropods motility is achieved by contraction of striated fibers (skeletal muscles).

Ciliary Motility

Ciliary motility involves beating of the cilia lining the digestive tract.

Cilia generate fluid currents that propel food within the digestive tube.

Ciliary motility is the only mechanism used to translocate food along the alimentary canals of annelids, many mollusks, tunicates, and cephalochordates.

Muscular digestive tracts are lined by cilia in echinoderms and most mollusks. So, they use both mechanisms simultaneously.

Topic-201 Peristalsis

Peristalsis—Definition

The continuous contractions and relaxations of the circular and longitudinal muscles that produce alternating waves of constriction and distension in the lumen of the alimentary canal are known as peristalsis.

Peristalsis starts from the buccal cavity and continues along the esophagus to the stomach and then along the whole alimentary canal till the rectum.

Peristalsis—Mechanism

The arrangement of the alimentary smooth musculature in vertebrates consists of an inner circular layer and an outer longitudinal layer.

The contraction of the inner circular layer is coordinated with relaxation of the outer longitudinal layer. This coordination produces constriction in the lumen.

Alternatively, contraction of the longitudinal layer with relaxation of the circular layer produces distension.

Peristalsis—Roles

1. **Push the food**
   Peristalsis results in pushing the food along the alimentary canal in the direction of the peristaltic wave.
   It pushes the food from mouth to anus, inducing swallowing and culminating in defecation.
2. **Regurgitation and Vomiting**
   Regurgitation occurs when peristalsis takes place in the reverse direction, moving the luminal contents back into the buccal cavity.
   Ruminants regularly use regurgitation to bring up the unchewed food for further chewing.
   Reverse peristalsis in non-ruminants results in emesis (vomiting).

Topic-202 Control of Alimentary Canal Motility

Alimentary canal motility in vertebrates involves coordinated contractions of circular and longitudinal smooth muscle layers.

This coordination is regulated by a combination of intrinsic and extrinsic (neural and hormonal) mechanisms.

Intrinsic Mechanisms

- The smooth muscle tissue in the wall of the alimentary tract is myogenic.
- It is capable of producing an intrinsic cycle of electrical activity that leads to muscle contraction without external neural stimulation.
- This cycle occurs due to rhythmic depolarizations and repolarizations called the basic electric rhythm (BER).
- This rhythm consists of spontaneous slow waves of depolarization that progress slowly along the muscle layers.
- Some of these slow waves give rise to action potentials (APs) produced by an inward current carried by calcium ions.
- These calcium currents lead to contractions of the smooth muscle cells.
- The amplitude of the slow-wave BER is modulated by:
  - local influences such as stretching of the muscle tissue by contents in its lumen.
  - chemical stimulation of the mucosa by substances in the chyme.

Extrinsic Hormonal Mechanisms

A chemical stimulant in the chyme causes the release of a local gastrointestinal peptide hormone which, in turn, modulates the motility of the muscle tissue.

Extrinsic Neural Mechanisms

Postganglionic neurons of sympathetic and parasympathetic autonomic nervous system along with aminergic and peptidergic neurons diffusely innervate throughout the smooth muscle layers of intestinal tract and influence the intestinal motility.

Parasympathetic Network

The parasympathetic network is made up of cholinergic neurons and receives input via branches of the vagus nerve.

Acetylcholine released by their nerve endings mediates excitatory actions i.e., increased motility and gastrointestinal secretion.

Sympathetic Network

Adrenergic postganglionic neurons of the sympathetic division directly innervate all the tissues of the gut wall.

These innervations have inhibitory effect on motility of the stomach and intestine.

They release norepinephrine which inhibits the smooth muscle cells to develop action potentials.

Role of Aminergic and Peptidergic Neurons

Smooth muscles in the alimentary canal of vertebrates are also regulated by non-adrenergic, non-cholinergic neurons that release a variety of peptides and purine nucleotides.

These neurons include aminergic neurons that release ATP, 5-HT, dopamine and GABA.

Peptidergic neurons release enkephalins, vasoactive intestinal polypeptide (VIP), substance P, gastrin-releasing peptide, neurotensin, cholecystokinin (CCK), and neuropeptide Y (pancreatic polypeptide).

This host of transmitter substances allows very fine control over the numerous interacting functions of the alimentary canal.

Topic-203 Gastrointestinal Secretions

The alimentary canal in many animals has been described as the "largest endocrine and exocrine gland of the body" as its secretions include both endocrine and exocrine secretions.

Endocrine Secretions of GI Tract

Endocrine secretions of digestive tract include many gastrointestinal hormones that are liberated directly into the bloodstream.

These hormones are produced by the enteroendocrine cells spread throughout the digestive tract e.g. in the stomach, pancreas, and small intestine.

These hormones act on tissues of the alimentary canal and control various functions of the digestive organs.

Gastrointestinal Hormones

The gastrointestinal hormones include.

1. Gastrin
2. Cholecystokinin
3. Secretin
4. Glucagon
5. Vasoactive intestinal peptide (VIP)
6. Gastric inhibitory peptide (GIP)
7. Somatostatin
8. Motilin
9. Substance P
10. Ghrelin—a peptide hormone released from stomach and liver and is often referred to as the "hunger hormone". Its high levels are found in individuals that are fasting.

Many gut peptides, such as secretin, cholecystokinin and substance P are also found to play a role of neurotransmitters and neuromodulators in the central and peripheral nervous systems.

Exocrine Secretions of GI Tract

Exocrine tissues of the gastrointestinal tract include the salivary glands, secretory cells in the stomach and intestinal epithelium, and secretory cells of the liver and pancreas.

Exocrine gastrointestinal secretions are not composed of a single species of molecule. They usually consist of aqueous mixtures of substances including enzymes, water, mucous, inorganic salts and ions.

The digestive tract of human secretes about seven liters of exocrine fluids in a day.

Enzymes in Exocrine Secretions of GI Tract

Exocrine secretions of GI tract contain digestive enzymes which catalyze the hydrolysis of large food molecules into simpler compounds that can be absorbed through the cells lining the intestine.

The digestive enzymes belong to three major groups: proteases, carbohydrases, and lipases corresponding to the three major types of foodstuffs.

Some others classes of enzymes are present in the secretions of intestinal cells and are specialized to digest specific components of food e.g.

- Nucleases, nucleotidases and nucleosidases which hydrolyze nucleic acids and their residues.
- Esterases which hydrolyze esters present in ripe fruits.

Topic-204 Exocrine Secretions—Water, Mucus and Ions

There are large variations in the composition of exocrine secretions from different glands of the alimentary canal.

However, all these secretions usually contain large amount of water, mucus, salts and ions in addition to enzymes.

Water

Water makes most of the volume of these secretions, as these are water-based fluids.

To prevent a huge water loss, the distal portions of the gut reabsorb most of this secreted water.

Mucus

Mucus is also an important constituent of all GI secretions.

It is composed of a specific family of glycoproteins termed mucins and is generally very viscous.

Mucus is made by two types of specialized cells termed mucus cells in the stomach and goblet cells in the intestines.

The salivary glands and pancreas secrete a thinner mucoid solution while the mucus produced in the stomach and intestinal secretions is thicker.

Mucus serves to lubricate and prevent mechanical and enzymatic injury to the lining of the gut.

Ions and Electrolytes

The ions secreted in exocrine fluids are primarily H^+, Cl^-, and HCO_3^- along with Na^+ and K^+.

H^+ and Cl^- are secreted by the parietal cells into the lumen of the stomach creating acidic conditions with a low pH of 1.

HCO_3^- ions are present in the salivary and pancreatic secretions. They neutralize the acid secretions entering the duodenum of small intestine.

Pancreatic HCO_3^- comes from acinar cells in the form of $NaHCO_3$. $NaHCO_3$ dissociates and results in the high concentration of both HCO_3^- and Na^+ present in the pancreatic duct.

Topic-205 Exocrine Secretions—Bile and Bile Salts

The vertebrate liver secretes bile that does not contain any digestive enzymes. However, it contains many salts which are essential for the digestion of fats.

Composition of Bile

Bile consists of water and a weakly basic mixture of cholesterol, lecithin, inorganic salts, organic salts, and bile pigments.

The bile organic salts are manufactured by the liver from cholesterol and amino acids complexed with sodium.

The bile pigments are derived from biliverdin and bilirubin, which are products of the breakdown of hemoglobin coming from ruptured red blood cells.

Storage and Concentration of Bile

Bile produced in the liver is transported via the hepatic duct to the gallbladder.

Gallbladder concentrates and stores bile.

Concentration involves osmotic removal of water, followed by active transport of Na^+ and Cl^- from the bile across the gallbladder epithelium.

Functions of Bile

Bile serves numerous functions important to digestion.

- In the initial segment of small intestine, high alkalinity of bile neutralizes the highly acidic food coming from stomach.
- Bile salts act as a detergent for the emulsification of fats dispersing them in aqueous solution. This process facilitates enzymatic fat digestion
- Bile salts also disperse lipid-soluble vitamins for transport in the blood.
- Bile fluid contains waste substances removed from the blood by the liver, such as hemoglobin pigments, cholesterol, steroids, and drugs. These substances are either broken down by digestive enzymes or excreted in the feces.

Topic-206 Digestive Enzymes: Proteases

Proteases

Proteases are proteolytic enzymes which digest proteins by breaking the peptide bonds of proteins and polypeptides.

Types of Proteases

Proteases are categorized as:

- Endopeptidases
- Exopeptidases

Endopeptidases confine their attacks to bonds well within the protein molecule, breaking large peptide chains into shorter polypeptide segments. These shorter segments provide a much greater number of sites of action for the exopeptidases.

Exopeptidases attack only peptide bonds near the end of a peptide chain, releasing free amino acids, dipeptides and tripeptides.

Specificity of Proteases

Some proteases exhibit marked specificity for particular amino acid residues located on either side of the bonds they attack.

Examples

The endopeptidase trypsin attacks only those peptide bonds in which the carboxyl group is provided by arginine or lysine, regardless of where they occur within the peptide chain.

The endopeptidase chymotrypsin attacks peptide bonds containing the carboxyl groups of tyrosine, phenylalanine, tryptophan, leucine, and methionine.

Proteases in Digestive Tract

Pepsin

In mammals, protein digestion usually begins in the stomach by the action of the gastric protease pepsin.

Pepsin functions best at a low pH value of 1-2, maintained by the secretion of gastric HCl.

Action of pepsin results in the hydrolysis of proteins into polypeptides and some free amino acids.

Proteases in Pancreatic Juice

Pancreas produces several proteases which are secreted in the pancreatic juice.

These include trypsin, chymotrypsin and carboxypeptidase.

These enzymes continue the proteolytic process, yielding a mixture of free amino acids and small peptide chains.

Intestinal Proteases

Many proteolytic enzymes (e.g. Erepsin) are secreted by the epithelium of the intestinal wall.

These proteases act on polypeptides and hydrolyze them into short oligopeptides consisting of two to three amino acids residues, and then further break them into individual amino acids.

Topic-207 Digestive Enzymes: Carbohydrases and Lipases

Carbohydrases

Carbohydrases are the enzymes which digest carbohydrates by hydrolysis.

They can be divided functionally into two groups:

- Polysaccharidases
- Glycosidases

Polysaccharidases

Polysaccharidases hydrolyze the glycosidic bonds of long-chain carbohydrates such as cellulose, glycogen, and starch.

The most common polysaccharidases are the amylases which are secreted in vertebrates by the salivary glands and pancreas. They hydrolyze all but the terminal glycosidic bonds within starch and glycogen, producing disaccharides and oligosaccharides.

Glycosidases

Glycosidases act on disaccharides such as sucrose, maltose and lactose.

They hydrolyze the alpha-1,6 and alpha-1,4 glycosidic bonds present in the disaccharides, releasing their constituent monosaccharides.

Lipases

Lipases are lipid and fat digesting enzymes, secreted in the pancreatic juice and intestinal epithelium.

They act on emulsified fat droplets and hydrolyze fat molecules into monoglycerides and diglycerides and finally into free fatty acids and glycerol.

Emulsification by bile salts is necessary for lipases to hydrolyze lipids. If there are insufficient bile salts, fats in food are not emulsified and fat digestion by the lipases remains incomplete.

Topic-208 Proenzymes

Proenzymes

Certain digestive enzymes, in particular proteolytic enzymes, are synthesized, stored, and released in an inactive molecular form known as a proenzyme or zymogen.

Packaging of the enzyme in an inactive form prevents self-digestion of releasing tissues/cells by the enzyme.

The proenzyme is activated by the removal of a portion of the molecule, either by the action of another enzyme specific for this purpose or through a rise in ambient acidity.

Examples

Pepsinogen-Pepsin

Pepsin is the gastric proteolytic enzyme.

It is produced by the chief cells of stomach in its inactive form known as pepsinogen, which is a proenzyme.

Pepsinogen is activated in the lumen of stomach through a rise in acidity by HCl into its active form, pepsin.

Trypsinogen-Trypsin

Trypsinogen is secreted in pancreatic juice as a proenzyme.

The proenzyme trypsinogen is a 249-residue inert polypeptide. It is activated in the duodenum when a 6-residue segment is cleaved from the amino terminal end.

This cleavage is achieved either by the action of an intestinal proteolytic enzyme, enterokinase or another activated trypsin molecule.

Chymotrypsinogen-Chymotrypsin

Chymotrypsinogen is also secreted as proenzyme in pancreatic juice.

It is activated by trypsin into its hydrolytic active form chymotrypsin.

Topic-209 Control of Digestive Secretions

The secretory rate and quantity of digestive secretions is regulated through several interacting features.

These include:

1. Complex involuntary neural and hormonal controls
2. Presence of food
3. Cognition or thought processes

1. Involuntary neural and Hormonal Controls

- Salivary secretions are very rapid and entirely under involuntary neural control.
- Gastric secretions are under hormonal as well as neural control.
- Intestinal secretions are slower and are primarily under hormonal control.

As in other systems, neural control predominates in rapid reflexes, whereas endocrine mechanisms are involved in reflexes that develop over minutes or hours.

2. Role of Presence of Food

Among vertebrates, the primary stimulus for secretion of digestive juices in a given part of the digestive tract is the presence of food.

The presence of food molecules stimulates chemosensory endings, which leads to the reflex activation of autonomic efferents that activate or inhibit motility and exocrine secretion.

Appropriate food molecules also directly stimulate epithelial endocrine cells by contact with their receptors, causing reflex secretion of gastrointestinal hormones into the local circulation.

These reflexes permit secretory organs outside the alimentary tract (liver and pancreas) to be properly coordinated with the need for digestion of food.

3. Role of Cognition or Thought Processes

In mammals, cognition or thought processes also control the digestive secretions.

Cephalic influences such as mental images of food as well as learned behaviors stimulate digestive secretions.

Topic-210 Control of Salivary and Gastric Secretions

Control of Salivary Secretions

Salivary glands produce a slow flow of saliva in the absence of food.

Secretion of saliva is stimulated by the presence of food in the mouth.

Food stimulates cholinergic parasympathetic nerves in the salivary glands which result in increase in salivary secretion.

Cognitive awareness of food also results in rapid secretion of saliva.

Gastric Secretions

The gastric juice contains HCl, Pepsin and mucus which are synthesized and secreted separately by three different types of cells present in the gastric glands located in the wall of stomach.

- HCl is secreted by parietal or oxyntic cells.
- Mucus is synthesized and secreted by goblet cells.
- Pepsin is secreted in its inactive form pepsinogen by the chief cells.

Control of Gastric Secretions

Gastric secretion in mammals occurs in three distinct phases:

- Cephalic phase
- Gastric phase
- Intestinal phase

Cephalic Phase

In the cephalic phase, gastric secretion occurs in response to the sight, smell, and taste of food, or in response to conditioned reflexes.

This phase is mediated by the brain (hence the term cephalic) and is blocked by section of the vagus nerve.

Gastric Phase

The gastric phase is mediated by the hormone gastrin and the compound histamine.

The gastric secretion is stimulated directly by the presence of food in the stomach, which stimulates both chemoreceptors and mechanoreceptors.

Secretagogues in food, such as caffeine, alcohol, and active ingredients of spices increase gastric secretion.

Intestinal Phase

The intestinal phase of gastric secretion is more complex.

It is controlled by gastrin, as well as the hormones secretin, vasoactive intestinal peptide (VIP), and gastric inhibitory peptide (GIP).

As food enters the duodenum of the small intestine, partially digested proteins in acidic chyme directly stimulate the duodenum's mucosa to secrete enteric gastrin.

Enteric gastrin has the same action as stomach gastrin, stimulating the gastric glands to increase their rate of secretion.

GIP is liberated by endocrine cells in the mucosa of the upper small intestine in response to the entry of fats and sugar into the duodenum. GIP inhibits secretion of gastric glands.

Topic-211 Control of Intestinal and Pancreatic Secretions

Intestinal Secretions

The epithelium of the mammalian small intestine secretes intestinal juice which is a mixture of two fluids secreted by Brunner's glands and crypts of Lieberkuhn (mucosal glands).

Brunner's glands Secretion

Brunner's glands are present in the first part of the duodenum between the pyloric sphincter and the pancreatic duct.

They secrete a viscous, enzyme-free, alkaline mucoid fluid.

This fluid enables the duodenum to withstand acidic chyme coming from the stomach until it can be neutralized by the alkaline pancreatic and biliary secretions coming from the pancreatic duct.

Secretion of Crypts of Lieberkuhn

Crypts of Lieberkuhn (mucosal glands) in the intestinal folds produce a thinner, enzyme-rich alkaline fluid.

This fluid mixes with duodenal secretions.

Control of Intestinal Secretions

The secretion of intestinal juice is regulated by several hormones, including secretin, gastric inhibitory peptide (GIP), and gastrin.

It is also under neural control as vagus nerve innervation stimulates secretion.

Distension of the wall of the small intestine also elicits a local secretory reflex.

Secretions of Pancreas

The exocrine tissue of pancreas produces pancreatic juice that enters the duodenum through the pancreatic duct.

Pancreatic juice contains several digestive enzymes in an alkaline, bicarbonate-rich fluid.

The pancreatic enzymes include: alpha-amylase, trypsin, chymotrypsin, elastase, carboxypeptidases, aminopeptidases, lipases, and nucleases. These enzymes show optimum activity at a neutral or slightly alkaline pH.

Control of Pancreatic Secretions

Pancreatic secretions are controlled by a variety of hormones including peptide hormones, gastrin, cholecystokinin, somatostatin and enkephalins.

Effect of Peptide Hormones

Acid chyme reaching the small intestine from the stomach stimulates the release of peptide hormones secretin and vasoactive intestinal peptide (VIP) by endocrine cells in the upper small intestine.

These peptides stimulate secretion of pancreatic enzymes.

Effect of Gastrin

Gastrin secreted from the stomach lining also elicits a small flow of pancreatic juice in anticipation of the food coming in the duodenum.

Effect of Cholecystokinin

Secretion of pancreatic enzymes is also elicited by the upper intestinal hormone cholecystokinin that is secreted in response to fatty acids and amino acids in the intestinal chyme.

This hormone also stimulates contraction of the smooth muscle wall of the gallbladder, forcing bile into the duodenum.

Effect of Somatostatin and Enkephalins

Somatostatin inhibits gastric and pancreatic secretions.

Enkephalins inhibit gastric acid secretion but stimulate pancreatic enzyme secretion.

Topic-212 Absorption of Food in Intestine

The breakdown products of digestion (amino acids, glucose, fatty acids etc.) are absorbed from the gut in to the bloodstream and transported to animal's tissues and cells.

Most absorption occurs through ileum of the small intestine. Ileum is specialized for absorptive functions as it has villi. Each villus is richly supplied with blood capillaries and a lacteal of lymphatic system. A villus also has a covering of absorptive epithelial cells with microvilli in their apical membrane.

The microvilli have specialized features including a glycocalyx, digestive enzymes associated with the membrane, and specific intramembrane transporter proteins. All these features make them specialized structures to absorb all types of digested food products.

Absorption of Sugars and Amino Acids

Microvilli take up simple sugars and amino acids by diffusion or active transport into the epithelial cells from where they are diffused into the blood capillaries of villi.

Absorption of Fatty Acids and Glycerol

As the fats and lipids are digested in the small intestine, bile salts aggregate around the products of digestion and form micelles.

Micelles pass by simple diffusion into the epithelial cells.

In the epithelial cell, fatty acids and glycerol recombine to form triacylglycrerol. Proteins combine with triacylglycrerol to form chylomicrons (lipoproteins). These chylomicrons leave the epithelial cells and enter the lacteals of villi and are transported by the lymphatic system.

These lipoproteins are passed into the blood stream via thoracic lymphatic duct.

In the blood, these lipoproteins are hydrolyzed by plasma enzymes into fatty acids which can enter the tissue cells.

Topic-213 Absorption Mechanisms

Absorption of products of digestion in small intestine involves several transfer processes.

These include:

- Simple passive diffusion
- Carrier mediated facilitated diffusion
- Co-transport and counter-transport
- Active transport
- Endocytosis

The type of transfer mechanism used depends on the type of molecule being transported during the absorption process.

Simple Diffusion

Simple diffusion can take place across the plasma membrane of epithelial cells either through lipid bilayer or through water-filled pores.

Lipid soluble substances e.g. fatty acids, monoglycerides and cholesterol can diffuse through the lipid bilayer.

Substances that pass through water-filled pores include water and water soluble nonelectrolytes and alcohols. Their net diffusion rate is proportional to their chemical concentration gradient.

Certain electrolytes are also absorbed through diffusion. Their net diffusion rate is proportional to their electrochemical gradient.

Carrier Mediated Transport

Some monosaccharides (e.g. fructose) and few amino acids cannot pass through water-filled pores by simple diffusion, although they are water soluble. It is because they bear charges and are usually larger than the pore size.

They are absorbed across the absorptive cell membrane by carrier mediated transport (facilitated diffusion) with the help of specific channel proteins located in the membranes.

Topic-214 Absorption by Active Transport and Endocytosis

Absorption by Active Transport

Many nutrients, including amino acids, vitamins and most glucose molecules, are pumped actively against concentration gradients by the epithelial cells of the villus.

Active Transport of Glucose and Amino Acids

The energy for the transport of glucose, amino acids, dipeptides and tripeptides is supplied by a sodium co-transport mechanism.

The molecules to be absorbed bind to a specific transport protein present in the membrane of microvilli.

This transport protein requires sodium binding before transport can occur.

After binding, the sodium ion moves down its electrochemical gradient to the interior of the cell and pulls the glucose, amino acid or peptide along with it.

In this way, both the sodium ion and glucose molecule are transported together to the interior of the cell. This is called co-transport (or secondary active transport).

At least five types of transport proteins have been found in the luminal membranes of intestinal epithelial cells for the transport of amino acids, peptides and glucose. Each type is specific for diverse binding properties of different substances that are to be carried into the cell.

Endocytosis

Some oligopeptides are taken up by absorptive cells through endocytosis.

In newborn mammals this process is responsible for the uptake in the intestine of immunoglobulin molecules derived from the mother's milk.

Once inside the absorptive cell, nutrients pass through the basolateral membranes of the absorptive cell into the interior of the villus and then move from the interstitial fluid into the circulatory system.

Topic-215 Transport of Nutrients

Digestion products absorbed enter either the blood or the lymphatic circulation in the villi.

Blood Transport of Sugars and Amino Acids

Sugars and amino acids primarily enter the capillaries of the villi.

These capillaries are drained by venules that lead into the hepatic portal vein.

The hepatic portal vein takes the blood from the intestine directly to the liver.

In the liver, much of the glucose is taken up by hepatocytes and, under the influence of insulin, is converted into glycogen granules for storage. Only a controlled quantity is released into the circulation to be utilized by tissues.

Lymphatic Transport of Fats

In vertebrates about 80% of the chylomicrons enter the lacteals of villi and are carried by lymphatic vessels. The remaining 20% enter the blood directly.

The lymph is poured into the bloodstream via thoracic lymphatic vessel which delivers fats to the blood for circulation.

Topic-216 Water and Electrolyte Absorption

Release of Water in Alimentary Secretions

In the process of producing and secreting various digestive juices, the exocrine tissues of alimentary canal and accessory organs release a large amount of water and electrolytes.

In humans, this amount is over 7 liters per day that is about 1.5 times the total blood volume.

This huge quantity of water along with electrolytes cannot be lost from the body with the feces.

Reabsorption of Water

Nearly all secreted water and electrolytes are recovered by uptake in the intestine.

Water is reabsorbed throughout the intestine; however, most of the reabsorption takes place in the lower part of the small intestine and colon.

Osmotic pressure is the motive force leading to net water movement from the intestinal lumen to the interior of the villus, this movement is entirely passive.

The osmotic gradient driving water from the lumen into the villus is set up primarily by the active transport of substances from the lumen into the villus, in particular the transport of salt, sugar, and amino acids.

The elevated osmotic pressure within the villus that results from this active transport draws water osmotically.

Control of Water Absorption

Excessive uptake of water from the lumen across the intestinal wall results in abnormally dry lumen contents that result in constipation.

This situation is normally prevented by an inhibitory action on electrolyte and water uptake by some of the gastrointestinal hormones.

Gastrin acts indirectly to inhibit water absorption from the small intestine, while secretin and CCK reduce the uptake of Na^+, K^+, and Cl^- in the upper jejunum.

Bile acids and fatty acids also inhibit the absorption of water and electrolytes.

Absorption of Na^+ and Cl^-

Most of Na^+ are actively absorbed by cells located at the tip of each villus.

Cl^- ion follow passively Na^+ ions.

The absorption of Na^+ and Cl^- into the villus is enhanced by high concentrations of glucose in the intestinal lumen, which stimulate sodium-sugar co-transport.

Absorption of Ca^{2+}

Ca2+ requires a special active transport mechanism for absorption from the gut.

The calcium ion is first bound to a calcium-binding protein in the microvillus and is then transported as a complex into the absorptive cell by an energy-consuming process.

From the absorptive cell the Ca^{2+} then passes into the blood.

The presence of calcium-binding protein is regulated by the hormone calcitriol, vitamin D_3.

The release of Ca^{2+} from the absorptive cell into the blood is accelerated by parathyroid hormone.

Absorption of Vitamin B_{12}

Vitamin B_{12} (molecular weight of 1357) is the largest water-soluble essential nutrient taken up intact across the intestinal lumen in the distal ileum.

Thermoregulation

Topic-217 Homeotherms and Poikilotherms

On the basis of stability of body temperatures, animals have been traditionally classified into two types:

- Homeotherms
- Poikilotherms

Homeotherms

Homeotherms can regulate their body temperatures within a narrow physiological range by controlling heat production and heat loss.

They are able to maintain body temperatures in spite of variation in ambient (environmental) temperatures.

Examples

- **Mammals:** can maintain their core body temperature between 37°C to 38°C.
- **Birds:** can maintain their core body temperature closer to 40°C.
- Some other vertebrates and some invertebrates can control their body temperatures during periods of activity or rapid growth.

Poikilotherms

Poikilotherms are the animals which lack high rates of heat production and their body temperatures tend to fluctuate with the ambient temperature.

Warm Blooded and Cold Blooded

The general terms "warm blooded" and "cold blooded" have been traditionally used for homeotherms and poikilotherms respectively.

Early comparative physiologists considered all fishes, amphibians, reptiles and invertebrates to be poikilotherms (cold blooded) and birds and mammals to be homeotherms (warm blooded).

However these terms are not considered physiologically correct because thermal behaviors of animals are quite variable. Many poikilotherms can become quite warm and endotherms may become cold regionally or temporally. So the use of these terms has been abandoned in scientific literature.

Topic-218 Endotherms and Ectotherms

Physiologists classify animals into endotherms and ectotherms on the basis of source of body heat.

Endotherms

Endotherms are animals that generate their own body heat through metabolism and elevate their body temperatures considerably above ambient temperatures.

They also have relatively low thermal conductivity because of good insulation with fur, feathers or sub-cutaneous fat. This insulation enables them to conserve heat in spite of a high temperature gradient between body and environment.

Homeothermic and Heterothermic Endotherms

- Mammals and birds regulate their temperatures within relatively narrow limits. They are said to be homeothermic endotherms.
- A few large fishes (sharks and tuna) and some flying insects maintain regions of their body above ambient temperatures, for short periods of time under specific circumstances. They are known as regional heterothermic endotherms.

Significance of Endothermy

As endotherms can maintain their body temperatures well above the ambient temperatures in cold climates, they have been able to invade habitats that are too cold for most ectotherms.

Ectotherms

Ectothermic animals produce metabolic heat at comparatively low rates and rely almost entirely on environmental sources of heat for warming their bodies.

They also have high thermal conductance as their bodies are poorly insulated. As a result, heat derived from metabolic processes is quickly lost to cooler surroundings.

However, high thermal conductance allows ectotherms to absorb heat readily from their surroundings.

Behavioral Temperature Regulation in Ectotherms

Ectotherms regulate their body temperatures principally by means of behavioral temperature regulation.

In colder environment, many ectotherms behave in a way that facilitates heat absorption from the environment

In hotter environments, they tend to dissipate heat to the environment or minimizes heat uptake from the environment.

Examples

A lizard or a snake may bask in the sun with its body oriented for maximal warming until it achieves a temperature suitable for efficient muscular function.

Small ectotherms in hot environments (lizards, ants) often elevate their bodies to avoid hot surfaces over which they are moving.

The most effective thermoregulatory action taken by ectotherms is movement into a suitable microclimate in the environment. For example, keep hiding in the burrows having moderate temperatures during the heat of the day and coming out at night.

Topic-219 Heterotherms

Heterotherms are animals capable of varying degrees of endothermic heat production, but generally do not regulate body temperatures within a narrow range.

They may be divided into two groups:

- Temporal heterotherms
- Regional heterotherms

Temporal Heterotherms

Temporal heterotherms are animals whose temperatures vary widely over time on daily basis or on annual basis.

These animals have accurate temperature control mechanisms and so are basically homeothermic. Yet, they behave like temporal heterotherms because they allow their body temperatures to undergo daily cyclical fluctuations, having endothermic temperatures during periods of activity and lower temperatures during periods of rest.

Examples

- Monotremes (egg-laying mammals) such as the echidna are temporal heterotherms.
- Some mammals (e.g. bats) and birds (e.g. hummingbirds) which undergo hibernation and daily torpor are also temporal heterotherms.
- Some large endotherms (e.g. ground squirrels) resort to a long winter torpor with reduced body temperature to save energy.

Regional Heterotherms

Regional heterothermy describes organisms that are able to maintain different temperature zones in different regions of the body.

Regional heterotherms include both ectotherms and endotherms.

Many ectotherms can achieve high core temperatures through muscular activity, while their peripheral tissues and extremities approach the ambient temperature.

In many endotherms, temperature of limbs is allowed to fall near the ambient temperature, while keeping the core temperature nearly constant.

Examples

- Regional heterothermy is shown by many flying insects, pythons, and some fishes (mako sharks and tuna) which can raise the temperature of parts of their bodies well above ambient temperature by virtue of heat generated as a by-product of intense muscular activity.
- Some insects prepare for flight by exercising their flight muscles for a time to raise their temperatures before takeoff.
- Some large billfin fishes (e.g., marlin) use specialized ocular muscle called "heater tissue" to elevate brain temperature.
- Another special example of regional heterothermy is seen in the scrotums of some mammals, including canines, cattle, and human beings, which hold the testes outside the body core to keep them at a slightly lower temperature to prevent overheating of the testes which has a harmful effect on sperm production.

Topic-220 Ectotherms in Cold and Freezing Environments

Challenges of Cold Environments

The body temperature of most ectotherms depends on the ambient temperature.

Ectotherms occupying cold environments face two major problems:

1. Maintaining metabolism
2. Freezing threat

Maintaining metabolism in Cold Environments

For animals living in cool environments, survival requires maintaining adequate metabolism. At low temperatures enzyme activity becomes very slow.

Many animals living in cold environments have evolved enzymes that show maximal activity at temperatures many degrees below those of homologous enzymes in animals living in warmer environments.

Freezing Threat

Ectotherms living in freezing environments face freezing threat.

The formation of ice crystals within cells is usually lethal because, as the crystals grow in size, they rupture and destroy the cells.

No animal is known to survive complete freezing of its tissue water.

Animal Strategies to Avoid freezing

If ice crystals form and grow within cells, they damage the tissue by breaking the cells.

In contrast, ice crystals that form outside the cells do little damage.

The adaptiveness of animals living in freezing environments lies in crystal formation in the extracellular space where little tissue damage is caused.

Such animals employ three methods to survive freezing environments:

- Ice-nucleating Agents (INAs)
- Cryoprotectants (antifreeze substances)
- Supercooling

Ice-nucleating Agents (INAs)

Certain animals contain ice-nucleating agents (INAs) in the extracellular fluid.

INAs include many inorganic substances, microorganisms, proteins, and organic residues which can accelerate the process of crystal formation (nucleation).

INAs cause the extracellular fluids to freeze more readily than the intracellular fluids.

Ice Nucleation

As ice forms in the extracellular fluid, the unfrozen extracellular fluid becomes more concentrated with solutes.

This process draws water out of the cells, lowering the intracellular freezing point.

In this way, animals protect their cells from making ice crystals and can withstand freezing temperature.

Example

The freshwater larvae of the midge (fly) Chironomus, can survive freezing at temperatures as low as -32°C.

Antifreeze Substances

The body fluids of some cold-climate ectotherms contain cryoprotectants (antifreeze substances) e.g. glycerol, sorbitol and antifreeze proteins. These substances lower the freezing point of fluids. E.g. glycerol acting as an antifreeze can lower the freezing point of solute to as low as -17°C.

Supercooling

Ice crystals formation requires ice crystals or nuclei as seeds.

The animals which can undergo supercooling keep their bodies free of ice-nucleating agents INAs in a freeze-avoidance strategy.

So, their body fluids can be cooled below freezing temperature, yet remaining unfrozen because ice crystals fail to form.

Examples

- Certain Arctic bottom dwelling fishes live in a continually supercooled state and normally do not freeze unless they brush against frozen ice on the water surface.
- The larvae of parasitic wasp *Brachon cephi* can supercool to -47°C without ice crystal formation.

Topic-221 Ectotherms in Hot Environments

Ectotherms occupying hot environments are particularly vulnerable to warming. They may face daytime summer surface temperatures of nearly 40°C or above.

Most ectotherms have a critical thermal maximum below 45°C, above which long-term survival is not possible.

Above critical thermal maximum, affinity of the respiratory pigment for oxygen is decreased that affects most tissue functions. Moreover proteins are denatured and enzymes fail to function at higher temperatures.

Strategies to Live in Hot Environments

Most ectotherms actively respond to prevent elevation of their body temperatures to the critical thermal maximum. They tend to eliminate excess heat gained from the environmental by behavioral and physiological mechanisms.

Physiological Adaptations

Most ectotherms have high heat conductance that allows them to radiate heat rapidly on moving to less warm places.

Above 40°C many reptiles start panting, just like birds and mammals to increase heat loss through respiratory evaporation that causes cooling.

The marine iguanas can regulate heart rate and flow of blood to the surface tissues to either absorb or eliminate heat.

The Gila monster (*Heloderma suspectum*) evaporates water from its cloaca to cool its internal body temperature—a process similar to sweating in mammals.

Behavioral Adaptations

To avoid heat of the sun, most ectotherms move to shades or burrows that provide cooler microclimates.

The desert iguanas which are efficiently adapted to survive in harsh desert environments exhibit bipedal locomotion to run across hot sands. This behavior allows the abdomen to be raised from the ground, removing its contact from the hot substratum.

When crossing larger stretches of hot sands, they also dig sand at intervals to get cooling from lower cooler sand.

Topic-222 Costs and Benefits of Ectothermy

Ectothermy has its respective advantages and disadvantages relative to endothermy.

The biological costs and benefits of being an ectotherm are the inverse of those of being an endotherm.

In considering the costs and benefits of both, ectothermy and endothermy, we can make certain generalizations as follows.

Benefits of Ectothermy

1. Ectotherms use less energy in maintaining metabolic rate. They can use a large proportion of energy in growth and reproduction (over 50%). Endotherms have to consume major portion of energy (up to 98%) obtained from food in maintaining high metabolic rate and body temperature.
2. They require less food, they need less time spending in foraging. In contrast, endotherms are big consumers, spending more time on obtaining food. Example: A 300 g lizard needs 17 times lesser food per day as does a 300 g rodent living in the same habitat and having the same diet.
3. As they live at cooler places, they face less evaporative water loss, so need not to be massive for reducing surface-to-volume ratio.
4. Low respiratory rates also ensure less water loss.
5. In the tropics, ectotherms (e.g. reptiles) outcompete mammals both in abundance of species and in numbers of individuals. This competitive success is due to greater energy economy of the ectotherms.

Costs of Ectothermy

1. The inability to metabolically regulate body temperature requires behavioral thermoregulatory activities. Behavioral activities are time consuming and have certain limits e.g. basking in the sun is time wasting and is possible only if there is sufficient solar radiation.
2. In energy terms, ectotherms generate and expend lower energy. So, they cannot sustain longer periods of intense activity. In comparison, endotherms have high rates of aerobic respiration and energy yield. So, they can sustain longer periods of intense activity.
3. While constancy of body temperature in endotherms allows enzymes to function more efficiently, ectotherm's enzymes functions intermittently.
4. Ectotherms are more sluggish and less competent as predators in cold climates. So they are less abundant in temperate regions and cold climates. Endotherms have a significant competitive edge over ectotherms in the cold regions because their tissues are kept warm.
5. Because ectotherms tend to lower their activity levels periodically, they are vulnerable to predation. Whether they are basking in the sun or falling asleep because of dropping body temperature, they are at risk of predation.

Topic-223 Temperature Relations of Heterotherms

Heterotherms are animals intermediate between pure ectotherms and endotherms.

Heterotherms generally do not regulate their body temperatures within a narrow range. They mostly depend on external source of heat, like ectotherms. However, they are capable of endothermic heat production through muscular activity to varying extent.

They include certain insects, reptiles and fishes.

Heat Generation in Heterothermic Insects

Some flying insects (e.g. locusts, beetles, arctic flies) behave strictly as ectotherms when inactive.

However, they can raise the core temperatures in their thoracic parts to more or less regulated levels when preparing to fly.

Without this warm up, their flight muscles contract too slowly to produce sufficient power for flight and are unable to take off and fly.

As flight starts, the large thoracic flight muscles (which are among the most metabolically active tissues known) produce enough heat to maintain elevated muscle temperatures.

The thoracic temperature reaches about 40°C and is maintained during flight. At this level of heat generation the insects even have to employs heat-dissipating mechanisms to prevent overheating.

Heat Generation in Heterothermic Reptiles

Muscle-generated heat has been observed in some heterothermal reptiles including the female Indian python.

The brooding female Indian python elevates its body temperature with shivering thermogenesis to provide warmth for the group of eggs around which it coils itself.

The rate of muscle contractions increases with declining ambient temperature and an increased difference between the ambient and body temperatures.

Heat Generation in Heterothermic Fishes

Unlike terrestrial ectothermic vertebrates, which can bask in the sun to warm up, marine ectotherms cannot obtain radiant energy of sun as a source of heat.

Fishes can raise above ambient temperature only through intensive metabolic activity.

Many teleosts are strictly ectothermic, operating at core temperatures close to ambient temperatures.

However, some fishes, such as tunas and Mako sharks, have specializations for generating and retaining sufficient heat to raise the temperature of body muscle, brain, and eyes some 10 degrees or more above their surroundings.

These fishes can therefore be classified as regional heterotherms.

Anatomical Features of Fishes for Heterothermy

Two anatomical features permit heterothermic fishes to maintain temperature of their swimming muscles.

First, the red (dark) aerobic swimming muscles are located relatively deep in the core of a fish's body.

Second, escape of heat through skin and gills is retarded by a countercurrent heat-exchange system composed of rete mirabile.

Topic-224 Thermal Neutral Zone, Lower Critical Temperature (LCT) and Upper Critical Temperature (UCT) for Endotherms

The degree of thermoregulatory activity required by homeothermic endotherms to maintain a constant core temperature varies with environmental temperature.

Thermal Neutral Zone

The range of temperatures at which the basal rate of heat production balances heat loss to the environment is termed as the thermal neutral zone.

Within this range of temperatures, an endotherm does not need to expend energy to maintain its body temperature. It can regulate its body temperature by adjusting the rate of heat loss through alterations in the thermal conductance of the body surface.

These adjustments include:

- Vasomotor responses
- Postural changes to alter exposed areas of surface
- Regulation of insulating effectiveness of the pelage by raising or lowering hairs or feathers.

Lower Critical Temperature (LCT)

Lower critical temperature is the ambient temperature below which an endotherm’s basal metabolic rate becomes insufficient to balance heat loss.

Below this temperature, an endotherm must increase heat production by thermogenesis above basal levels to offset heat loss.

Heat production rises linearly with decreasing temperature below the lower critical temperature. This is termed as the zone of metabolic regulation.

If the environmental temperature drops below the zone of metabolic regulation, compensating mechanisms fail, the body cools, and the metabolic rate drops.

If an animal's body temperature falls below its normal values, the animal enters a state of hypothermia. If this condition persists, the animal grows progressively cooler, its metabolic rate falls and it soon dies.

Upper Critical Temperature (UCT)

Temperature zone which activates evaporative heat-dissipating mechanisms.

Animals react to higher temperature by losing heat by active heat-dissipating mechanisms such as sweating or panting.

temperatures above the upper critical temperature lead to hyperthermia. Hyperthermia occurs as the heat production and gain exceeds the rate of heat loss.

If evaporative heat loss mechanisms fail, hypothermia becomes lethal.

Topic-225 Temperature Relations of Endotherms

Maintaining Body Temperature

Homeothermic endotherms include most mammals and birds. They maintain their body temperatures relatively constant within a narrow range that is independent of the surrounding environmental temperature.

Core and Peripheral Temperatures

The core temperatures are maintained nearly constant between 37°C and 38°C in mammals and about 40°C in birds.

The temperatures of peripheral tissues and extremities are held less constant and are sometimes allowed to approach environmental temperatures.

Mechanisms of Endothermy

Endotherms use a variety of mechanisms for heat production (thermogenesis), heat conservation and heat dissipation.

These mechanisms include:

- Homeostatic mechanisms
- Muscular mechanisms (shivering thermogenesis)
- Hormonal mechanisms (non-shivering thermogenesis)
- Counter current exchange mechanisms
- Evaporative cooling mechanisms
- Cooling behaviors

Basal Heat Production and Metabolism

Basal heat production for different homeotherms of a given size is about the same, and the basal metabolic rate can be 10 times as high as the standard metabolic rate of ectotherms of comparable size measured at similar body temperatures.

This elevated basal metabolism, in conjunction with heat conserving and heat-dissipating mechanisms, allows homeotherms to maintain constant body temperatures as much as 30°C or more above ambient temperatures.

Topic-226 Thermogenesis in Endotherms

Endotherms Mechanisms for Thermogenesis

When the ambient temperature drops below the lower critical temperature, an endothermic animal responds by generating large amounts of additional heat, thereby preventing a decrease in the core temperature.

There are two primary means of extra heat production:

- Shivering thermogenesis
- Non-shivering thermogenesis

Both processes convert chemical energy into heat by a normal energy-converting metabolic mechanism involved in primary heat production.

However, all the chemical-bond energy released in this process is essentially used to generate heat rather than to chemical or mechanical work.

Shivering Thermogenesis

Shivering is a means of using muscle contraction to liberate heat.

The nervous system activates groups of antagonistic skeletal muscles so that there is little net muscle movement other than shivering.

The activation of muscle causes ATP to be hydrolyzed. The chemical energy released during contraction appears as heat.

Non-shivering Thermogenesis

In non-shivering thermogenesis, enzyme systems for the metabolism of fats are activated throughout the body, so conventional fats are broken down and oxidized to produce heat.

Non-shivering thermogenesis is activated by the sympathetic nervous system through the release of norepinephrine and thyroid hormones.

Brown Fat

A specialization found in a few mammals for fat-fueled thermogenesis is brown fat, also called brown adipose tissue (BAT).

This fat contains extensive vascularization and so many mitochondria that it is brown (owing largely to mitochondrial cytochrome oxidase).

Oxidation is triggered by thyroid hormones and takes place within the brown fat containing cells, which are richly endowed with fat-metabolizing enzyme systems.

In ordinary body fat, deposits must first be reduced to fatty acids, which enter the circulation and eventually are taken up by other tissues, where they are oxidized.

In this way, brown fat is an adaptation for rapid, massive heat production.

Brown fat heats up significantly during thermogenesis. This newly produced heat is rapidly dispersed to other parts of the body by blood flowing through the extensive vasculature of the brown fat tissue.

Topic-227 Endothermy in Cold Environments

Retaining Body Heat in Cold

Endotherms adapted to cold environments have evolved temporary and permanent mechanisms to retain body heat. These mechanisms involve anatomical and physiological features e.g. pelage, plumage, blubber and vasoconstriction.

Role of Pelage and Plumage

Fur and feathers provide heat insulation to mammals and birds respectively.

They lower specific heat conductances by reducing convection and dissipation of body heat by the wind.

The fluffed hairs and feathers trap air which is warmed by body heat, protecting the animal from cold.

The size and thickness of pelage varies in animals with insulation needs arising in face of environmental cold and seasonal variation.

Arctic mammals have very thick pelage, which provides more endurance to cold.

Animals living in the temperate zone exhibit seasonal variations by shedding fur or feathers in summer and growing new bodily cover in winter. This provides thick insulation during the winter, and prevents overheating during the summer.

Role of Blubber

Marine mammals (seals, walruses, sirenians, cetaceans—whales) have thick layers of insulating subcutaneous fat which is known as blubber.

Blubber is a good insulator because it has a lower thermal conductivity.

Fatty tissues are metabolically very inactive and have little blood vascularization. This prevents heat loss through body surface.

Vasoconstriction

An important means of controlling heat loss from the surface is the diversion of blood flow away from the skin.

Vasoconstriction of arterioles leading to the skin keeps warm blood from perfusing cold skin and conserves the heat of the body core.

Topic-228 Countercurrent Heat Exchange Systems

Heat Loss through Extremities

The limbs and extremities (e.g. gills, ears, nose) of endotherms are well vascularized but not massively insulated.

They are also thin and have large surface areas.

So they are major avenues of heat loss.

Reduction of Heat Loss—Countercurrent Exchange System

In many homeothermic animals living in cold climates excessive heat loss from extremities is reduced drastically by countercurrent heat exchange systems. Such systems have been found in cetaceans, seals, wading birds, arctic land mammals and many heterothermic fishes

A countercurrent exchange system is based on the fact that arterial blood, originating in the animal's core, is warm. Conversely, the venous blood returning from peripheral tissues may be very cold.

In the countercurrent exchange systems of appendages, arteries bringing blood towards extremities, pass just parallel to the out-going veins.

As the blood flows from the core and enters the limbs, the warm arterial blood gives up heat to the returning venous blood. So, arterial blood becomes successively cooler as it enters the extremity. By the time it reaches the periphery, this blood is precooled, only a few degrees above the ambient temperature. On reaching the extremity, it causes little heat loss.

Conversely, the returning venous blood becomes warmed by the arterial blood, so it is nearly at core temperature before it enters the core.

Rete Mirabile

Rete mirabile is a highly evolved system of countercurrent heat exchange. In this system, the artery carrying warm blood flowing toward the extremity is completely encased in a circlet of veins carrying cold blood back from the extremity.

Rete mirabile is found widely in homeothermic and heterothermic vertebrates

Advantage of Countercurrent Heat Exchange Systems

This system reduces heat loss into the environment without reducing blood flow, oxygen and nutrients supply.

This is the best alternative arrangement to vasoconstriction that reduces flow of blood to the periphery.

Due to this system, the extremities of cold-climate endotherms are maintained at temperatures that are far below the core temperature and often approach ambient temperature.

Topic-229 Endothermy in Hot Environments

Dissipation of Body Heat from Surface

Endotherms adapted to hot environments have evolved mechanisms to lose body heat.

Heat is always lost from the body surface. It is transferred from core to the surface by circulation. The rate of heat loss to the environment is regulated by the blood vessels supplying blood to the surface.

Heat Windows

Endotherms use various heat windows on their body surface to lose body heat. Opening or shutting these windows occurs by regulating blood flow.

These heat windows permit the loss of heat by radiation, conduction and convection.

Examples of Heat Windows

Ears:

The thin, membranous and lightly furred ears of many mammals with extensively branched blood vessels act as temperature-regulating windows.

Horns

Horns of various mammals (e.g. goats and cattle) are highly vascularized by a network of blood vessels that, under conditions of heat load, vasodilate and act as radiators of heat.

Limbs and Snout

The limbs and snout, having large surface area to volume ratios, are also used as thermal windows for the dissipation of heat by regulation of the rate of blood flow through the arterioles serving the skin of the appendages.

Lightly Furred or Naked Areas on Body

Some mammals living under conditions of intense solar radiation or high temperatures have certain areas of the body surface exceptionally lightly furred or even naked to facilitate heat loss by radiant, evaporative, or conductive means.

Such areas generally include the axilla (armpit), groin, scrotum, and parts of the ventral surface.

Some of these areas, such as the udder and scrotum, carry additional temperature sensors that are used to detect changes in air temperature with minimal interference from the core temperature.

Evaporative Heat Loss

Evaporating water from the surface takes away heat and causes cooling effect. Many vertebrates use sweating or panting to produce evaporative cooling.

Behavioral Mechanisms

Variations in posture or body orientation can increase rate of heat loss.

By adjusting the posture and orientation of body with respect to solar radiation, endothermic animals can adjust the degree to which their thermal windows are open or shut, permitting a change in thermal conductance.

This posturally controlled flexibility in surface insulation permits variability in heat transfer across the surface of endotherms that is independent of surface-to-mass ratio.

Topic-230 Evaporative Cooling

Evaporation of Water

Evaporation of water from the surface causes cooling effect.

For animals, it is the most effective means of removing excess body heat.

Evaporative Cooling by Animals

- Animals with naturally moist skin, such as amphibians, keep a body temperature lower than ambient temperature because of evaporative cooling.
- Certain reptiles, birds and some mammals take available body water (saliva and urine) or water from the environment and spread it on various body surfaces, allowing it to evaporate and take away body heat.
- Some vertebrates use sweating or panting to produce evaporative cooling.

Sweating

- Sweating is a mechanism for evaporative cooling in some mammals having sweat glands in the skin.
- Sweat glands actively extrude water through pores onto the surface of the skin.
- Sweating is under autonomic control.

Panting

- Panting involves use of respiratory system to lose heat by evaporative cooling. It occurs in mammals, birds and some reptiles.
- During panting, mammals inhale through the nose and exhale through the mouth, exposing the tongue and other mouth structures to encourage further water evaporation and therefore heat loss.
- It also involves increase in breathing rate and depth (hyperventilation).
- Heat is carried away in exhalant air because the dimensions of the mouth are such that exhalant air retains the heat absorbed in the lungs.
- In contrast, nasal passages and their vascularization are effective in retaining both water and body heat.

Topic-231 Thermostatic Role of Hypothalamus

A Thermostatic Control System

Body temperature control system of homeothermic endotherms is similar to the mechanized thermostatic control systems. Such a system has a temperature sensor with a set-point temperature, a thermostat and the heating unit.

Endothermic Heat Production

This analogy is especially applicable to endothermic heat production and control in the zone of metabolic regulation, in which heat production increases with decreasing ambient temperature.

Thermostatic Mechanism

The regulation of body temperature, T_b, works along the principles of negative feedback.

Most animals have many temperature sensors in various regions of the body.

Furthermore, to maintain T_b at about T_{set}, homeothermic animals can call on several heat-producing and heat-exchanging mechanisms, so the thermostat controls heat conserving and heat-loss mechanisms as well as heat production.

Temperature Sensors in Body

Mammalian body temperatures can vary widely (as much as 30^oC) between the periphery and the body core, with the extremities undergoing far more variation than the core.

Temperature-sensitive neurons and nerve endings exist in the brain, spinal cord, skin and sites in the body core, providing input to thermostatic centers in the brain.

Hypothalamus as Mammalian Thermostat

Mammals may have several thermoregulatory centers, the most important one is located in the hypothalamus and is considered to be the body's "thermostat".

Mammalian hypothalamic thermostat is highly sensitive to temperature.

Cooling the hypothalamus produces an increase in metabolic rate and a rise in Tb, whereas heating it evokes panting and a drop in Tb.

Temperature Sensitive Neurons of Hypothalamus

Highly temperature sensitive neurons are located in the anterior part of the hypothalamic thermostat.

Some of these neurons respond to increased hypothalamic temperature. These neurons activate heat dissipating responses such as vasodilation and sweating.

Other neurons respond when the brain temperature drops below the set-point temperature. They control the activation of heat-conserving responses (e.g., pilomotor) and heat-producing responses (e.g., shivering, nonshivering thermogenesis, brown-fat metabolism).

Topic-232 Thermoregulatory Centers in Non-mammals

Thermoregulatory Centers in Fishes

- Fishes have a temperature-sensitive center in the hypothalamus.
- It senses temperature rise of the body to regulate rate of respiration.
- A rise in temperature leads to an increased rate of metabolism, so need for oxygen is also increased. Fishes have adapted to a temperature-determined adjustment in the rate of respiration.

Thermoregulatory Centers in Reptiles

Reptiles also have a temperature-sensitive center in the hypothalamus.

The reptilian response to cooling of the hypothalamus is to engage in thermophilic (i.e., heat-seeking) behavior, whereas heating of the hypothalamus elicits thermopbobic (i.e., heat-avoiding) behavior.

Thermoregulatory Centers in Birds

Thermostatic control of body temperature in birds is more complex.

The region of the hypothalamus that serves as the thermoregulatory center in mammals is virtually insensitive to temperature changes in birds.

The spinal cord has a site of central temperature sensing in many birds.

Temperature receptors outside the central nervous system send signals to the thermostat in spinal cord, which in turn integrates the input and activates the thermoregulatory effectors.

Topic-233 Fever

Pyrogens

The hypothalamic thermoregulatory center is sensitive to certain chemicals collectively termed as pyrogens (fever-producing substances).

Categories of Pyrogens

There are two general categories of pyrogens, based on their origins.

- Exogenous pyrogens
- Endogenous pyrogens

Exogenous Pyrogens

- Exogenous pyrogens are toxins produced by gram negative bacteria.
- These are heat-stable, high-molecular-weight polysaccharides.
- They are so potent that a mere 10^{-9} g of purified toxin injected into a large mammal causes an elevation of body temperature.

Endogenous Pyrogens

- Endogenous pyrogens arise from animal's own tissues.
- They are heat-labile (heat sensitive) proteins.
- Leukocytes release endogenous pyrogens in response to circulating exogenous pyrogens produced by infectious bacteria.
- Thus, exogenous pyrogens cause a rise in body temperature indirectly by stimulating the release of endogenous pyrogens that act directly on the hypothalamic center.

Fever

The sensitivity of the hypothalamic temperature sensing neurons to pyrogen molecules leads to an elevation in the set point to a higher temperature than normal. As a result, body temperature rises several degrees. This rise in temperature is called as fever.

Significance of Fever

The adaptive significance of endogenous pyrogens and of their production of fever may be related to the bacteriostatic effects of elevated body temperature.

Topic-234 Thermoregulation during Exercise

Muscle Contraction and Heat Generation

For every joule of chemical energy converted into mechanical work during muscle contraction, 3 J of energy is degraded to heat. So, the energy efficiency of muscle contraction is only about 25%.

Exercise Generates Excess heat

Exercise involves muscle contraction. So, extra heat is produced during exercise that is added to the heat produced by basal metabolism. It causes a rise in body temperature above the set-point.

Rise in Body Temperature

To avoid a rise in core temperature, the extra heat generated during exercise is dissipated to the environment simultaneously. However, a rise in temperature does occur during exercise, indicating incomplete removal of the excess heat.

The level to which the core temperature rises in homeotherms is proportional to the rate of muscular work.

During light or moderate exercise in cool environments, body temperature rises to a new level and is regulated at that level as long as the exercise continues. Thus, temperature appears to remain under the control of the body's thermostat.

During heavy exercise, especially in warm environments, the core temperature can rise to high levels as heat-dissipating mechanisms are not able to balance heat production.

Elevations of 4 to 5 Celsius degrees in core temperature are commonly observed in human beings after strenuous, sustained running and in race horses, greyhounds, and sled dogs after racing.

Dissipation of Excess Heat

The rise in T_b is kept small by the high sensitivity of the feedback control of heat-dissipating mechanisms.

For example, a small increase in Tb above the set-point temperature produces a strong and steep increase in the rate of sweating. The effectiveness of evaporative heat loss is affected by humidity of the surrounding air-the higher the humidity, the less effective the heat loss.

Topic-235 Dormant States: Sleep and Torpor

Dormancy

Dormancy is a general term for reduced body activities, including reduced metabolic rate.

Dormancy can be classified based on its duration, ability for arousal and decrease in body temperature.

Dormancy includes: sleep, torpor, hibernation, winter sleep, and estivation.

All these forms of dormancy are related to physiological processes.

Sleep

Sleep is a naturally recurring state of mind and body, characterized by altered consciousness, decreased ability to react to stimuli, inhibition of nearly all voluntary muscles, and reduced interactions with surroundings

Sleep has been the most thoroughly investigated form of dormancy because it is also found in humans.

The time course and extent of sleep varies greatly among animals, e.g.

- Seals resting on ice sleep for only a few minutes at a time.
- Human beings and many other mammals sleep for hours at a time.
- Many of the big carnivores (e.g., lions and tigers) sleep for as long as 20 hours a day, especially after a meal.

Triggers of Sleep

There may be a variety of triggers of sleep.

In mammals, certain sleep-inducing substances build up in extracellular fluids of the central nervous system during wakefulness. Their concentration induces sleep.

The Sleep Process

Sleep process has been divided into two phases:

- Slow wave sleep (deep sleep)
- Rapid eye movement (REM) sleep

Slow Wave Sleep

In mammals, slow wave sleep is associated with:

- A drop in body temperature
- A drop in hypothalamic temperature sensitivity
- Changes in respiratory and cardiovascular reflexes

Rapid-Eye-Movement (REM) Sleep

A smaller portion of total sleep time associated with desynchronized and fast brain waves, eye movements, loss of muscle tone and suspension of homeostasis.

Hypothalamic temperature control is under suspension.

Torpor

Small endotherms can not maintain their high rates of metabolic rates when they are not feeding during periods of inactivity.

During those periods, some of these animals enter a state of torpor, in which body temperature and metabolic rate decrease.

The lower basal metabolism lowers the rate of conversion of energy stores into body heat.

Thus, it is generally advantageous to allow body temperature to decrease during periods of nonfeeding and inactivity.

As the animal comes out of torpor and becomes active, it undergoes a burst of metabolic activity, through shivering or oxidation of brown-fat and its body temperature rises.

Daily torpor is practiced by many terrestrial birds.

The hummingbird is a classic example, allowing its body temperature to fall from a daytime level of about 40°C to a nighttime level as low as 13°C.

Several species of small mammals also undergo torpor (e.g., shrews).

Topic-236 Hibernation, Winter Sleep and Estivation

Hibernation

During the winter, various endotherms go into hibernation that may last for weeks or months.

Hibernation is a time of decreased metabolism and lowered body temperature as well as the heart and breathing rates.

The set point of a hibernator's thermoregulatory center drops to about 20° C, but thermoregulation is not suspended.

True hibernation occurs in some small mammals belonging to orders, Rodentia (ground squirrels), Insectivora (shrews), and Chiroptera (bats).

Preparation for Hibernation

Hibernation occurs only in those mammals which can store sufficient energy reserves to survive the periods of nonfeeding.

Hibernating animals prepare for hibernation by building up fat reserves and growing long winter pelts.

All hibernating animals have brown fat.

Decreasing day length stimulates both increased fat deposition and fur growth.

Arousal from Hibernation

The rate of arousal from hibernation is often much higher than the rate of entry into hibernation. For example in ground squirrel, the transition to the dormant state is completed within 12 to 18 hours, whereas arousal requires less than 3 hours.

Arousal depends on rapid heating initiated by intensive oxidation of brown fat, accompanied by shivering. This leads to a large surge in metabolic rate and rise in body temperature.

Many hibernators arouse periodically (as often as once a week or as infrequently as every 4-6 weeks) to empty their bladders and defecate.

Winter Sleep

Winter sleep is a state of prolonged inactivity in winter that occurs in some larger mammals such as badgers, bears, opossums, raccoons, and skunks.

Body temperatures do not drop substantially (drops only a few degrees), and sleeping animals can wake and become active very quickly. As the body temperature remains near normal, this is not called hibernation.

Large energy reserves sustain these mammals through periods of winter inactivity.

Estivation

Aestivation is a period of inactivity in some animals to withstand extended hot and dry periods.

The animal enters a burrow as its environment begins to dry.

It generally does not eat or drink, its metabolism becomes slow and becomes inactive.

The animal emerges from estivation when moisture returns in environment.

Aestivation is common in many invertebrates (especially mollusks), lung fishes, reptiles, and amphibians.

Animal Behavior

Topic-237 Animal Behavior: Basics

Animal Behavior

The sum of responses made by an animal to the external and internal stimuli is called behavior.

Behavior is based on physiological systems and processes of the animal.

Ethology

The scientific study of animal behavior in their natural environment is called as ethology.

Examples of Behavior

The behaviors may be as simple as moving towards or away from light to the complex behaviors e.g.

- Territory defense
- Courtship and mating
- Liking, disliking and fighting or caring
- Migration and navigation
- Building hives, nests or dens
- Producing voices or songs
- Recognition and communication
- Procuring food
- Learning and memorizing

Behavior is Subject to Natural Selection

Most of the animal behaviors are essential for survival and reproduction.

So the development of behavior has remained subject to substantial natural selection.

Natural selection contributed to the anatomical development of nervous system as well as enhancing the physiological features that contributed to the evolution of complex behavioral patterns.

Objectives of Ethology

1. Understanding the stimuli that elicit behavior.
2. Understanding the physiological mechanisms that mediate the response.
3. Understanding how an animal's experience during growth and development influence the response.
4. Understanding how the behavior aids in survival and reproduction.
5. Understanding the behavior's evolutionary history.

Topic-238 Scientific Approaches to Study Animal Behavior

Behavioral scientists have taken two complementary approaches to understand animal behavior.

1. Neuroethological approach
2. Ethological approach

Neuroethological Approach

The neuroethological approach involves bringing the animal into the laboratory and observing its behavior in a very simplified set of well-defined circumstances.

Simplified Set of Circumstances

In the laboratory:

- there are no predators
- the number and sex of conspecifics (members of the same species) are controlled by the experimenter
- there are unusual lights, smells and sounds
- the animal is often limited to a relatively small and confined space

Study of Behavior with Neuroethological Approach

- The neuroethological approach focuses on the study of neuronal responses generated during the manifestation of an animal’s behavior.
- In some cases, the nervous system of animal is exposed surgically to allow the experimenter to record from neurons while behavior is going on.
- Studies under these reduced conditions are useful for obtaining answers to many questions about behavior.
- However, the results from such experiments can be difficult to translate into an understanding of how animals deal with challenges in their everyday lives in natural environment.

Ethological Approach

- The ethological approach is to go into the field and observe the animal’s behavior in its natural environment.
- Observing animals in their normal state is an ancient practice.
- However, for a physiologist, these natural conditions raise severe problems that differ from those created by laboratory conditions.
- While observing an animal in its natural setting, it is impossible to record activity from the animal's nervous system that is involved in producing behavior.

Topic-239 Neurosensory Basis of Behavior

Role of Nervous System in Behavior

Nervous system plays an essential role for the human and other animal's behaviors.

The behavior of an organism is manifested in response to the stimuli from the environment. The stimuli act as sensory input for the nervous system. In response to the sensory input, nervous system produces motor output that controls the contraction of muscles.

All behavioral acts are generated by this motor output of the nervous system.

Complexity of Behaviors

Some of these input-output relations are simple and predictable reflexes. Other kinds of behavior are highly dependent on information stored from past experience, and therefore are less predictable to an observer.

Neuronal Networks Underlie All Behaviors

The hardware underlying all behavior is composed of neuronal networks i.e. interconnecting circuits of neurons.

Lower animals with simple behavioral patterns have simple neural network.

The neural circuitry is more complex in animals with a centralized nervous system.

As the interneurons of the CNS increase, the behavioral potential of animal also increases.

Higher animals with complex CNS have developed the ability to learn from experience and to associate combinations of stimuli.

The complex behaviors, learning and memory have neurophysiological basis, rooted in the vast array of neural circuitry lying between the afferent sensory pathways and efferent motor pathways.

Topic-240 Neuronal Plasticity

Although the basic architecture of the nervous system is established during embryonic development, the behavioral patterns can modify with experience. This capacity of nervous system to change with experience is known as neuronal plasticity.

Neuronal plasticity is of premier importance for the survival of any organism.

Examples of Neural Plasticity

- Learning, memory and development of motor skills and habits.
- Neural plasticity lies behind human intelligence.
- The ability of all higher animals to respond to stimuli in diverse ways, different from fixed and programmed reflexes.
- Behavioral plasticity, virtually demonstrated by all animals.

Mechanism of Neuronal Plasticity

Much of the reshaping of the nervous system occurs at synapses.

The mechanisms that underlie synaptic plasticity are currently the subject of many experiments.

Modifying Synapses—Synaptic Plasticity

Synaptic plasticity may takes place as a result of developmental events over the course of a lifetime.

Synaptic connections that are established in embryos are later refined into adult patterns.

Later, changes in synaptic strengths result in learning and memory at mature synapses.

Changes in mature synapses involved in learning and memory depend on a retrograde (reverse) signal that is sent from the postsynaptic neuron to the presynaptic neuron.

Mechanisms of Synaptic Plasticity

Change in mature synapses i.e. synaptic plasticity occurs by two types of mechanisms:

(1) Presynaptic mechanisms:
Involve changes in the functioning of presynaptic terminals.
For example, a change in the amount of transmitter released from presynaptic terminals.

(2) Postsynaptic Mechanisms:
Involve changes in the postsynaptic neurons.
For example a change in the postsynaptic apparatus that results in altered amplitude of depolarizations.

Topic-241 Inborn, Innate or Instinctive Behaviors

Inborn, Innate or Instinctive Behaviors

The inborn or innate behaviors are developmentally fixed particular complex behaviors exhibited by most members of a species.

These behaviors are built-in part of the nervous system at the time of birth or develop at an appropriate point in maturation.

Such behaviors are also known as instinctive behaviors which are different from reflexes, which are simple responses of an organism to a specific stimulus.

Genetic Basis of Instinctive Behaviors

Innate behaviors are closely controlled by genes with little or no environmental influence.

The responses to particular stimuli or situations are determined by neural programmes which are encoded by the hereditary material. These programmes contain information for anatomical and physiological organization of the nervous system.

Exhibition of Instinctive Behaviors

Innate behaviors do not have to be learned or practiced.

They are performed in a reasonably complete form even when they are exhibited first time by the individual, without prior exposure or experience.

The innate behaviors can often be modified through experience but only to a limited extent.

Examples

- A human new born will turn to suckle when brought near the mother’s breasts.
- Sea turtles, newly hatched on a beach, will automatically move toward the ocean.
- A marsupial climbs into its mother's pouch upon being born.
- Animal fighting, animal courtship behaviors, internal escape functions and building of nests are also instinctive behaviors.

Importance of Innate Behaviors

Innate behaviors usually involve basic life functions that are important in the survival.

They are especially important for those animals that have short life spans and poorly developed nervous coordination.

Innate Behaviors in Higher Animals

The role of instincts in determining the behavior of animals varies from species to species.

In animals with more complex neural systems, there is greater role of cerebral cortex and social learning, so instincts play a lesser role.

Topic-242 Fixed Action Patterns

Fixed action patterns are a type of instinctive behaviors that are relatively fixed within the species and, once triggered, inevitably run to completion.

They consist of a sequence of unlearned acts that is essentially unchangeable and, once initiated, is usually carried to completion.

Fixed action patterns are triggered in response to external cues known as sign stimuli or key stimuli.

The key stimuli are also called as releasers, because they appear to release a prepatterned behavioral response within an animal.

Example

A classic example of a sign stimulus and fixed action pattern is observed in the male three-spined stickleback fish.

Male sticklebacks have red bellies. A male attacks other males with red bellies, coming to their nesting territories, perceiving them as invaders.

A male stickleback will not attack a fish lacking red coloration but will behave aggressively and attack even unrealistic models and objects having areas of red color on the underside.

So red underside is a sign stimulus for the male sticklebacks and aggressive attacking behavior is the fixed action pattern.

Properties of Fixed Action Patterns

1. They are not simple reflexes. They are relatively complex motor acts, each consisting of a specific temporal sequence of components.
2. They are typically elicited by specific key stimuli rather than by general stimuli.
3. Fixed action patterns have all-or-none property i.e. if the stimulus is removed after the animal's response has initiated, the behavior continues to completion. It means that the key stimulus is required to turn the pattern on; but, once begun, it plays out independently of further stimulation.
4. The stimulus threshold for fixed action patterns may have quite large variation. It varies with the state of the animal. For example, if an animal has copulated once to an arousing stimulus, it cannot be induced to copulate again without intense stimulation.
5. All members of a species perform a given fixed action pattern nearly identically.
6. Fixed action patterns are inherited genetically. They are exhibited in a recognizable form even by the members that have had no prior experience with the key stimuli.

Topic-243 Animal Orientation, Kinesis and Taxis

Animal Orientation

Orientation is the ability of an animal to determine its position in space and move predictably with respect to specific stimuli.

The best examples of animal orientation are observed in response to gravity. A human who has fell down on ground orients himself by standing up. Some animals orient themselves towards the incoming light. Other examples include acoustic and olfactory orientations.

Sensory Motor Basis of Orientation

Animal orientation is a complex process that requires the integration of sensory input and the coordination of motor output.

It includes receiving information through the sensory receptor neurons, processing and correlating it in the central nervous system and forming a response through the muscles that cause the body of animal to move.

Types of Orienting Movements

1. Kinesis
2. taxis

Kinesis

Kinesis is a change in activity or turning rate in response to a change in location.

For example, sow bugs (terrestrial crustaceans) exhibit a kinesis in response to variation in humidity. They become more active in dry areas and less active in humid areas. They do not move toward or away from specific conditions, but their increased movement under dry conditions makes it more likely that they will leave a dry area and encounter a moist area, where they survive better.

Taxis

A taxis is an oriented movement toward (positive taxis) or away (negative taxis) from some stimulus.

For example, trout and many other river fishes automatically swim or orient themselves in an upstream direction (toward the current). This taxis keeps the fish from being swept away and keeps it facing the direction from which food will come.

Cockroaches move away from the stimulus of light, which is called negative phototaxis.

An animal that turns toward the light is said to exhibit positive phototaxis.

Topic-244 Animal Navigation and Migration

Migration is a regular, long-distance change in location. It involves animal navigation over long distances through unfamiliar territories and environments they have not previously encountered.

Many animals seasonally migrate in response to various environmental stimuli involving seasonal droughts or harsh weather conditions of summer and winter.

Migratory behavior is observed in a wide variety of birds, fishes, butterflies, grey whales and many other animals.

Cues Used in Migration and Navigation

The navigational abilities of animals rely on several different sensory cues that guide in orientation and finding their way to the destination.

Many of these cues are difficult to detect and study as they are generally unavailable to human beings.

During navigation, animals may use:

- Sun compass
- Celestial cues
- Earth's magnetic field
- Visual, auditory and olfactory cues

Sun Compass

Many migrating animals track their position relative to the sun.

The sun's position relative to Earth changes throughout the day.

Animals adjust for these changes by means of a circadian clock, an internal mechanism that maintains a 24-hour activity rhythm or cycle.

Celestial Cues
Nocturnal navigators can use the North Star, which has a constant position in the night sky.

Earth's Magnetic Field
Many long-distance migrants, e.g. birds and fishes, navigate by sensing their position relative to Earth's magnetic field.

Visual, Auditory and Olfactory Cues
Many birds also use particular landmarks, odors, sounds, and additional visual cues such as the plane of polarized light to find their way to reach their destination.

Topic-245 Behavioral Rhythms

Biological Clocks

Animals respond rhythmically to cyclic fluctuations in their environment like changes in seasons and daily light and dark periods. The rhythmic behaviors of animals are controlled by the biological clocks which regulate the daily and seasonal cyclic activities of animals. These clocks have a molecular genetic basis as genes for keeping time sense have been identified in many animals including human beings.

These biological clocks may be:

- Circadian clocks
- Circannual clocks
- Lunar clocks

These clocks are influenced by dark and light periods and lunar cycles.

Circadian Clock and Circadian Rhythms

The most important biological clock is the circadian clock that regulates and coordinates daily behavioral activities of all animals.

The output of this clock is a circadian rhythm that is exhibited as a daily cycle of rest and activity.

The clock is normally synchronized with the light and dark cycles of the environment. However it can also maintain rhythmic activity under constant environment conditions, such as during hibernation.

Circannual Rhythms

Some behaviors, such as hibernation, migration and reproduction, reflect biological rhythms linked to the yearly cycle of seasons are called circannual rhythms.

A circannual rhythm spans events that repeat on a 12 month basis corresponding to the tilt of earth's spin axis resulting in seasonal changes.

Circannual rhythms influence the animal physiology by changing the periods of day light and darkness as well as temperature in the environment.

Lunar Cycles and Tidal Rhythms

The lunar cycles are controlled by the phase of the moon whose gravitation pull determines the timing of tidal rhythms in marine environments.

The courtship and reproductive behaviors of many marine animals are linked to the timing of the new and full moon.

Timing the behavior to the lunar cycle links their reproduction to the times of greatest tidal movement.

The tides disperse larvae to deeper waters, where they complete early development in relatively safer environment.

Topic-246 Animal Signals and Communication

Animal Signals

A stimulus transmitted from one animal to another is called a signal.

The signals may be visual, chemical, tactile, or auditory.

Animal Communication

The transmission and reception of signals constitute animal communication.

Animal communication is an essential element of interactions between individuals.

The communication patterns are correlated with the signals and result in:

- Visual communication
- Olfactory communication
- Tactile communication
- Auditory communication

The form of communication used to transmit information is closely related to an animal's lifestyle and environment. For example, in nocturnal terrestrial mammals visual displays are relatively ineffective. Instead, these species use olfactory and auditory signals, which work as well in the dark as in the light. In contrast, the diurnal (active mainly in daytime) animals communicate primarily by visual and auditory signals.

Examples of Animal Communication

1. Courtship Behavior of Drosophila

Courtship behavior in fruit fly *Drosophila melanogaster* constitutes a stimulus-response chain, in which the response to each stimulus is itself the stimulus for the next behavior.

- The courtship begins with a male identifying and orienting toward a female of the same species.
- When the male sees the female, he relies on visual communication.
- In addition, the male's sense of smell, or olfactory system, detects chemicals released into the air by the female. This is an example of chemical communication.
- Having recognized the female, the male approaches and taps the female with a foreleg. This touching is the tactile communication.
- Now the male extends and vibrates his wing, producing a specific courtship song. This singing is an example of auditory communication, informs the female that the male is of the same species.

If all of these forms of communication are successful, the female will become receptive and allow the male to attempt copulation.

2. Communication for Information Transfer

The information content of animal communication varies considerably.

One remarkable example is the symbolic language by which honeybees share information about the location of food sources.

A honeybee which finds a food source returns to its hive quickly becomes the center of attention for other bees, called followers. If the food source is close to the hive (less than 50 m away), the returning bee

moves in tight circles while waggling its abdomen from side to side. This behavior, called the "round dance;' motivates the follower bees to leave the hive and search for nearby food.

When the source of food is farther from the nest, the returning bee instead performs a "waggle dance." This dance, consisting of a half-circle swing in one direction, a straight run, and a half-circle swing in the other direction, communicates to the follower bees both the direction and distance of the food source from the hive.

Topic-247 Pheromones and Chemical Communication

Many animals communicate by releasing chemical substances called pheromones.

Pheromones are:

- Chemical messengers with particular odors
- Released to the environment
- Spreads through air
- Effective at remarkably low concentrations
- Detected by the olfactory receptors of the receiving animal
- Affect the behavior of individuals of the same species

Chemical communication is well developed in insects, fishes, salamanders and mammals.

Roles of Pheromones

1. **Recognize members of species:**
   - Unicellular organisms recognize members of their own species.
   - Many insects identify members of their colony through chemical communication.
2. **Reproductive Roles:**
   - Pheromones may attract members of opposite sex for breeding.
   - They induce courtship behaviors in the members of a species
   - Pheromones have adaptive value for successful external fertilization. In many aquatic invertebrates and fishes, spawning by one individual triggers spawning in other individuals of the population because pheromones are liberated along with the gametes. These pheromones induce spawning in other individuals.
3. **Territorial behaviors:**
   - Many male mammals mark their territories with pheromones to warn other males of their occupied area.
4. **Insect's Social Order**
   - In a honeybee colony, pheromones produced by queen maintain hive's complex social order.
5. **Alarm Calls**
   - Members of some species produce pheromones on getting injured to warn other members of presence of danger.
6. **Repel Predators**
   - Pheromones also are used to repel predators.
   - A common example is the foul-smelling musk that makes skunks unpalatable to their enemies.

Advantages of Pheromones as Chemical Signal

The chemical signals:

1. Usually provide a simple message that can last for hours or days
2. Are equally effective during night or day
3. May be transported over long distances by air
4. Take relatively little energy to produce

Disadvantages of Chemical Signals

Chemical signals are slow to act and cannot be changed quickly.

Topic-248 Genes & Environment in Behavior Development

Interaction of Genetic & Environmental Factors

- In past, the scientists have debated whether the behavior of an animal is due to genes (nature) or to environment (nurture).
- It is now known that animal behavior is governed by complex interactions between genetic and environmental factors.
- Although the inherited DNA sequence of an individual provides instructions for the development of behavior, the expression of genes is influenced by many factors.

Environmental Factors in Behavior Development

These include:

- Environment of the fertilized egg
- Animal's diet
- Animals social interactions
- Environmental conditions surrounding the animal

Environment's Effect on Behavioral Gene Expression

Scientists have used a number of methods to determine the extent of environmental effects on the expression of genes determining an animal's behavior.

1. **Cross-fostering study**

In this method, the young of one species are placed in the care of adults from another species.

The extent to which the offspring's behavior changes in such a situation is one measure of how the social and physical environment influences behavior.

The males of certain mouse species have behavioral differences that are well suited for cross-fostering studies.

2. **Twin Studies**

For humans, the influence of genetics and environment on behavior can be explored by a twin study.

In such a study, researchers compare the behavior of identical twins raised apart with those raised in the same household.

The results of twin studies carried out for human behavioral disorders, such as schizophrenia, anxiety disorders, and alcoholism have revealed that in different environments the expression of disorder varies between 20 to 80 % in separate environmental conditions.

Both environment and genetics therefore contribute significantly to the behaviors that characterize these disorders in humans.

Topic-249 Habituation and Imprinting

Learning Behaviors

Most behaviors are variable with experience. Such behaviors are known as learning behaviors. Habituation and imprinting are two of the many examples of learning behaviors.

Habituation

Habituation is one of the simplest forms of learning.

In habituation, an animal learns to ignore a repeated, irrelevant stimulus.

It is defined as a loss of responsiveness to stimuli that convey little or no new information.

Example

- Many mammals and birds recognize alarm calls of members of their species.
- However, if these calls prove fake repetitively, they stop responding (the "cry-wolf" effect).
- An example of learning by habituation is observed in squirrels.
- When one squirrel feels threatened, the others hear its signal and go to the nearest refuge.
- However if the signal comes from an individual who has caused many false alarms, its signal is ignored.

Significance

By habituation to unimportant stimuli, an animal conserves energy and time that are better spent on other important functions.

Imprinting

Imprinting is the formation of a long-lasting behavioral attachment to a particular individual or object at a specific stage in life.

It is a type of learning in which a very young animal fixes its attention on the first object with which it has visual, auditory or tactile experience and thereafter follows that object.

Sensitive or Critical Period

Imprinting usually forms only during a specific sensitive period, also called a critical period, a limited developmental phase when certain behaviors can be learned.

During the sensitive period, the young imprint on their parent and learn the basic behaviors of their species, while the parent learns to recognize its offspring.

Examples

Among gulls, the sensitive period for a parent to bond with its young lasts one to two days. If bonding does not occur, the parent will not care for the infant, leading to death for the offspring.

Another example comes from the classic experiment of Konard Lorenz who conducted experiments with geese. He made the goslings to imprint on him. These goslings followed him as though he was their mother.

Topic-250 Spatial Learning and Cognitive Maps

Spatial Learning

Every natural environment shows some spatial variation, e.g. in the locations of nest sites, hazards, food, and prospective mates. Consequently, an organism's fitness may be enhanced by the capacity for spatial learning, the establishment of a memory that reflects the environment's spatial structure.

Example

The behavior of a female digger wasp, which nests in small burrows dug into sand mounds. When a wasp lefts her nest to go hunting, she covers the entrance with sand. Upon her return, she flies directly to her hidden nest, despite the presence of hundreds of other burrows in the area. The wasp locates her nest by learning its position relative to visible landmarks, or location indicators.

This visual learning of object maps is proved by experimentally disturbing the landmarks that confuses the wasp.

Cognitive Maps

Some animals guide their activity by a cognitive map, a representation in the nervous system of the spatial relationships between objects in an animal's surroundings. Rather than relying solely on moving from landmark to landmark, animals using cognitive maps can navigate more flexibly and efficiently by relating landmark positions to one another.

Example

The Clark's nutcracker birds store thousands of pine seeds in fall at thousands of hiding places called caches, distributed over an area as large as 35 km^2. During the winter, the birds relocate many of their caches.

By experimentally varying the distance between landmarks, researchers demonstrated that birds can identify the halfway point between landmarks. Such behavior suggests that nutcrackers employ an abstract geometric rule, which we can approximate as "Seed caches are found halfway between particular landmarks". Such rules are a fundamental property of cognitive maps and reduce the amount of detail required to remember an object's location.

Topic-251 Associative Learning

A learning that involves making associations between experiences.

Example

If a white-footed mouse bites a brightly colored caterpillar that tastes bad, the mouse avoids attacking insects of similar appearance in future.

The ability to associate one environmental feature (such as a color) with another (such as a foul taste) is called associative learning.

Types of Associative learning

Associative learning can be divided into two types:

- Classical conditioning
- Operant conditioning

Classical Conditioning

In classical conditioning, an arbitrary stimulus becomes associated with a particular outcome.

Example

Russian physiologist Ivan Pavlov carried out early experiments in classical conditioning, demonstrating that if he always rang a bell just before feeding a dog, the dog would eventually salivate at the bell's sound alone, in anticipation of food.

Operant Conditioning

In operant conditioning, also called trial-and-error learning, an animal learns to associate one of its own behaviors with a reward or punishment and then tends to repeat or avoid that behavior.

Example

A predator may learn to avoid certain kinds of potential prey if they are associated with painful experiences.

Similarly, an animal may repeat an action that has resulted in a surprise finding of food. For example, in a classic experiment, a rat was trained through repeated trials to obtain food by pressing a lever.

Topic-252 Cognition and Problem Solving

Cognition

Cognition is the process of intellect, represented by awareness, understanding, reasoning, recalling memories and judgments.

It is the most complex form of learning.

Cognition is exhibited characteristically by primates. Many mammals, cephalopods and insects also exhibit cognition.

Problem Solving

Problem solving is a cognitive activity of devising a method to proceed from one state to another in the face of apparent obstacles.

Problem solving capacity of an animal depends on the information processing ability of a nervous system.

Examples

- If a chimpanzee is placed in a room with several boxes on the floor and a banana hung high out of reach, the chimp can “size up" the situation and stack the boxes, enabling it to reach the food.
- Such problem solving behavior is highly developed in some mammals, especially primates and dolphins.
- Notable examples have also been observed in some bird species, especially ravens, crows, and jays.
- Many animals learn to solve problems by observing the behavior of other individuals. Young wild chimpanzees, for example, learn how to crack oil palm nuts with two stones by copying experienced chimpanzees.

Topic-253 Foraging Behavior

Foraging Behavior

The behavior that includes all activities of an animal related to search, recognize and capture food items is called foraging behavior.

Foraging Choices for Animals

Foraging behavior of animals includes the following choices:

1. What items should be included in the diet?
2. If food occurs in patches, what path should an animal take between patches, and how should it locate new patches of food?
3. If the food in a patch is depleted, when should the animal depart from that location and seek another patch of food?

- Hummingbirds and various species of bees that visit clumps of flowers to obtain nectar must make each of these decisions.
- Owls that forage for small rodents in different habitats, including fields and forests, must make similar decisions.

Evolution of Foraging Behavior

As nutrition is essential for an animal's survival and reproductive success, natural selection has operated to refine behaviors that enhance the efficiency of feeding.

Example

- The fruit fly (*Drosophila melanogaster*) provides an example of simple genetic variation that explains the evolution of foraging behavior.
- Variation in a gene called *forager* (*for*) dictates the food search behavior of fruit fly larvae.
- The larvae carrying the for^R ("Rover") allele travel nearly twice as far while feeding as larvae with the for^s ("sitter") allele.

- Experiments revealed that the enzyme encoded by the *forager* locus is involved in signal transduction pathways and is more active in for^R than in for^s larvae.
- These results indicate that changes in processing of environmental information can substantially alter behavior.

Topic-254 Foraging Model: Risk and Reward Balance

Gains and Costs of Foraging

There are energy gains and costs in finding and consuming food:

- Energy gain from food
- Energy expenditure in foraging
- Risk of predation while foraging

Optimal Foraging Model

Behavioral ecologists apply a model of cost-benefit analysis to study the foraging strategies of animals.

This model concludes that animal foraging behaviors reflect a compromise between competing selective pressures i.e. the benefits of nutrition and the costs of obtaining food.

Optimal Energy Expenditure

- This model analyzes the energy needed to search food, energy needed to pursue and handle food and energy required to digest the food.
- For an animal, the energy gain from a given food item must exceed the energy costs.
- According to the optimal foraging model, natural selection favors a foraging behavior that minimizes the costs of foraging and maximizes the benefits.

Balancing Risk and Reward

- One of the most significant potential costs to a forager is risk of predation.
- Maximizing energy gain and minimizing energy costs are of little benefit if the behavior makes the forager a likely meal for a predator.
- Animal's foraging behaviors reflect more safety from predation risk and not the hunger for more food gain

Topic-255 Mating Behaviors and Systems

Mating Behavior

Mating behaviors of animal species evolved through sexual selection, a form of natural selection.

Mating behavior enhances reproductive success of a species.

Mate Choice

The mating behavior involves mate choice. Mate choice includes:

- Seeking or attracting mates
- Choosing among potential mates
- Competing for mates

Mating Systems

Mating systems of different species vary.

They are of three types:

- Monogamous mating system

- Polygamous mating system
- Promiscuous mating system

Monogamous Mating Systems

- In monogamous relationship, one male mates with one female during a breeding season.
- The mates usually remain together for a longer period.
- Monogamy is rare in most animal groups. It is specifically found in groups exhibiting nesting behaviors and parental care.
- In birds it is most frequent as more than 90% are monogamous.
- In a few bird species such as swans and geese, partners are chosen for life and often remain together throughout the year.
- In others, seasonal monogamy is more common

Polygamous Mating Systems

In polygamous system, an individual of one sex mates with several of the other.

Polygamous relationships are of two types:

- Polygynous Systems
- Polyandrous Systems

Polygynous Systems

- Polygynous systems involve an exclusive relationship of one male with two or more females
- Polygynous species are generally dimorphic, with males being showier and often larger than females

Polyandrous Systems

- In polyandrous systems, one female has an exclusive relationship with two or more males.
- Polyandrous species are also dimorphic with females generally more ornamented and larger than males.

Promiscuous Mating System

- In this system, a member of one sex within the social group mates with any member of the opposite sex.
- This system involves no strong pair-bonds or lasting relationships.
- Among promiscuous species, males and females are often so much alike morphologically that they may be difficult or impossible to distinguish based on external characteristics.

Topic-256 Parental Care

Parental Care

- Parental care caters the needs of the newborn and young offspring.
- It is more evolved in animals whose newborns and young cannot care for their selves.
- So, it has been an important factor in the evolution of mating systems.

Parental Care in Birds

Birds present an excellent example of parental care.

- Most newly hatched birds cannot care for themselves.
- They require a large, continuous food supply that is difficult for a single parent to meet.
- To enhance the chances of viability of offspring, most birds are monogamous.
- The male stays with and helps its mate till the offspring are able to fulfill their own needs.

Parental Care in Mammals

- In mammals, lactating female nourishes the young and is involved in most of the caring behaviors. Males usually play no role in raising the young.
- In some mammalian species, such as lions, males protect the females and young. In such species a male or small group of males takes care of many females at once in a harem.

Polygyny When no Parental Care

- Mammals and birds whose young ones can feed and care for themselves almost immediately after hatching, males do not stay with their partner.
- Males of these species can maximize their reproductive success by seeking other mates.
- Therefore, polygyny is relatively common in such animals.

Male Parental Care: Certainty of Paternity

- Certainty of paternity is an important factor influencing mating behavior and parental care of males.
- Certainty of maternity of young born to or eggs laid by a female is not doubtful. However, certainty of paternity is relatively low in most species, even within a monogamous relationship.
- Male parental care is found in very few species of birds and mammals. In such species, males engage in behaviors that appear to increase their certainty of paternity. These behaviors include guarding females, removing any sperm from the female reproductive tract before copulation, and introducing large quantities of sperm to displace the sperm of other males.

Topic-257 Sexual Selection and Mate Choice

Sexual Selection

Sexual selection among the members of a species can take the form of:

- Intersexual selection
- Intrasexual selection

Intersexual Selection

- Intersexual selection involves mate choice.
- In this selection, members of one sex choose mates on the basis of particular characteristics of other sex, such as appearance, color or courtship songs.
- Intersexual selection has led to sexual dimorphism in secondary sexual characteristics, such as the ornate plumage of birds, or the antlers of deer, or the manes of lions.
- Such selective characters correlate in general with the mate's health and vitality.
- Purpose of mate choice is to choose a healthy partner that is likely to enhance the reproductive success.
- Mate choice patterns vary among species. In some, females make choice for their mates while in others, males make selections among the available pool of females.

Intrasexual Selection

- Intrasexual selection involves competition between members of the same sex for access to members of the opposite sex.
- Intrasexual selection usually occurs between males and may take the form of male-to-male combat or fight.
- Such species or individuals have developed better weapons for competition e.g. horns, antlers etc.
- There are two main types of competition over females, scramble and contest competition.
- Scramble involves a race to get to the female first and grab her. Male body size and vigor contributes to success.

- Contest is a more typical form of competition where the male with the best fighting technique, largest body size or the largest weapons will win the female.

Topic-258 Social Behaviors

Social behaviors consist of a set of interactions among individuals of the same species.

Sociality in Animals

- A wide range of sociality occurs among animals.
- Some animals rarely interact with one another. Such animals are called asocial animals e.g. mosquitoes and polar bears.
- Social animals live together in large groups or form tightly knit colonies.

Examples of Social Animals

- Ants and termites
- Bees and wasps
- Many birds
- Wilderbeests
- Wolves and Lions
- Monkeys and gorillas
- Humans

Characters of Social Behaviors

- Most social behaviors are based on cooperation, competition, conflict, exchange, or coercion or sacrifice.
- Most social animals often cooperate with one another to conduct many tasks.
- Many social behaviors are adaptive and increase animal's fitness and reproductive success.
- Animals in groups show joint aggregation against predators.
- Many social behaviors are agonistic and based on competition. Such behaviors are selfish and individuals benefit at the expense of others
- Even in the absence of agonistic behavior, most adaptations that benefit one individual may indirectly harm others.

  For example, superior foraging ability of one individual may leave less food for others.

- Some behaviors are altruistic or unselfish in which an animal favors other at the expense of its own benefits.

Topic-259 Social Behavior: Living in Groups

Animal populations are often organized into groups.

Groups may be:

- Simple aggregation of individuals
- Complex animal societies

Aggregation

- An aggregation is any form of gathering of animals or the process of coming together.
- There is minimal interaction between members of an aggregation.
- For example, several Drosophila flies living on a piece of rotting fruit.

Animal Societies

- A stable group of individuals of the same species that maintains a close-knit cooperative social relationship is called animal society.
- This association typically has complex social organization and extends beyond the level of mating and taking care of young.
- Such societies are observable both in invertebrates and vertebrates.

Advantages of Social Grouping

- Belonging to a group gives protection against predators.
- Group members may warn each other about an intruder.
- Cooperative hunting and search of food increases the feeding efficiency.
- Social huddling in cold weather protects its members from harsh environment.
- Members of a social groups help each other in finding mate and rearing of young.
- Many insects have developed social grouping. These social groups have division of labor. Specific individuals perform specialized tasks of defense, food procurement and feeding of young.

Disadvantages of Social Groupings

- Competition for resources (food, mates) develops between members of a social group.
- Diseases and parasites spread more rapidly in a group of animals.
- They interfere each other for reproduction and rearing of young.
- Large congregations of animals are more susceptible to predation. During attacks, large groups have difficulty seeking hiding places.

Topic-260 Agonistic Behaviors

Agonistic Behavior

- Agonistic behaviors are confrontation behaviors involving aggression, threat displays, attacks and fights between animals.
- Agonistic behaviours are seen in many animal species because resources including food, shelter, and mates are often limited.
- These behaviors are usually used only to intimidate the enemies and are rarely lethal
- Agnostic behaviors seem to be antisocial but these are necessary to maintain social order.

Types of Agonistic Behaviors

Agnostic behaviors are important in the maintenance of territories and dominance hierarchies. So they are of two types:

- Territorial Behaviors
- Dominance hierarchies

1. Territorial Behaviors

Territorial ownership and defense is an important aspect of sociality in animal populations e.g. insects, crustaceans, fishes, amphibians, lizards, birds, and mammals, including humans.

A territory is a fixed area in which an animal spends most of its time and defends it from intruders of the same species.

Purposes of Territorial Behaviors

Territorial behavior has multiple purposes:

Breeding place: Many male birds and mammals occupy a breeding territory. The male actively defends his area against other males. He attracts a female in his territory and performs courtship and mating without interference.

Food source: Some animals occupy territories that contain abundant food supply.

Shelter: Purpose of some territories is to provide shelter to the animal from predators and unfavorable climate.

2. Dominance Hierarchies

Dominance hierarchies exist in many vertebrate groups.

- The organization of group of animals in such a way that some members of the group have greater access to resources like food or mates than others is called dominance hierarchies.
- Some animals are present near the top of the order. They have first choice of resources.
- The animals present near the bottom do not get sufficient resources.

Topic-261 Altruism

Definition

Altruism is a selfless behavior in which an individual endangers his own survival or sacrifices some of its own reproductive potential to benefit another individual.

Evolutionary Basis

Altruism is an innate trait developed through the process of evolution in many animal species particularly in species with complex social structures e.g. primates, some other mammals and insects. However, there are wide individual variations in tendencies toward altruism.

Negates Darwinism Fitness Principle

Altruism is a different kind of social behavior which has been hard to explain within the framework of Darwinism and natural selection because it reduces the fitness of the altruist.

Significance of Altruism

- In ethology, the costs and benefits of a behavior are measured in terms of reproductive fitness, or expected number of offspring.
- So by behaving altruistically, an organism reduces the number of offspring it is likely to produce itself. However, by sacrificing its own favors to save others, it indirectly boosts the would-be number of offspring that other members of species are likely to produce.

Examples of Altruism

- A belding's ground squirrel that sees a predator gives a high-pitched alarm call that alerts unaware individuals to retreat to their burrows. The squirrel that warns others becomes conspicuous and increases the risk of being killed by the predator. Similar behaviors are observed in crows and monkeys.
- In social insect colonies (ants, wasps, bees and termites), sterile workers devote their whole lives to caring for the queen, constructing and protecting the nest, foraging for food, and tending the larvae. Such behaviour is maximally altruistic
- Similarly, worker bees sting intruders. This behavior helps defend the hive but results in the death of those workers.
- Vampire bats regurgitate blood and donate it to other members of their group who have failed to feed that night, ensuring they do not starve.

Topic-262 Kin Selection and Inclusive Fitness

Kin Selection

Kin selection is an altruistic evolutionary strategy that favors the reproductive success of an organism's close relatives, even at a cost of organism's own survival and reproduction.

Kin selection is an instance of inclusive fitness, which combines the number of offspring produced by an individual itself and the number of individuals it can ensure by supporting other close relatives (siblings).

William Hamilton proposed the idea of kin (relatives) selection to explain how selection acting on related animals can affect the fitness of an individual.

Through kin selection, a gene carried by a particular individual passes to the next generation through a related individual.

The fitness of an individual is based on the genes it passes on as well as on those common genes that its relatives pass on.

For kin selection to be effective, the individuals of a group must be able to identify their relatives. This is the reason that it is found in animals with advanced social structures i.e. primates and social insects.

The altruistic kin selection behavior is most readily apparent in the act of parents sacrificing for their offspring. When parents sacrifice their own well-being to produce and aid offspring, this actually increases the fitness of the parents because it maximizes their genetic representation in the population.

Inclusive Fitness

Inclusive fitness is the measure of fitness of whole group of members of a species which may not be close relatives.

An organism's inclusive fitness is determined by its personal fitness, plus its weighted effects on the fitness of every other individual in the population. The weights are determined by the coefficient of relationship.

It is favored by natural selection that acts to maximize the fitness of group of individuals in the population, rather than thinking in terms of selfish genes trying to maximize their future representation in the gene-pool.

Helping behavior of individuals of a species for those who are not their close relatives is an example of inclusive behavior.

Topic-263 Reciprocal Altruism

Reciprocal Altruism

- Some animals occasionally behave altruistically toward others who are not relatives.
- A baboon may help an unrelated companion in a fight, or a wolf may offer food to another wolf even though they share no kinship.
- Such behavior can be adaptive if the aided individual returns the favor in future.
- This sort of exchange of aid is called reciprocal altruism.
- Reciprocal altruism is limited largely to species with social groups stable enough that individuals have many chances to exchange aid (e.g. chimpanzees, humans).
- It generally occurs when individuals are likely to meet again.

Cheating

- Reciprocal altruism is also associated with negative consequences if favors are not returned to individuals who had been helpful in the past.
- Such a pattern of behavior that does not reciprocate favors is referred to as “cheating” by behavioral ecologists.
- It occurs as cheating may benefit the cheater substantially.

Game Theory and Tit for Tat Strategy

- Game theory explains how reciprocal altruism dominates over cheating behavior. It involves a behavioral strategy called tit for tat.
- In the tit-for-tat strategy, an individual treats another in the same way it was treated the last time they met.
- Individuals adopting this behavior are always altruistic, or cooperative, on the first encounter with another individual and will remain so as long as their altruism is reciprocated.
- When their cooperation is not reciprocated, individuals employing tit for tat will retaliate immediately but return to cooperative behavior as soon as the other individual becomes cooperative.
- The tit for tat strategy has been used to explain the few apparently reciprocal altruistic interactions observed in animals-ranging from blood sharing between nonrelated vampire bats to social grooming in primates.

Topic-264 Social Learning

Social Learning

The type of learning that involves observing and copying others behavior is called social learning.

Example: young chimpanzees learn to crack palm nuts by copying the behavior of more experienced individuals.

Social Learning and Culture

- Social learning forms the roots of culture. Culture is defined as a system of information transfer through social learning or teaching that influences the behavior of individuals in a population.
- Cultural transfer of information can alter behavioral phenotypes and thereby influence the fitness of individuals.
- Culture based changes in phenotype occur on a much shorter time scale than changes resulting from natural selection.

Social Learning Behaviors

Social learning behaviors are prevalent in animal kingdom, specially vertebrates.

Types:

1. Copying behaviors
2. Fine-tuned social learning

Copying Behaviors: Mate-Choice Copying

- Mate-choice copying is a form of social learning behavior in which individuals in a population copy the mate choice of others.
- In mate choice copy, an individual is attracted to a mate that is also attractive to most other individuals of opposite sex.
- This behavior has been demonstrated in several fish and bird species.

Fine-tuned Social Learning

- Fine-tuned learning with experience in humans is well known. A human baby starts to learn expressive words and accents that modify and improve with age.
- Studies on vervet monkeys have demonstrated fine-tuned learning in animals.

Fine-Tuned Learning of Alarm Calls

- Vervet monkeys produce a complex set of alarm calls. Their repertoire of calls includes giving distinct alarm calls for predators like leopards, eagles or snakes.
- When a vervet sees a leopard, it gives a loud barking sound. When it sees an eagle, it gives a short double-syllabled cough. The alarm call for a snake is a “chutter”.

- Upon hearing a particular alarm call, other vervets behave in an appropriate way: They run up a tree on hearing the alarm for a leopard. Look up on hearing the alarm for an eagle. Look down on hearing the alarm for a snake.
- Infant vervet monkeys give alarm calls, but in a relatively undiscriminating way. For example, they give the "eagle" alarm on seeing any bird, including harmless birds. With age, the monkeys improve their accuracy. Infants learn how to give the right call by observing other members of the group and receiving social confirmation of correct calls from learned members.

Topic-265 Evolution and Human Culture

Genetic Basis of Human Culture

- Human culture and certain social behaviors are due to expression of genes that have been perpetuated by natural selection. So, human culture is related to Darwinian evolutionary theory.
- However, an irrational interpretation of the genetic and evolutionary basis of human behavior may be misleading. It may be used to justify the status quo in human society, thus rationalizing social injustices and animal behaviors of human.

Plasticity of Human Behavior

- Actually, the evolutionary explanation of human behavior does not make human behavior a rigid genetic model.
- Just as the anatomical features of individuals vary extensively, so do the inherent behaviors.
- Just like environment intervenes in the phenotypic expression of a genotype for physical traits, it is more significantly involved in the expression of behavioral traits.
- As human have greater capacity for learning, human behavior is probably more plastic than that of any other animal.

Control of Instinctive Behavior by Culture

- In the human cultural history, humans have built up a diversity of structured societies with institutionalized governments, laws, cultural values, and religions.
- These institutions define what behavior is acceptable and what is not. Many of the instinctive behaviors that might enhance an individual's Darwinian fitness are considered unacceptable by law and are prohibited.
- Perhaps these are the social and cultural institutions that make human behavior distinct and controlled than that of other animal societies.